ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

Control Theory

هذا العمل صدقة جارية عن أرواح موتى وشهداء المسلمين عامة وشهداء فلسطين خاصة

اللهم إنهم في ذمتك وحبل جوارك، فَقِهِم من فتنة القبر وعذاب النار، وأنت أهل الوفاء والحق، اللهم اغفر لهم وارحمهم، فإنك أنت الغفور الرحيم

اللهم اغفر لهم وارحمهم واعف عنهم وعافهم، وأكرم نزلهم ووسع مدخلهم، واغسلهم بالماء والثلج والبرد، ونقهم من الخطايا كما ينقى الثوب الأبيض من الدنس، وأبدلهم داراً خيراً من دارهم وأهلاً خيراً من أهلهم وأزواجاً خيراً من أزواجهم، وقهم فتنة القبر وعذاب النار

اللهم إنهم عبيدك وابناء عبيدك وابناء إماءك كانوا يشهدون أن لا إله إلا أنت، وأن محمداً عبدك ورسولك، وأنت أعلم بهم، اللهم إن كانوا محسنين فزدهم في إحسانهم، وإن كانوا مسيئين فتجاوز عن سيئاتهم، اللهم لا تحرمنا أجرهم ولا تفتنا بعدهم

اللهم سدد رمي المقاومة

اللهم لا ترفع لليهود راية ولا تحقق لهم غاية واجعلهم لمن خلفهم عبرة وآية

قال تعالى:{﴿ سُبْحَانَ الَّذِي أَسْرَى بِعَبْدِهِ لَيلاً مِنَ الْمَسْجِدِ الْحَرَامِ إِلَى الْمَسْجِدِ الأَقْصَا الَّذِي بَارَكْنَا حَوْلَهُ لِنُرِيَهُ مِنْ آيَاتِنَا إِنَّهُ هُوَ السَّمِيعُ الْبَصِيرُ * وَآتَيْنَا مُوسَى الْكِتَابَ وَجَعَلْنَاهُ هُدىً لِبَنِي إِسْرَائِيلَ أَلاَّ تَتَّخِذُوا مِنْ دُونِي وَكِيلاً * ذُرِّيَّةَ مَنْ حَمَلْنَا مَعَ نُوحٍ إِنَّهُ كَانَ عَبْداً شَكُوراً * وَقَضَيْنَا إِلَى بَنِي إِسْرَائِيلَ فِي الْكِتَابِ لَتُفْسِدُنَّ فِي الأَرْضِ مَرَتَيْنِ وَلَتَعْلُنَّ عُلُواً كَبِيراً * فَإِذَا جَاءَ وَعْدُ أُولاهُمَا بَعَثْنَا عَلَيْكُمْ عِباداً لَنَا أُولِي بَأْسٍ شَدِيدٍ فَجَاسُوا خِلالَ الدِّيارِ وَكَانَ وَعْداً مَفْعُولاً * ثُمَّ رَدَدْنَا لَكُمُ الْكَرَّةَ عَلَيْهِمْ وَأَمْدَدْنَاكُمْ بِأَمْوَالٍ وَبَنِينَ وَجَعَلْنَاكُمْ أَكْثَرَ نَفِيراً * إِنْ أَحْسَنْتُمْ أَحْسَنْتُمْ لِأَنْفُسِكُمْ وَإِنْ أَسَأْتُمْ فَلَهَا فَإِذَا جَاءَ وَعْدُ الآخِرَةِ لِيَسُوؤا وُجُوهَكُمْ وَلِيَدْخُلُوا الْمَسْجِدَ كَمَا دَخَلُوهُ أَوَّلَ مَرَّةٍ وَلِيُتَبِّرُوا مَا عَلَوْا تَتْبِيراً * عَسَى رَبُّكُمْ أَنْ يَرْحَمَكُمْ وَإِنْ عُدْتُمْ عُدْنَا وَجَعَلْنَا جَهَنَّمَ لِلْكَافِرِينَ حَصِيراً ﴾} سورة الإسراء

Eng. Mohannad Akram Mohammed Al Manasra, BAU-FET Electrical Power Engineering Graduate

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

والكمال لله وحده لهذا من الممكن أن تتواجد بعض الأخطاء في الحلول لذلك نرجو منكم الاستفسار عن الأسئلة التي تريدون السؤال عنها بالتواصل مع مُنشأ هذا العمل

Gmail **Telephone** **Linkedin**

Telegram **Pinterest** **Facebook**



ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل منشأ هذا العمل

Contents

1. Previous Exams Questions 4
1.1 Introduction to Control Systems 4
1.2 Laplace Transforms 7
1.3 Modeling of Mechanical, Liquid-Level and Electrical Control Systems 16
1.4 Block Diagram Reduction and Obtaining Transfer Function 22
1.5 Calculation of Time Response of First-Order & Second-Order 30
1.6 Control Action: Applying Different Types of Controllers (P, I, D, PI, PD and PID) 51
1.7 Analyzing System Stability Using Routh's and Nyquist Criteria 56
1.8 Drawing Bode Plot and Nyquist Plot for a Given Transfer Function 60
1.9 Applying Root-Locus Technique for Control System Analysis 73
2. Additional Questions 75
2.1 Introduction to Control Systems 75
2.2 Laplace Transforms 76
2.3 Modeling of Mechanical, Liquid-Level and Electrical Control Systems 78
2.4 Block Diagram Reduction and Obtaining Transfer Function 80
2.5 Calculation of Time Response of First-Order & Second-Order 93
2.6 Control Action: Applying Different Types of Controllers (P, I, D, PI, PD and PID) 142
2.7 Analyzing System Stability Using Routh's and Nyquist Criteria 145
2.8 Drawing Bode Plot and Nyquist Plot for a Given Transfer Function 158
2.9 Applying Root-Locus Technique for Control System Analysis 173
Appendix I 174
Appendix II 174
Appendix III 175
References 176

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

Previous Exams Questions

Introduction to Control Systems

1. Feedback is an operation used to:

a. To maintain a constant output
b. To change the type of the output
c. Reduce the difference between the actual output and the desired output
d. Reduce the difference between the actual output and some reference input

Correct answer: Reduce the difference between the actual output and some reference input

2. In steam engine speed control system, the feedback element is:

a. Gear box
b. Pilot valve
c. Control valve
d. Power cylinder

Correct answer: Gear box

3. In temperature control system, the feedback path is:

a. Analog to digital converter
b. Interface
c. Thermometer
d. All above

Correct answer: All above

4. The output of the controller is called:

a. Manipulated variable
b. Feedback signal
c. Reference signal
d. Error signal

Correct answer: Manipulated variable

5. Open loop control can be used only if:

a. If the relationship between the output and the input is well known
b. There are no internal disturbances
c. There are no external disturbances

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

d. All above

Correct answer: All above

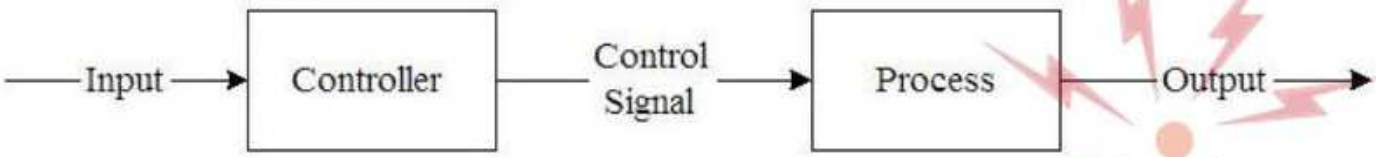
Open-loop control can be used, in practice, only if the relationship between the input and output is known and if there are neither internal nor external disturbances.

6. Which of the following is an example of an open loop system?

a. Household refrigerator
b. Washing machine programs
c. Air condition
d. Not mentioned

Correct answer: Washing machine programs

❖ Open-loop control system:



Open Loop System

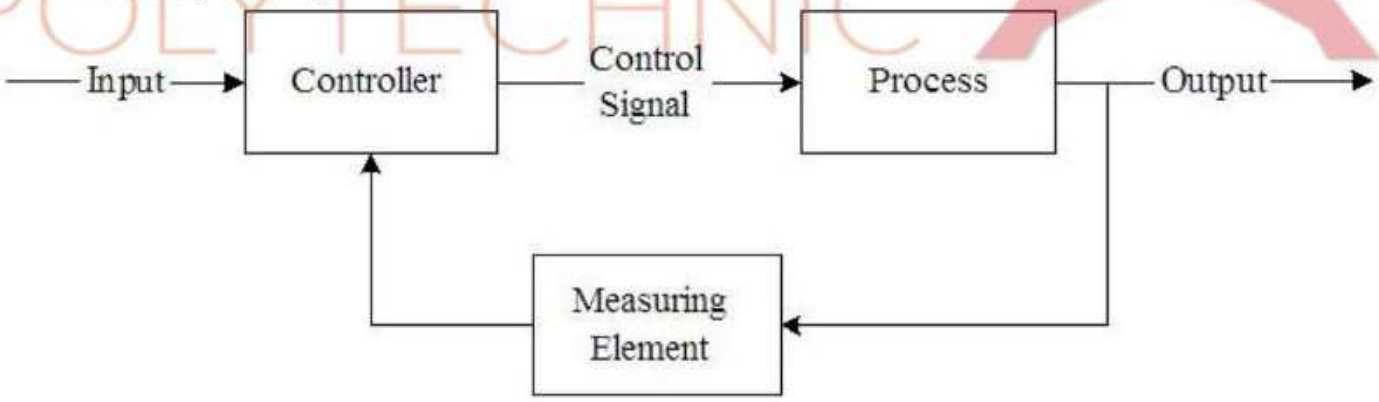
➢ These are the systems in which the control action is independent of output.
➢ Examples: Traffic signals, Bread toaster, Sprinkler, Ordinary washing machine, Execution of a program by computer, and Systems having no sensor, etc.
➢ Advantages:

1. Simple construction and design.
2. Cost is less.
3. Maintenance is easy.
4. No problem of instability.
5. Convenient to measure when the output is difficult to measure.

➢ Disadvantages:

1. These are less accurate, and their accuracy depends on the calibration.
2. Inaccurate results are obtained with parameter variations within the system.
3. Re-calibration of the controller is required from time to time to maintain accuracy.

❖ Closed-loop control system:



Closed Loop System

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

- These are the systems in which the control action depends on the output. These systems have a tendency to oscillate.
- Examples: Temperature controllers, Speed control of the motor, Systems having sensors, Human eye, Household refrigerator, Respiratory system of an animal, Stabilization of air pressure entering into a mask, etc.
- Advantages:

1. More accurate.
2. Non-linear distortions are less.
3. The output is less sensitive to parameter changes within the system.
4. Bandwidth increases.

- Disadvantages:

1. Design is complicated.
2. More expensive.
3. May become unstable if there are malfunctions in the feedback.

7. An open-loop control system utilizes an actuating device to control a process _____:

a. Without using feedback
b. Using feedback
c. Not mentioned
d. Using sensor

Correct answer: Without using feedback

8. Which system has tendency to oscillate?

a. Closed loop system
b. Open loop system
c. Neither open loop system nor closed loop system
d. Both open loop system and closed loop system

Correct answer: Closed loop system

9. The major disadvantage of a feedback system may be:

a. Inaccuracy
b. Inefficiency
c. Instability
d. Unreliability

Correct answer: Instability

POLYTECHNIC

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

Laplace Transforms

10. The residues (constants) of partial fraction for $F(s) = \frac{1}{\{s(s+1)\}}$ are:

a. 1;-1
b. 0;-1
c. 0;1
d. 1;1

Correct answer: 1;-1

11. Forward Laplace transform of $f(t) = te^{-2t}$ is:

a. $\frac{1}{s^2}$
b. $\frac{1}{s}$
c. $\frac{1}{(s+1)^2}$
d. $\frac{1}{(s+2)^2}$
e. $\frac{1}{(s-1)^2}$

Correct answer: $\frac{1}{(s+2)^2}$

12. Forward Laplace transform of $f(t) = e^{+2t}$ is:

a. $\frac{1}{s-2}$
b. $\frac{1}{2s}$
c. $\frac{1}{s+2}$
d. s^2

Correct answer: $\frac{1}{s-2}$

13. A damped ramp function can be created multiplying ramp function by:

a. Constant
b. Step function
c. Exponential function (precisely, Damped exponential function)
d. Impulse function
e. Unit step function
f. Sinusoidal (sin) function

Correct answer: Exponential function (precisely, Damped exponential function)

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

14. The residues (constants) of partial fraction for $F(s) = \frac{1}{s^2}$ are:

a. $0; 1$
b. $1; -1$
c. $1; 1$
d. $0; -1$

Correct answer: $0; 1$

15. A damped sinusoidal function can be created multiplying sinusoidal by:

a. Step function
b. Ramp function
c. Impulse function
d. Exponential function

Correct answer: Exponential function

16. The damped cosine function can be created multiplying cosine function by:
Or
The damped sine function can be created multiplying cosine function by:

a. Sin function
b. Rise exponential function
c. Unit step
d. Damped exponential function

Correct answer: Damped exponential function

17. When the function F(s) has distinct poles, the type of the function f(t) is:
Or
When the function F(s) has all non-repeated simple roots, the type of the function f(t) is:

a. Exponential function
b. Critically damped sinusoidal
c. Damped sinusoidal
d. Damped ramp
e. (Undamped/ Sinusoidal function) (precisely, undamped sinusoidal function)

Correct answer: Exponential function

18. If $y(t) = e^{-t} - e^{t}$, Then Y(s) will be:

a. $\frac{2}{s^2+1}$
b. $-\frac{2}{s^2+1}$
c. $\frac{1}{s^2+1}$
d. None of the above

Correct answer: None of the above

$$Y(s) = -\frac{2}{s^2 - 1}$$

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

19. The output of the function of a system, find $y(t)$:

$$Y(s) = \frac{3}{s^2 + 8s + 7}$$

a. $0.5e^{-t} - 0.5e^{-7t}$
b. $-0.5e^{-t} + 0.5e^{-7t}$
c. $-0.5e^{-t} - 0.5e^{-7t}$
d. $0.5e^{-t} + 0.5e^{-7t}$

Correct answer: $0.5e^{-t} - 0.5e^{-7t}$

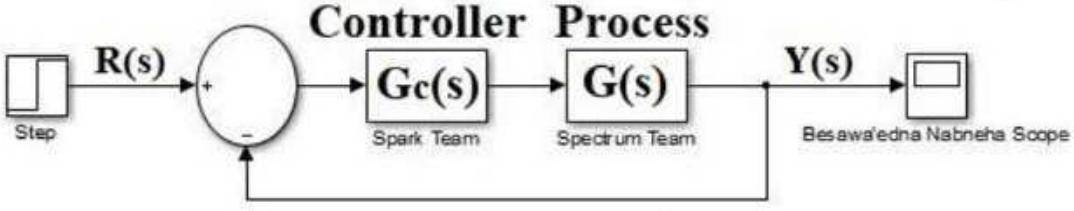
$$Y(s) = \frac{3}{s^2 + 8s + 7} = \frac{3}{(s+1)(s+7)} = \frac{A}{s+1} + \frac{B}{s+7} = \frac{As + 7A + Bs + B}{(s+1)(s+7)}$$

$$7A + B = 3$$
$$A + B = 0$$
$$B = -A$$
$$7A - A = 3$$
$$6A = 3$$
$$A = \frac{1}{2}$$
$$B = -A = -\frac{1}{2}$$

$$Y(s) = \frac{\frac{1}{2}}{s+1} + \frac{-\frac{1}{2}}{s+7}$$

$$y(t) = \mathcal{L}\{Y(s)\} = \mathcal{L}\left\{\frac{\frac{1}{2}}{s+1} + \frac{-\frac{1}{2}}{s+7}\right\} = \frac{1}{2}e^{-t} - \frac{1}{2}e^{-7t}$$

20. Consider the following system, the steady state output y_{ss} if the input is unit ramp $\frac{1}{s^2}$ will be:



$$G_c(s)G(s) = \frac{6}{s(s+3)}$$

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل منشأ هذا العمل

a. $-\frac{2}{9}+\frac{2}{3}t-t^2+\frac{2}{9}e^{-3t}$

b. $\frac{2}{9}-\frac{2}{3}t+t^2-\frac{2}{9}e^{-3t}$

c. $-\frac{2}{9}+\frac{2}{3}t-2t^2+\frac{2}{9}e^{-3t}$

d. $\frac{2}{9}-\frac{2}{3}t+2t^2-\frac{2}{9}e^{-3t}$

Correct answer: $\frac{2}{9}-\frac{2}{3}t+t^2-\frac{2}{9}e^{-3t}$

$$Y(s)=R(s)G_c(s)G(s)=\frac{1}{s^2}\frac{6}{s(s+3)}=\frac{A}{s}+\frac{B}{s^2}+\frac{C}{s^3}+\frac{D}{s+3}$$

$$Y(s)=\frac{6}{s^3(s+3)}=\frac{As^2(s+3)}{s^3(s+3)}+\frac{Bs(s+3)}{s^3(s+3)}+\frac{C(s+3)}{s^3(s+3)}+\frac{Ds^3}{s^3(s+3)}$$

$$Y(s)=\frac{As^3+3As^2+Bs^2+3Bs+Cs+3C+Ds^3}{s^3(s+3)}=\frac{6}{s^3(s+3)}$$

$$3C=6\cdots\cdots\cdots(1)$$

$$C=2$$

$$3B+C=0\cdots\cdots\cdots(2)$$

$$B=-\frac{C}{3}=-\frac{2}{3}$$

$$3A+B=0\cdots\cdots\cdots(3)$$

$$A=-\frac{B}{3}=-\frac{-\frac{2}{3}}{3}=\frac{2}{9}$$

$$A+D=0\cdots\cdots\cdots(4)$$

$$D=-A=-\frac{2}{9}$$

$$y(t)=\mathcal{L}\{Y(s)\}=\mathcal{L}\left\{\frac{A}{s}+\frac{B}{s^2}+\frac{C}{s^3}+\frac{D}{s+3}\right\}=\mathcal{L}\left\{\frac{\frac{2}{9}}{s}+\frac{-\frac{2}{3}}{s^2}+\frac{2}{s^3}+\frac{-\frac{2}{9}}{s+3}\right\}$$

$$y(t)=\frac{2}{9}-\frac{2}{3}t+t^2-\frac{2}{9}e^{-3t}$$

21. What are the zeros for the following transfer function?

$$\frac{Y(s)}{R(s)}=\frac{3}{s^2+4s+3}$$

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

a. 0
b. 3, 1
c. $-3, -1$
d. No zeros
e. None of the above

Correct answer: No zeros

22. The Laplace transform of $f(t) = te^{-9t}$ is:

a. $\frac{1}{s+9}$
b. $\frac{1}{(s+9)^2}$
c. $\frac{9}{s^2+9}$
d. $\frac{1}{s^2+9}$

Correct answer: $\frac{1}{(s+9)^2}$

23. The poles of the following transfer function $G(s) = \frac{2s+1}{s^2+4s+3}$ are:

a. $-1, -3$
b. $1, -3$
c. $1, 3$
d. $-1, 3$

Correct answer: $-1, -3$

The poles of the transfer function is the roots of the denominator.

$$as^2 + bs + c = 0$$

$$s^2 + 4s + 3 = 0$$

$$s = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$s = \frac{-4 \pm \sqrt{4^2 - 4 * 1 * 3}}{2 * 1} = -1, -3$$

24. The initial value $f(o)$ of the function $F(s) = \frac{s+2}{s(s+1)}$ is:

a. 0.5
b. 1
c. 2
d. 0

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

Correct answer: 1

$$f(o) = \lim_{s\to\infty} sF(s) = \lim_{s\to\infty} s * \frac{s+2}{s(s+1)} = \lim_{s\to\infty} \frac{s+2}{s+1}$$

Using L'Hôpital's rule:

$$f(o) = \lim_{s\to\infty} * \frac{1}{1} = 1$$

Another way to find the solution:
Using partial fractions:

$$\frac{s+2}{s(s+1)} = \frac{A}{s} + \frac{B}{s+1} = \frac{As + A + Bs}{s(s+1)}$$

$$s + 2 = As + A + Bs$$

By using the comparison method:

$$A = 2$$

$$A + B = 1$$

$$B = -1$$

$$F(s) = \frac{2}{s} - \frac{1}{s+1}$$

$$f(t) = 2 - e^{-t}$$

$$f(t = 0) = 2 - e^{0} = 2 - 1 = 1$$

25. The partial fraction expansion of $F(s) = \frac{2}{(s+2)(s+1)}$ is:

a. $\frac{a}{(s+1)} + \frac{b}{(s+2)}$
b. $\frac{2}{(s+2)(s+1)}$
c. $\frac{as+b}{(s+2)(s+1)}$
d. $\frac{2}{(s+1)} + \frac{2}{(s+2)}$

Correct answer: $\frac{a}{(s+1)} + \frac{b}{(s+2)}$

26. For $f(t) = e^{-t} + te^{-t} + e^{-2t}$ the system has _____ pole(s).

a. 3
b. 2
c. 4
d. 1

Correct answer: 3

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

$$\mathcal{L}\{f(t)\} = \mathcal{L}\{e^{-t} + te^{-t} + e^{-2t}\} = \frac{1}{s+1} + \frac{1}{(s+1)^2} + \frac{1}{s+2}$$

$$= \frac{(s+1)(s+2)}{(s+1)^2(s+2)} + \frac{s+2}{(s+1)^2(s+2)} + \frac{(s+1)^2}{(s+1)^2(s+2)}$$

$$= \frac{(s+1)(s+2) + s + 2 + (s+1)^2}{(s+1)^2(s+2)}$$

Poles are the roots of denominator, so the poles are:

$$s = -1, -1, -2$$

27. Given the following differential equation:

$$\frac{d^2y}{dt^2} + 3\frac{dy}{dt} + 2y = u(t)$$

$$Y(s) = ?$$

a. $\frac{1}{(s^2+3s-2)}$

b. $\frac{-1}{s(s^2+3s+2)}$

c. $\frac{-1}{s(s^2+3s-2)}$

d. $\frac{1}{s(s^2+3s+2)}$

Correct answer: $\frac{1}{s(s^2+3s+2)}$

$$Y(s)s^2 + 3Y(s)s + 2Y(s) = \frac{1}{s}$$

$$Y(s) = \frac{1}{s(s^2+3s+2)}$$

28. When the function F(s) has repeated poles, the type of the function f(t) is:
Or
When the function F(s) has multiple poles, the type of the function f(t) is:

a. Damped sinusoidal
b. (Undamped/ Sinusoidal function) (precisely, undamped sinusoidal function)
c. Critically damped sinusoidal
d. Damped ramp
e. Exponential function

Correct answer: Damped ramp

29. Laplace transform of unit step function u(t) is:

a. 1

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص او جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

b. s
c. t
d. $\frac{1}{s}$

Correct answer: $\frac{1}{s}$

30. Assume zero initial conditions, and r(t) is unit impulse function, the solution of the following equation is:

$$\frac{dy}{dt} + 2y(t) = r(t)$$

a. e^{-2t}
b. e^{2t}
c. e^{t}
d. e^{-t}
e. None of the above

Correct answer: e^{-2t}

Using Laplace Transform:

$$sY(s) + 2Y(s) = 1$$

$$Y(s) = \frac{1}{s+2}$$

Using inverse Laplace Transform:

$$y(t) = e^{-2t}$$

31. The s function $\frac{s+2}{(s+2)^2+9}$ is the forward Laplace transform of:

a. Cosine function
b. Sinusoidal function
c. Damped cosine function
d. Damped sinusoidal function

POLYTECHNIC

Correct answer: Damped cosine function

$$\mathcal{L}^{-1}\left\{\frac{s+2}{(s+2)^2+9}\right\} = e^{-2t}\cos(3t)$$

32. Laplace transform is not used to:

a. Solving linear differential equations
b. Determine transfer function
c. Solving nonlinear differential equations
d. Converting linear differential equations into algebraic equations

Correct answer: Solving nonlinear differential equations

33. Laplace transform of ramp function f(t)=At is:

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص او جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

a. $\frac{1}{s}$
b. s^2
c. s
d. $\frac{A}{s^2}$

Correct answer: $\frac{A}{s^2}$

34. The inverse Laplace transform of $F(s) = \frac{2}{(s+1)(s+2)}$ is:

a. $2e^{-t} + 2e^{-2t}$
b. $e^{t} - 2e^{2t}$
c. $2e^{-t} - 2e^{-2t}$
d. $2e^{t} + 2e^{2t}$

Correct answer: $2e^{-t} - 2e^{-2t}$

35. A pole on the real axis generates an exponential response:

a. True
b. False

Correct answer: True

36. Use the final value Theorem below to find the final value of $X(z)$ below:

$$x(\infty) = \lim_{z\to 1}[(1-z^{-1})X(z)]$$

$$X(z) = \frac{2}{1-z^{-1}} - \frac{1}{1-e^{-aT}z^{-1}}$$

a. 2
b. Zero
c. 1
d. None of the above

Correct answer: 2

$$x(\infty) = \lim_{z\to 1}[(1-z^{-1})X(z)] = x(\infty)$$

$$x(\infty) = \lim_{z\to 1}\left[(1-z^{-1})\left(\frac{2}{1-z^{-1}} - \frac{1}{1-e^{-aT}z^{-1}}\right)\right]$$

$$x(\infty) = \lim_{z\to 1}\left[2 - \frac{(1-z^{-1})}{1-e^{-aT}z^{-1}}\right] = 2$$

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

Modeling of Mechanical, Liquid-Level and Electrical Control Systems

37. A control system has three inputs and two outputs, the dimension of output matrix D is:

a. Two by three
b. Three by two
c. Two by two
d. One by one

Correct answer: Two by three
Where:
A(t): The state matrix, (n*n)
B(t): The input matrix, (n*r)
C(t): The output matrix, (m*n)
D(t): The direct transmission matrix, (m*r)
n: State variables
m: Outputs
r: Inputs

38. For standard translation mechanical system, the input signal to the system is:

a. Acceleration
b. Displacement
c. Velocity
d. Applied force

Correct answer: Applied force

39. In force-voltage analogy, velocity is analogous to:

a. Charge
b. Inductance
c. Current
d. Capacitance

Correct answer: Current
See Appendix I

40. The principle of superposition is applied to:

a. Linear time variant systems
b. Nonlinear time variant systems
c. Nonlinear time invariant systems
d. Linear time invariant systems

Correct answer: Linear time invariant systems

41. In RLC series circuit there are _____ storage elements:

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

a. 2
b. 4
c. 1
d. 3

Correct answer: 2

42. The parameters of translation mechanical system are: spring constant K, friction coefficient b and mass M The system variables are the applied force f(t), the displacement y(t), the speed v(t) and the acceleration a(t), the mass is given by:

a. $f(t) = M\,y(t)$
b. $f(t) = M\,v(t)$
c. $f(t) = M$
d. $f(t) = M\,a(t)$

Correct answer: $f(t) = M\,a(t)$

43. The parameters of translation mechanical system are: spring constant K, friction coefficient b and mass M The system variables are the applied force f(t), the displacement y(t), the speed v(t) and the acceleration a(t), the spring is given by:

a. $f(t) = K\,v(t)$
b. $f(t) = K$
c. $f(t) = K\,y(t)$
d. $f(t) = K\,a(t)$

Correct answer: $f(t) = K\,y(t)$

44. For standard translation mechanical system, the output and the input are related by:

a. First order differential equation
b. First Order algebraic equation
c. Second order algebraic equation
d. Second order differential equation

Correct answer: Second order differential equation

45. The transfer function of RL circuit, where the applied voltage is the input and the current is the output is:

a. $\frac{s}{RLs+1}$
b. $\frac{Ls}{RLs+1}$
c. $\frac{1}{Ls+R}$
d. $\frac{L}{RLs+1}$

Correct answer: $\frac{1}{Ls+R}$

46. In force-current analogy, mechanical force is analogous to:

a. Inductance
b. Current
c. Charge
d. Capacitance

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

Correct answer: Current
See Appendix I

47. An RC series circuit, if the applied voltage is the input, and the current is the output, then the transfer function is:

a. $\frac{1}{RCs+1}$
b. $\frac{Cs}{RCs+1}$
c. $\frac{Rs}{RCs+1}$
d. $\frac{R}{RCs+1}$

Correct answer: $\frac{Cs}{RCs+1}$

48. The transfer function of RL circuit, where the applied voltage is the input and the resistance voltage is the output is:

a. $\frac{1}{RLs+1}$
b. $\frac{R}{RLs+1}$
c. $\frac{R}{R+Ls}$
d. $\frac{L}{RLs+1}$
e. $\frac{s}{RLs+1}$

Correct answer: $\frac{R}{R+Ls}$

49. The relationship between the input and the output of a control system is given by a third order differential equation, the dimension of the state matrix A is:

a. Two by two
b. Three by three
c. One by one
d. Three by two

Correct answer: Three by three
Where:
A(t): The state matrix, (n*n)
B(t): The input matrix, (n*r)
C(t): The output matrix, (m*n)
D(t): The direct transmission matrix, (m*r)
n: State variables
m: Outputs
r: Inputs

50. For standard translation mechanical system, the output signal to the system is:

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

a. Applied force
b. Velocity
c. Acceleration
d. Displacement

Correct answer: Displacement

51. Block diagram representation cannot compute Gain of each component of the system:

a. False
b. Cannot determine
c. Not necessary
d. True

Correct answer: False

52. Linearize the nonlinear equation $y = 0.2x^3$ at $x = 1$:

a. $y = -0.6x - 0.4$
b. $y = 0.6x + 0.4$
c. $y = 0.6x - 0.4$
d. Not mentioned

Correct answer: $y = 0.6x - 0.4$

$$y = 0.2x^3$$

At $x = 1$, we have $y = 0.2$:

$\dot{y} = 0.6x^2$, so at $x = 1$, we have $\dot{y}(1) = 0.6$

The linearization is:

$$y = 0.2 + 0.6(x - 1) = 0.6x - 0.4$$

53. In force-voltage analogy, mass is analogous to:

a. Inductance
b. Capacitance
c. Current
d. Charge

Correct answer: Inductance

See Appendix I

54. Transfer function of a system is used to calculate which of the following?

a. The time constant
b. The order of the system
c. The output for any given input
d. The steady state gain

Correct answer: The output for any given input

55. In force-voltage analogy, damping coefficient is analogous to:

a. Inductance
b. Resistance
c. Capacitance

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

d. Charge

Correct answer: Resistance
See Appendix I

56. Poles of the input function determine the form of the transfer function:

a. True
b. False

Correct answer: False

57. The system given by $y''(t) + ay'(t) + y(t) = x(t)$ where a is constant can be represented by:

a. Simulation model
b. State space model
c. Transfer function
d. All mentioned

Correct answer: All mentioned
The differential equations can be represented by transfer function, simulation model and state space model.

58. In force-current analogy, velocity is analogous to:

a. Voltage
b. Resistance
c. Current
d. Capacitance
e. Inductance
f. Charge

Correct answer: Voltage
See Appendix I

59. The relationship between the input and the output of a control system is given by a third order differential equation, the number of state variable for this system is:

a. 3
b. 2
c. 4
d. 1

Correct answer: 3

60. An RC series circuit, if the applied voltage is the input, and the capacitance voltage is the output, then the transfer function is:

a. $\frac{Cs}{RCs+1}$
b. $\frac{R}{RCs+1}$
c. $\frac{s}{RCs+1}$
d. $\frac{1}{RCs+1}$

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

Correct answer: $\frac{1}{RCs+1}$

61. In two stages RC cascade circuit control system and unit step input, the number of poles of the output signal which is the voltage across the second capacitor are:

a. 3
b. 4
c. 1
d. 2

Correct answer: 3



بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

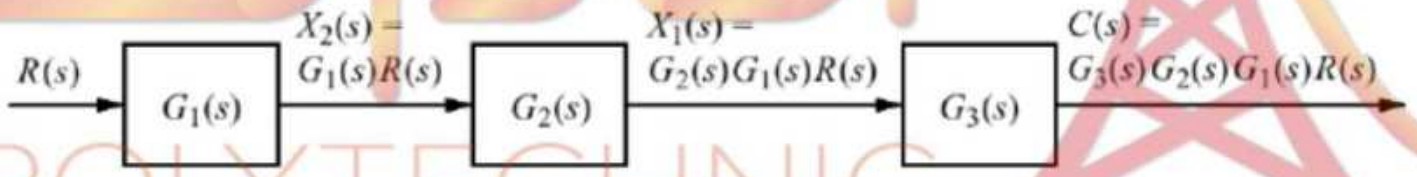
Block Diagram Reduction and Obtaining Transfer Function

62. The forward transfer function of two blocks G1 and G2 connected in series is:
Or
The equivalent transfer function of two blocks G1 and G2 connected in series is:

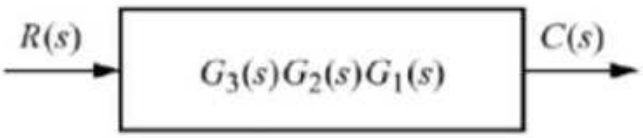
a. $G_1 G_2$
b. $\frac{G_1}{G_2}$
c. $G_1 + G_2$
d. $\frac{G_2}{G_1}$

Correct answer: $G_1 G_2$

Cascade (Series) Connections



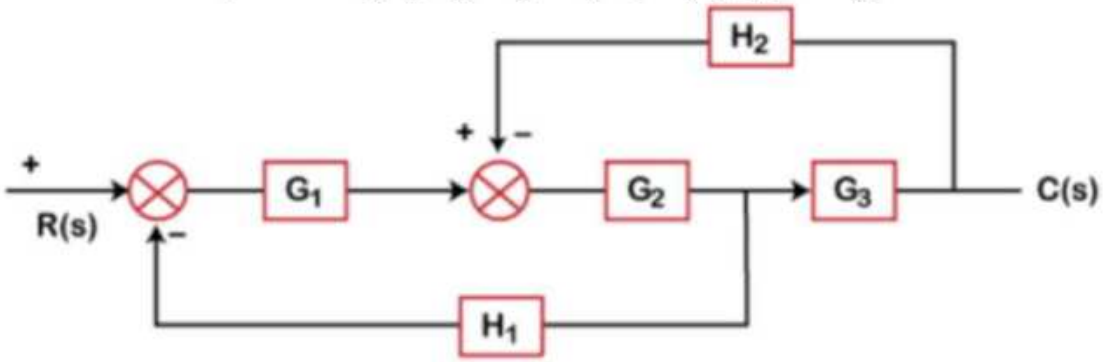
(a)



(b)

63. Determine the transfer function of the given system:

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل



a. $\frac{G1G2G3}{1+H2G2G3+G2G1H1}$
b. $\frac{G1G2G3}{1+G1G2G3H1H2}$
c. $\frac{G1G2G3}{1+G1G2G3H1}$
d. $\frac{G1G2G3}{1+G1G2G3H1+G1G2G3H2}$

Correct answer: $\frac{G1G2G3}{1+H2G2G3+G2G1H1}$

64. In closed loop system with positive feedback, the forward path is G(s), and the feedback path is H(s), the transfer function:

a. $\frac{G(s)}{[1+G(s)]}$
b. $\frac{G(s)}{[1-G(s)H(s)]}$
c. $\frac{G(s)}{[1-G(s)]}$
d. $\frac{G(s)}{[1+G(s)H(s)]}$

Correct answer: $\frac{G(s)}{[1-G(s)H(s)]}$

65. In closed loop transfer function, the numerator is:

a. Forward and open loop transfer function
b. Forward transfer function

Correct answer: Forward transfer function

66. The forward transfer function of two blocks G1 and G2 connected in parallel is:
Or
The equivalent transfer function of two blocks G1 and G2 connected in parallel is:

a. $G_1 - G_2$
b. $G_1 G_2$

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

c. $\frac{G_1}{G_2 s}$

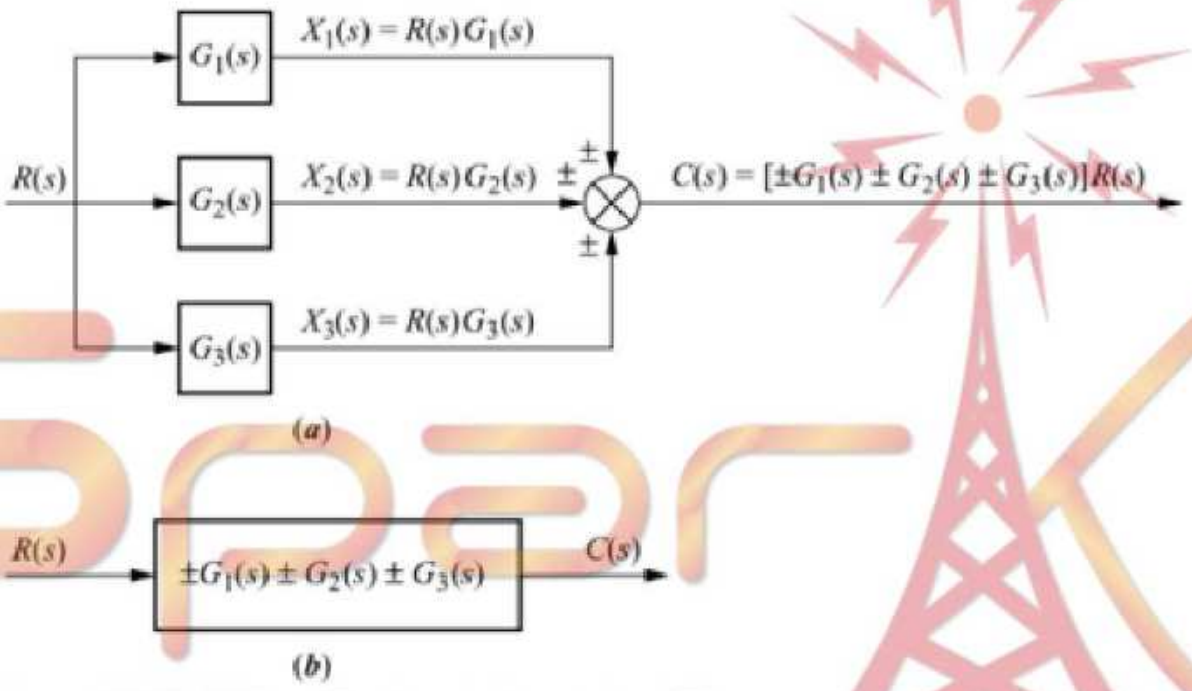
d. $\frac{G_2}{G_1}$

e. $G_1 + G_2$

f. $\frac{G_1}{G_2}$

Correct answer: $G_1 + G_2$

Parallel Connections



See Appendix III

67. Loops which do not possess any common node are said to be _____ loops.

a. Forward
b. Touching
c. Feedback
d. Non touching

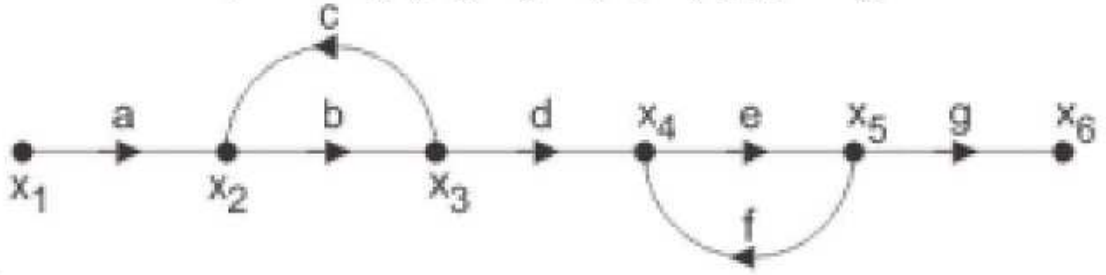
Correct answer: Non touching

Loop is the part of the network in which the branch starts from the node and comes back to the same node and non-touching loop must not have any node in common.

68. Use mason's gain formula to find the transfer function of the given signal flow graph:

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل منشأ هذا العمل



a. $\frac{abd}{1-ac}$
b. $\frac{abdeg}{1-(bc+ef)+bcef}$
c. $\frac{abd}{1-(bc+ef)+bcef}$
d. $\frac{adcdef}{1-(bc+ef)+bcef}$
e. Not mentioned

Correct answer: $\frac{abdeg}{1-(bc+ef)+bcef}$

Using mason's gain formula transfer function from signal flow graph can be calculated which relates the forward path gain to the various paths and loops.
The transfer function of the system is:

$$T(s) = \frac{\sum_k P_k \Delta_k}{\Delta} = \frac{P_1 \Delta_1}{\Delta}$$

P_1 is forward path transfer function for 1st path

The cofactor Δ_1, is the determinant with the loops touching the 1st path removed

The determinant $\Delta = 1 - (sum\ of\ all\ different\ loop\ gains) + (sum\ of\ the\ gain\ products\ of\ all\ combinations\ of\ two\ nontouching\ loops) - (sum\ of\ the\ gain\ products\ of\ all\ combinations\ of\ three\ nontouching\ loops) + \dots$

$$P_1 = adcdef$$
$$\Delta_1 = 1$$
$$\Delta = 1 - (bc + ef) + bcef$$
$$T(s) = \frac{P_1 \Delta_1}{\Delta} = \frac{adcdef * 1}{1 - (bc + ef) + bcef} = \frac{adcdef}{1 - (bc + ef) + bcef}$$

69. The relationship between an input and output variable of a signal flow graph is given by the net gain between the input and output node is known as the overall gain of the system.

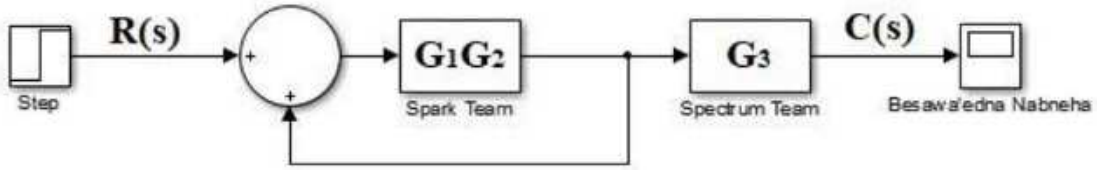
a. True
b. False

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

Correct answer: True

70. For the block diagram given in the following figure, the forward gain is:



a. $\frac{G_1G_2G_3}{1-G_1G_2}$

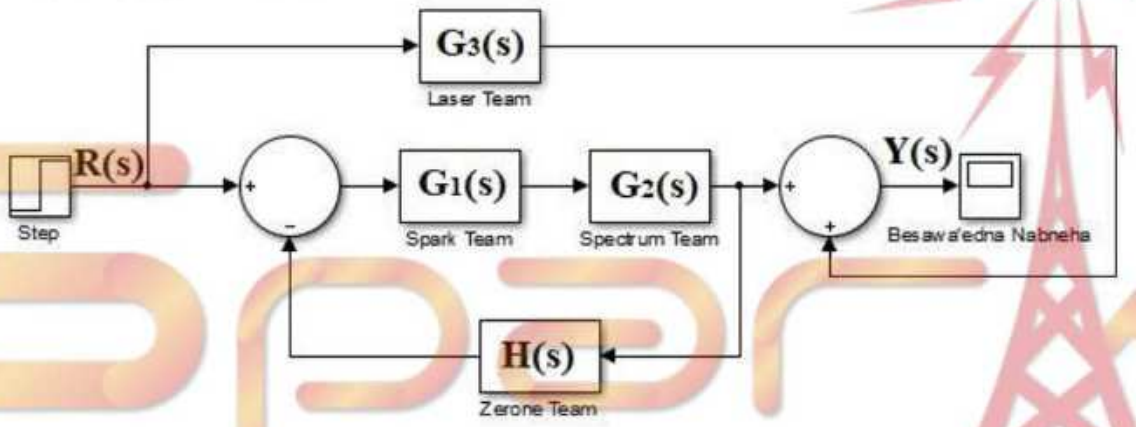
b. $G_1G_2G_3$

c. $\frac{G_1G_2}{1-G_1G_2G_3}$

d. None of the above

Correct answer: $G_1G_2G_3$

71. Find the transfer function:



a. $\frac{G_1G_2G_3}{1+G_1G_2H}$

b. $\frac{G_1G_2G_3}{1-G_1G_2H}$

c. $G_3 + \frac{G_1G_2}{1-G_1G_2H}$

d. $G_3 + \frac{G_1G_2}{1+G_1G_2H}$

e. None of the above

Correct answer: $G_3 + \frac{G_1G_2}{1+G_1G_2H}$

72. Cascade control means:

a. On-off control

b. Feed forward control

c. More than one feed-back loop

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

d. One feed-back loop

Correct answer: More than one feed-back loop

For more information about cascade control, see reference no.18

73. In closed loop system with negative feedback, the forward path is G(S) and the feedback path is H(s), the transfer function is:

a. $\frac{G(s)}{[1-G(s)H(s)]}$

b. $\frac{G(s)}{[1-G(s)]}$

c. $\frac{G(s)}{[1+G(s)]}$

d. $\frac{G(s)}{[1+G(s)H(s)]}$

Correct answer: $\frac{G(s)}{[1+G(s)H(s)]}$

74. In a signal flow graph, nodes are represented by small:

a. Pointers
b. Circles
c. Squares
d. Arrows

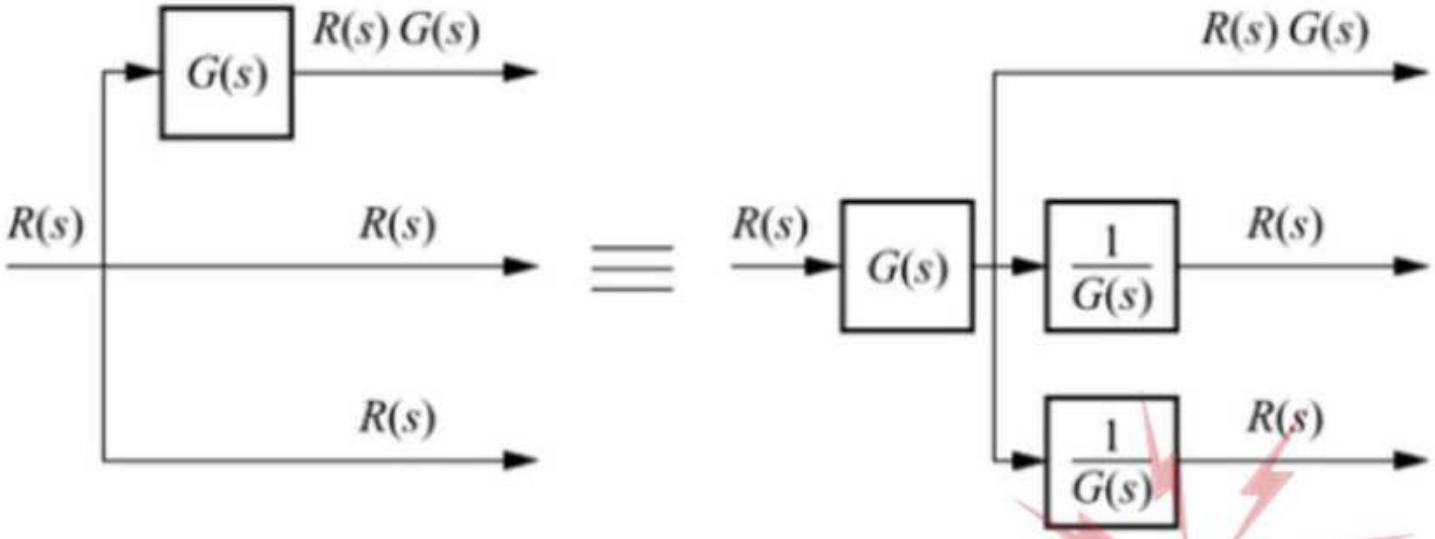
Correct answer: Circles

75. While moving a branch point after the block, which among the following should be added?
Or
While moving a pickoff point behind of the block, which among the following should be added?

a. Block of the same transfer function
b. Node
c. Summing point
d. Block of reciprocal transfer function

Correct answer: Block of reciprocal transfer function

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

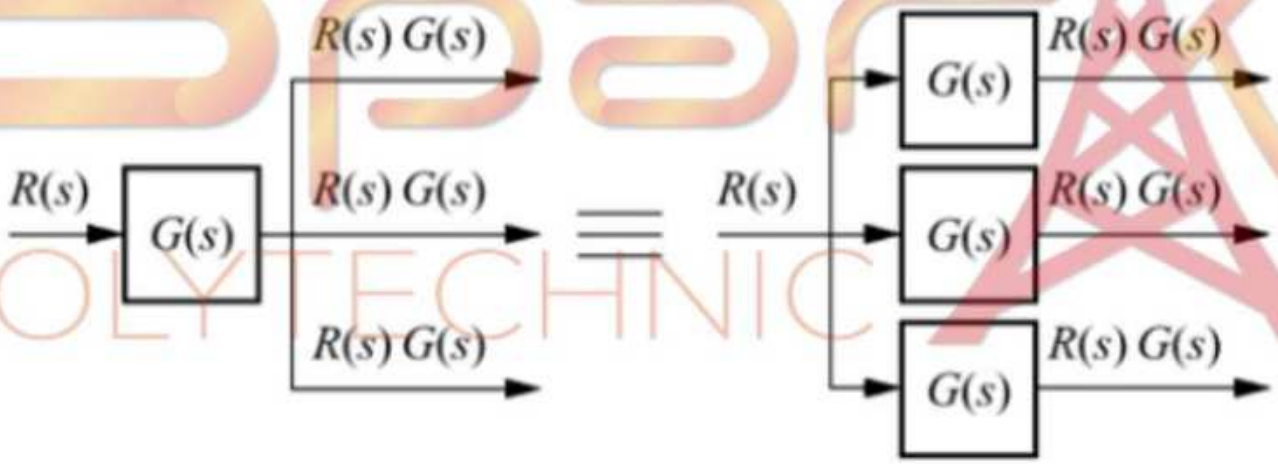


See Appendix III

76. While moving a branch point before the block, which among the following should be added?
Or
While moving a pickoff point ahead of the block, which among the following should be added?

a. Block of reciprocal transfer function
b. Summing point
c. Block of the same transfer function
d. Node

Correct answer: Block of the same transfer function



See Appendix III

77. The signal flow graph was used to obtain:

a. Closed loop transfer function
b. Feedback transfer function
c. Forward transfer function
d. Open loop transfer function

Correct answer: Closed loop transfer function

78. While moving a summing point after the block, which among the following should be added?

بسواعدنا نبنيها

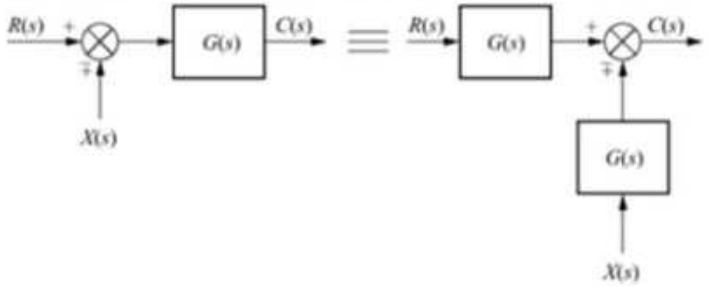
ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

Or

While moving a summing point behind of the block, which among the following should be added?

a. Summing point
b. Block of the same transfer function
c. Block of reciprocal transfer function
d. Node

Correct answer: Block of the same transfer function



$$C = G(+R \pm X)$$
$$= +GR \pm GX$$

See Appendix III

79. In signal flow graph, a node which has both incoming and outgoing branches is called:

a. Mixed node
b. Input node
c. General node
d. Output node

Correct answer: Mixed node

Nodes are the point by which the branches are outgoing and/or ingoing and this can be input, output or general node and mixed node is the node having both incoming and outgoing branches.

80. In general form, the closed loop control system is given by the transfer function:

a. $\frac{G(s)H(s)}{[1+G(s)H(s)]}$
b. $\frac{G(s)H(s)}{[1-G(s)H(s)]}$
c. $\frac{G(s)}{[1+H(s)]}$
d. $\frac{G(s)}{[1+G(s)H(s)]}$

Correct answer: $\frac{G(s)}{[1+G(s)H(s)]}$

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

Calculation of Time Response of First-Order & Second-Order

81. One of the following functions is not an input signal to control systems:

a. Unit ramp
b. Ramp function
c. Unit step
d. Damped exponential function

Correct answer: Damped exponential function

82. Given the following transfer function $G(s) = \frac{144}{s^2+6s+12}$. The exponential decay frequency equals to:

a. 0.25
b. 3
c. 144
d. 6
e. 12

Correct answer: 3

$$G(s) = \frac{{\omega^2}_n}{s^2 + 2\zeta\omega_n s + {\omega^2}_n} = \frac{144}{s^2 + 6s + 12}$$

$$\sigma_d = \zeta\omega_n = 3$$

83. Given the following transfer function $G(s) = \frac{144}{s^2+6s+12}$. The system is:

a. Overdamped
b. Underdamped
c. Critically damped
d. Undamped

Correct answer: Underdamped

$$G(s) = \frac{{\omega^2}_n}{s^2 + 2\zeta\omega_n s + {\omega^2}_n} = \frac{144}{s^2 + 6s + 12}$$

$${\omega^2}_n = 12 \rightarrow \omega_n = \sqrt{12}$$

$$\sigma_d = \zeta\omega_n = 3$$

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

$$\zeta = \frac{3}{\sqrt{12}}$$

84. Given the following transfer function $G(s) = \frac{80}{s+80}$. $T_c =$

a. 0.5
b. 0.05
c. 0.25
d. 0.0125

Correct answer: 0.0125

To determine the time constant T_c of the first order function by its transfer function, the system must be in the following form:

$$\frac{C(s)}{R(s)} = \frac{K}{Ts+1}$$

$$\frac{C(s)}{R(s)} = \frac{80}{s+80} = \frac{\frac{80}{80}}{\frac{1}{80}s + \frac{80}{80}} = \frac{1}{0.0125s+1}$$

Then, the time constant of this system is 0.0125

85. One of the following time response specifications is only a function of damping ratio:

a. Peak time
b. Delay time
c. Rise time
d. Maximum peak

Correct answer: Maximum peak

$$M_p = c(t_p) - 1$$

$$M_p = -e^{-\zeta\omega_n\left(\frac{\pi}{\omega_d}\right)}\left(\cos\pi + \frac{\zeta}{\sqrt{1-\zeta^2}}\sin\pi\right)$$

$$M_p = -e^{-\left(\frac{\sigma}{\omega_d}\right)\pi} = e^{-\left(\frac{\zeta}{\sqrt{1-\zeta^2}}\right)\pi}$$

86. If $c(t) = 3 + 5e^{-4t}$, the forced response is:

a. 5
b. 8
c. 3
d. None of the above

Correct answer: 3

87. Find the damped frequency W_d for the following transfer function in rad/s:

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

$$\frac{9}{s^2 + 6s + 9}$$

a. 0
b. 3
c. *Cannot determine*
d. 9

Correct answer: 0

$$\frac{C(s)}{R(s)} = \frac{{\omega^2}_n}{s^2 + 2\zeta\omega_n s + {\omega^2}_n} = \frac{9}{s^2 + 6s + 9}$$

$${\omega^2}_n = 9$$

$$\omega_n = 3$$

$$2\zeta\omega_n = 6$$

$$\zeta = 1$$

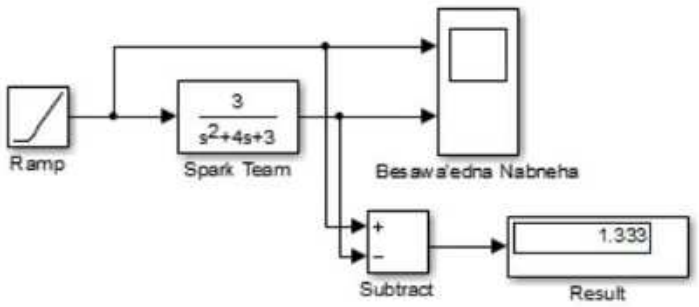
$$\omega_d = \omega_n\sqrt{1 - \zeta^2} = 3 * \sqrt{1 - 1^2} = 0$$

88. Steady state error for the following TF and unit ramp input $\frac{1}{s^2}$ is:

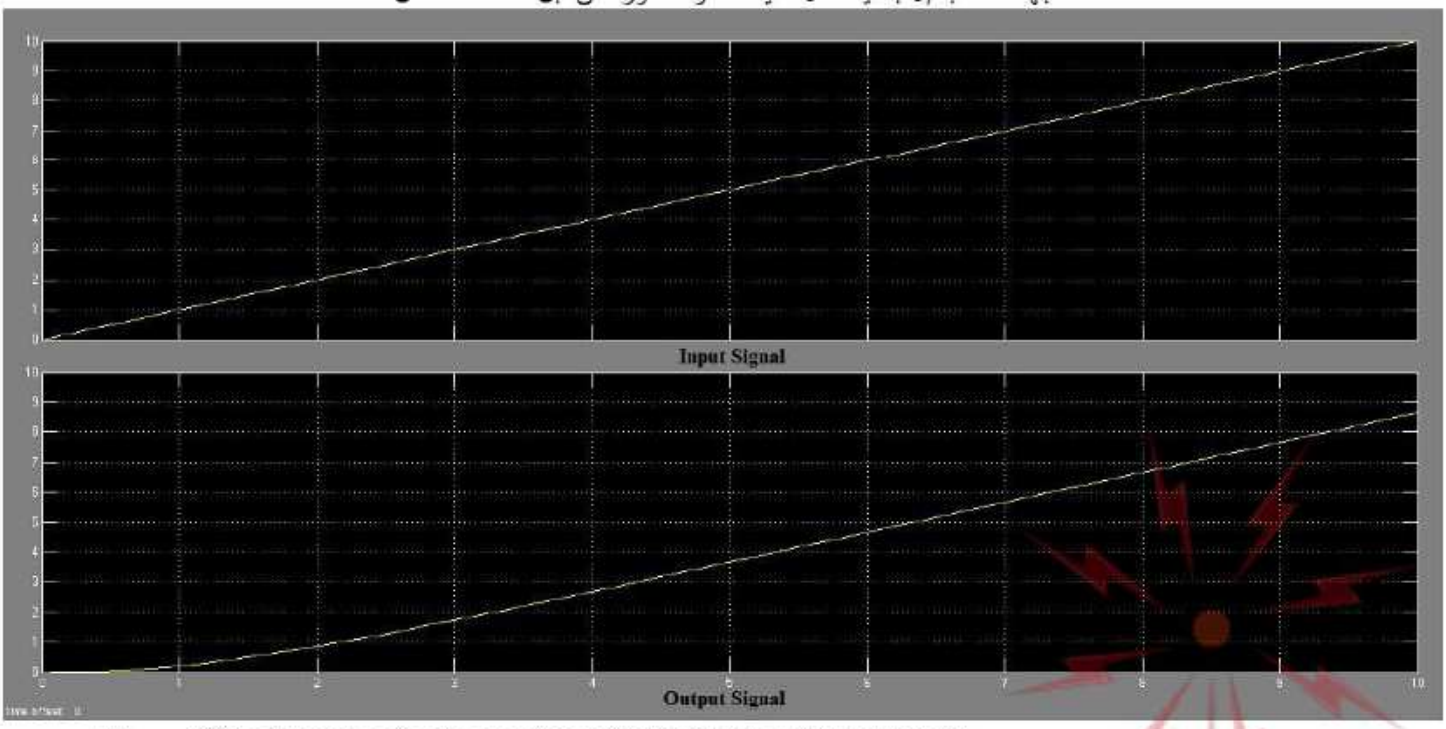
$$\frac{Y(s)}{R(s)} = \frac{3}{s^2 + 4s + 3}$$

a. 1.33
b. 1
c. 0
d. Infinity
e. Not mentioned

Correct answer: 1.33



ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل



89. If the forward gain of a negative unity feedback control system is:

$$\frac{100}{s(s+100)}$$

Find Settling time (2%) in seconds?

a. 0.08
b. 0.06
c. 0.02
d. Not mentioned

Correct answer: 0.08

The standard form for transfer function of second order system is:

$$\frac{C(s)}{R(s)} = \frac{{\omega^2}_n}{s^2 + 2\zeta\omega_n s + {\omega^2}_n} = \frac{\frac{100}{s(s+100)}}{1 + \frac{100}{s(s+100)}} = \frac{100}{s^2 + 100s + 100}$$

$$2\zeta\omega_n = 100$$

$$\zeta\omega_n = 50$$

The setting time for second order system is:

$$t_s = 4T = \frac{4}{\sigma} = \frac{4}{\zeta\omega_n} = \frac{4}{50} = 0.08 \quad (2\%\ criterion)$$

90. In first order system if we increase the time constant:

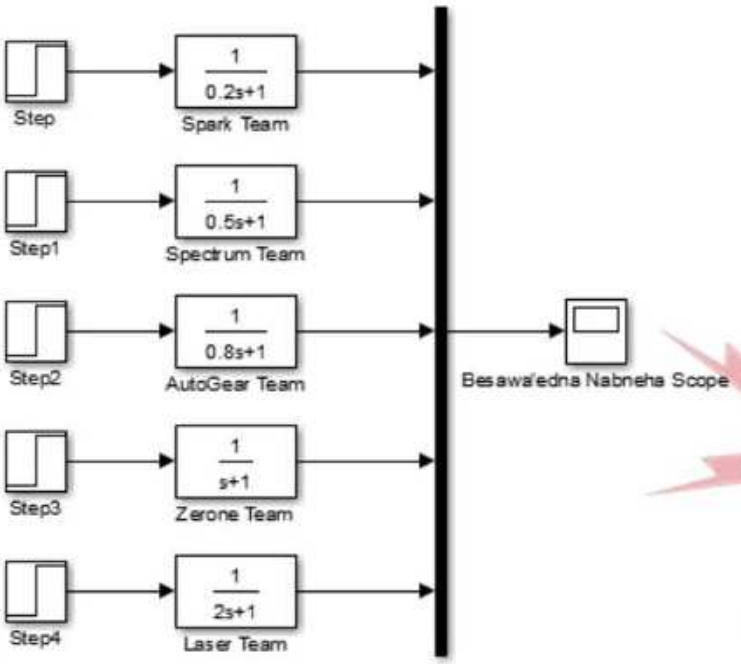
a. The steady state value increase

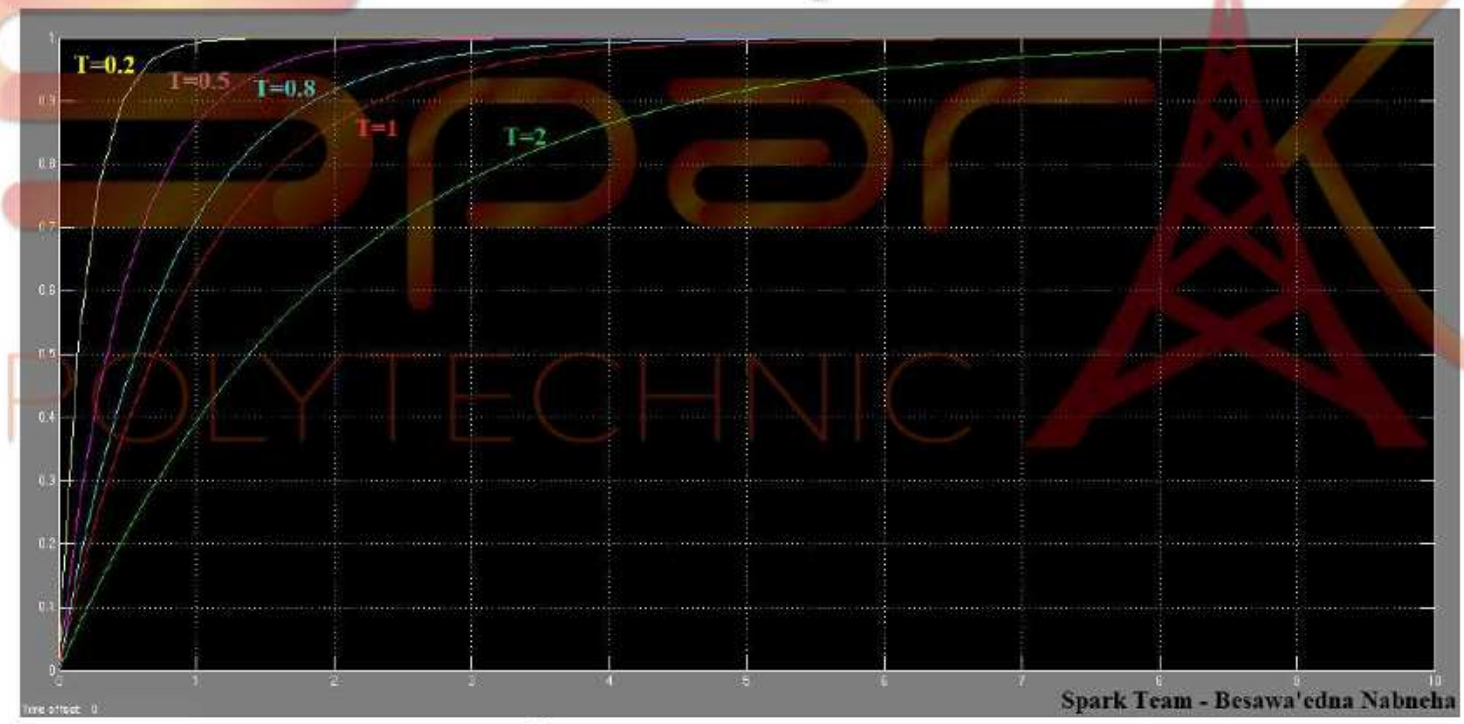
بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

b. The steady state value decrease
c. The steady state time increase
d. The steady state time decrease

Correct answer: The steady state time increase





91. In the Transfer function $\frac{K}{Ts+A}$. K represents:

a. Flexible feedback
b. Gain
c. Fixed Feedback

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

d. Time constant

Correct answer: Gain

92. In the Transfer function $\frac{K}{Ts+A}$. T represents:

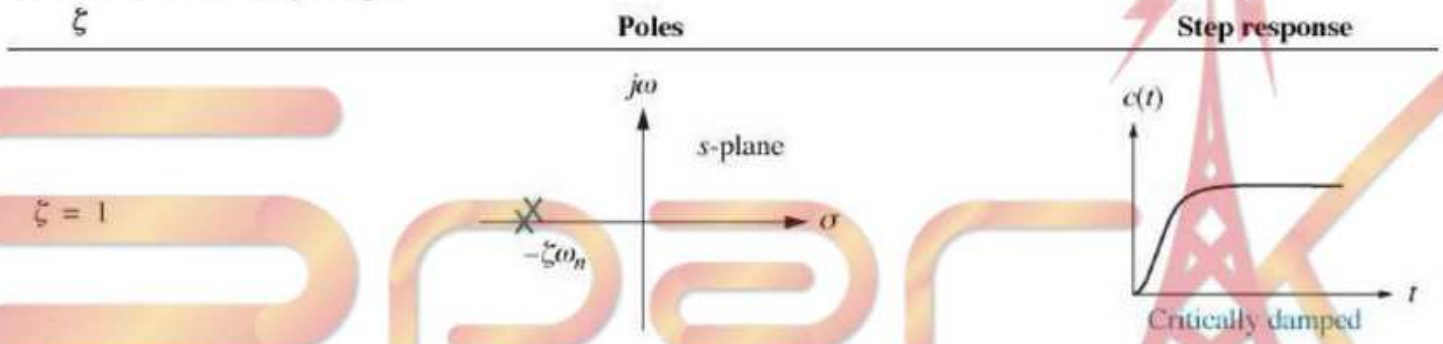
a. Fixed Feedback
b. Flexible feedback
c. Time constant
d. Gain

Correct answer: Time constant

93. If the damping ratio of second order system is unity, the time response is:
Or
For unity damping factor, the system will be:

a. Underdamped
b. Critically damped
c. Undamped (The transient response does not die out)
d. Overdamped

Correct answer: Critically damped



94. A second order system has $\frac{C(s)}{R(s)} = \frac{3}{s^2+4s+13}$ for 2% error the settling time of the system is:

a. 1
b. $\frac{1}{4}$
c. $\frac{1}{2}$
d. 2

Correct answer: $\frac{1}{2}$

The standard form for transfer function of second order system is:

$$\frac{C(s)}{R(s)} = \frac{{\omega^2}_n}{s^2 + 2\zeta\omega_n s + {\omega^2}_n} = \frac{3}{s^2 + 4s + 13}$$

$$2\zeta\omega_n = 4$$

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

$$\zeta\omega_n = 2$$

The time constant for second order system is:

$$T = \frac{1}{\zeta\omega_n} = \frac{1}{2}$$

95. Mathematically, the transient response time of first order system is:

a. T
b. 2T
c. 3T
d. 4T
e. 5T
f. Infinity

Correct answer: Infinity

96. The transfer function of first order system is $\frac{C(s)}{R(s)} = \frac{6}{8s+2}$. The D.C gain of this system is:

a. 3
b. 6
c. 4
d. 2
e. 8

Correct answer: 3

To determine the D.C gain or time constant of the first order function by its transfer function, the system must be in the following form:

$$\frac{C(s)}{R(s)} = \frac{K}{Ts+1}$$

$$\frac{C(s)}{R(s)} = \frac{6}{8s+2} = \frac{3}{4s+1}$$

Then, the D.C gain of this system is 3

97. The transfer function of second order system is $\frac{C(s)}{R(s)} = \frac{1}{s^2+2s+4}$. assume unit step input, the damping ratio is:

a. 2
b. 1
c. 4
d. 0.5

Correct answer: 0.5

The transfer function for second order system is:

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

$$\frac{C(s)}{R(s)} = \frac{{\omega^2}_n}{s^2 + 2\zeta\omega_n s + {\omega^2}_n}$$

Where:

ω_n: Damped natural frequency

ζ: The damping ratio

σ: Damping constant

ω_d: Un-damped natural frequency

VIN: The coefficient of S^2 must be one

$$2\zeta\omega_n = 2$$
$$2\zeta * 2 = 2$$
$$\zeta = \frac{1}{2}$$

98. The transfer function of second order system is $\frac{C(s)}{R(s)} = \frac{1}{s^2+2s+4}$. assume unit step input, the damping constant is:

a. 4
b. 1
c. 0.5
d. 2

Correct answer: 1

The transfer function for second order system is:

$$\frac{C(s)}{R(s)} = \frac{{\omega^2}_n}{s^2 + 2\zeta\omega_n s + {\omega^2}_n}$$

Where:

ω_n: Damped natural frequency

ζ: The damping ratio

σ: Damping constant

ω_d: Un-damped natural frequency

VIN: The coefficient of S^2 must be one

$$2\zeta\omega_n = 2$$
$$\zeta\omega_n = 1$$
$$\sigma = \zeta\omega_n = 1$$

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

99. The transfer function of second order system is $\frac{C(s)}{R(s)} = \frac{1}{s^2+2s+4}$. assume unit step input, the undamped natural frequency is:

a. 1
b. 4
c. 3
d. 2

Correct answer:
The transfer function for second order system is:

$$\frac{C(s)}{R(s)} = \frac{{\omega^2}_n}{s^2 + 2\zeta\omega_n s + {\omega^2}_n}$$

Where:
ω_n: Damped natural frequency
ζ: The damping ratio
σ: Damping constant
ω_d: Un-damped natural frequency

VIN: The coefficient of S^2 must be one

$${\omega^2}_n = 4 \rightarrow \omega_n = 2$$
$$2\zeta\omega_n = 2$$
$$2\zeta * 2 = 2$$
$$\zeta = \frac{1}{2}$$

$$\omega_d = \omega_n\sqrt{1-\zeta^2} = 2\sqrt{1-\left(\frac{1}{2}\right)^2} = \sqrt{3}$$

100. The transfer function of second order system is $\frac{C(s)}{R(s)} = \frac{1}{s^2+2s+4}$. assume unit step input, the setting time for 2% error.

a. 3
b. 2
c. 4
d. 1

Correct answer: 4
The transfer function for second order system is:

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

$$\frac{C(s)}{R(s)} = \frac{{\omega^2}_n}{s^2 + 2\zeta\omega_n s + {\omega^2}_n}$$

Where:

ω_n: Damped natural frequency

ζ: The damping ratio

σ: Damping constant

ω_d: Un-damped natural frequency

VIN: The coefficient of S^2 must be one

$$2\zeta\omega_n = 2$$

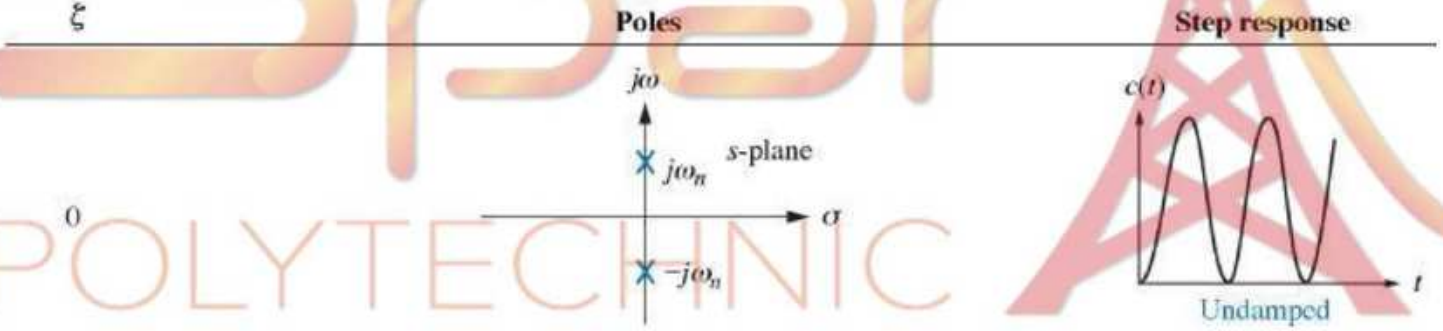
$$\zeta\omega_n = 1$$

$$t_s = 4T = \frac{4}{\sigma} = \frac{4}{\zeta\omega_n} = \frac{4}{1} = 4 \quad (2\%\ criterion)$$

101. If the damping ratio of second order system is zero, the time response is:
Or
For zero damping factor, the system will be:

a. Underdamped
b. Critically damped
c. Undamped (The transient response does not die out)
d. Overdamped

Correct answer: Undamped (The transient response does not die out)



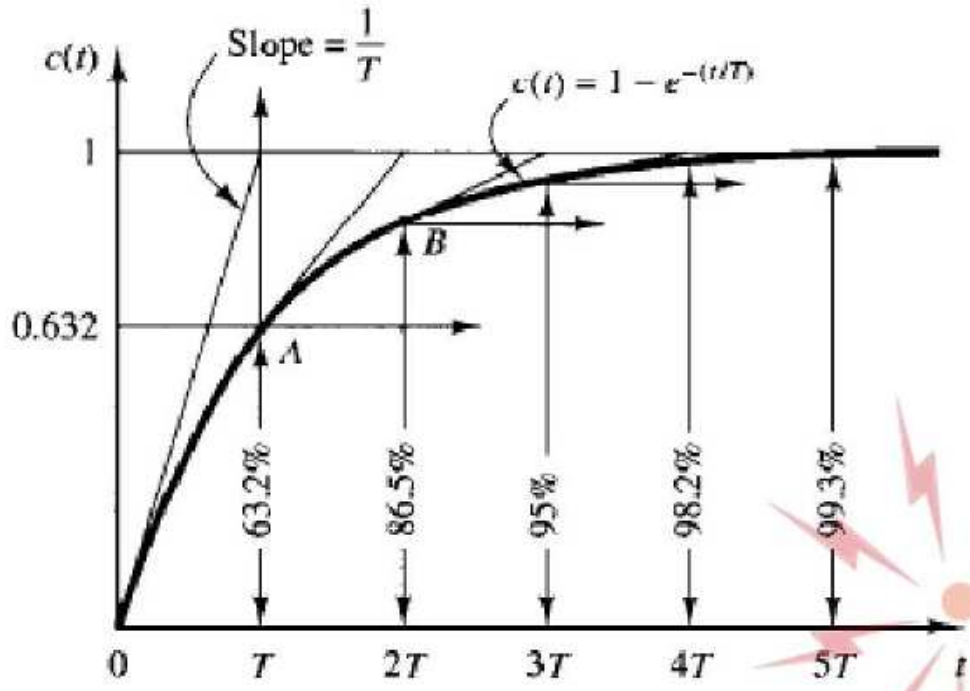
102. Practically, the transient response time needs to reach the 5% line of the final value of first order system is:
Or
The time required for the output of first order system to reach 95% its steady state value is:

a. T
b. 2T
c. 3T
d. 4T
e. 5T
f. Infinity

Correct answer: 3T

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل



103. The unit impulse response of standard first order system is:

a. $\frac{1+e^{t}}{T}$

b. $\frac{e^{-t}}{T}$

c. $\frac{1-e^{t}}{T}$

d. $\frac{-e^{t}}{T}$

Correct answer: $\frac{e^{-t}}{T}$

Since the Laplace transform of the unit-impulse function is 1, we obtain the output of the system as:

$$C(s) = \frac{1}{Ts+1} * 1 = \frac{1}{T}\frac{1}{s+\frac{1}{T}}$$

$$C(t) = \frac{e^{-t}}{T}$$

104. The unit ramp response of standard first order system is:

a. $\frac{1-e^{t}}{T}$

b. $\frac{e^{-t}}{T}$

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

c. $\frac{-e^{t}}{T}$

d. None of the above

Correct answer: None of the above

$$c(t) = t - T + Te^{-\frac{t}{T}}$$

Since the Laplace transform of the unit-ramp function is $\frac{1}{s^2}$, we obtain the output of the system as:

$$C(s) = \frac{1}{Ts+1} * \frac{1}{s^2}$$

And using partial fractions:

$$C(s) = \frac{1}{s^2} - \frac{T}{s} + \frac{T^2}{Ts+1}$$

$$C(t) = t - T + Te^{-\frac{t}{T}}, \quad for\ t \geq 0$$

105. The transfer function of first order system is $\frac{C(s)}{R(s)} = \frac{1}{4s+2}$. assume unit step input, the time constant is:

a. 4
b. 1
c. 0.5
d. 2

Correct answer: 2

$$\frac{C(s)}{R(s)} = \frac{1}{Ts+1} = \frac{1}{4s+2} = \frac{1}{2} * \frac{1}{2s+1}$$

So,

$$T = 2$$

The solution is being clearer by plotting the output wave and calculating the time in which the output wave reaches 63.2% of the steady state value, as this time represents the time constant.

$$\frac{C(s)}{R(s)} = \frac{1}{Ts+1}$$

Since the Laplace transform of the unit-step function is $\frac{1}{s}$, substituting R(s), we obtain:

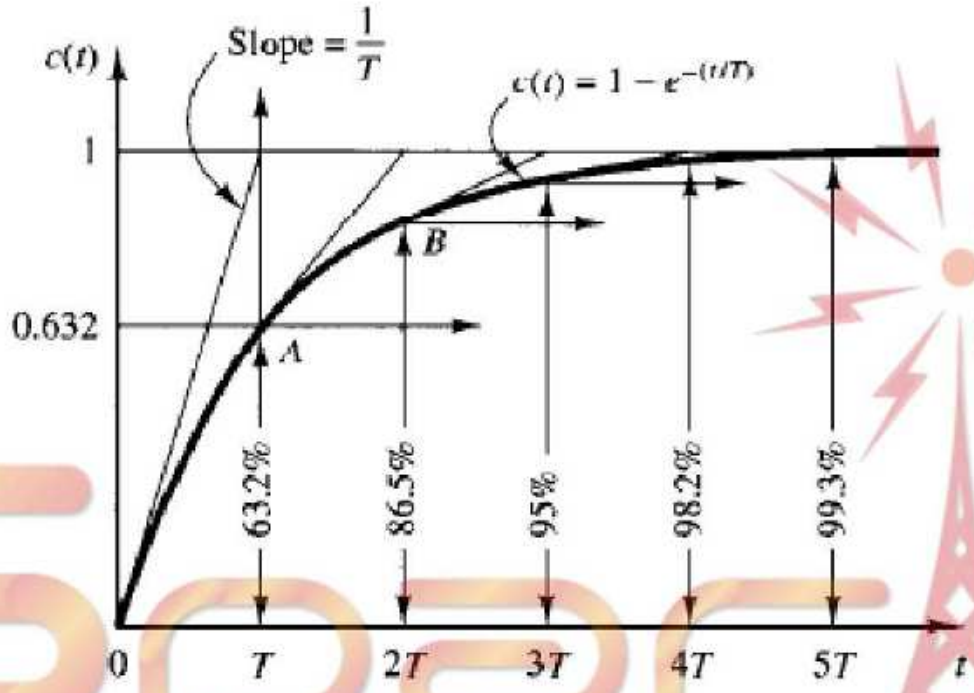
$$C(s) = \frac{1}{Ts+1} * \frac{1}{s} = \frac{1}{s} - \frac{T}{Ts+1} = \frac{1}{s} - \frac{1}{s+\frac{1}{T}} = 1 - e^{-\frac{t}{T}}, \quad for\ t \geq 0$$

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل منشأ هذا العمل

$$C(s) = \frac{1}{2}\frac{1}{2s+1} * \frac{1}{s} = \frac{1}{2}\left(\frac{1}{s} - \frac{2}{2s+1}\right) = \frac{1}{2}\left(\frac{1}{s} - \frac{1}{s+\frac{1}{2}}\right) = \frac{1}{2}\left(1 - e^{-\frac{t}{2}}\right), \quad for\ t \geq 0$$

t (sec)	0	T	2T	3T	4T	5T	inf
c(T)	0% = 0	63.2%	86.5%	95%	98.2%	99.3%	100% = 1



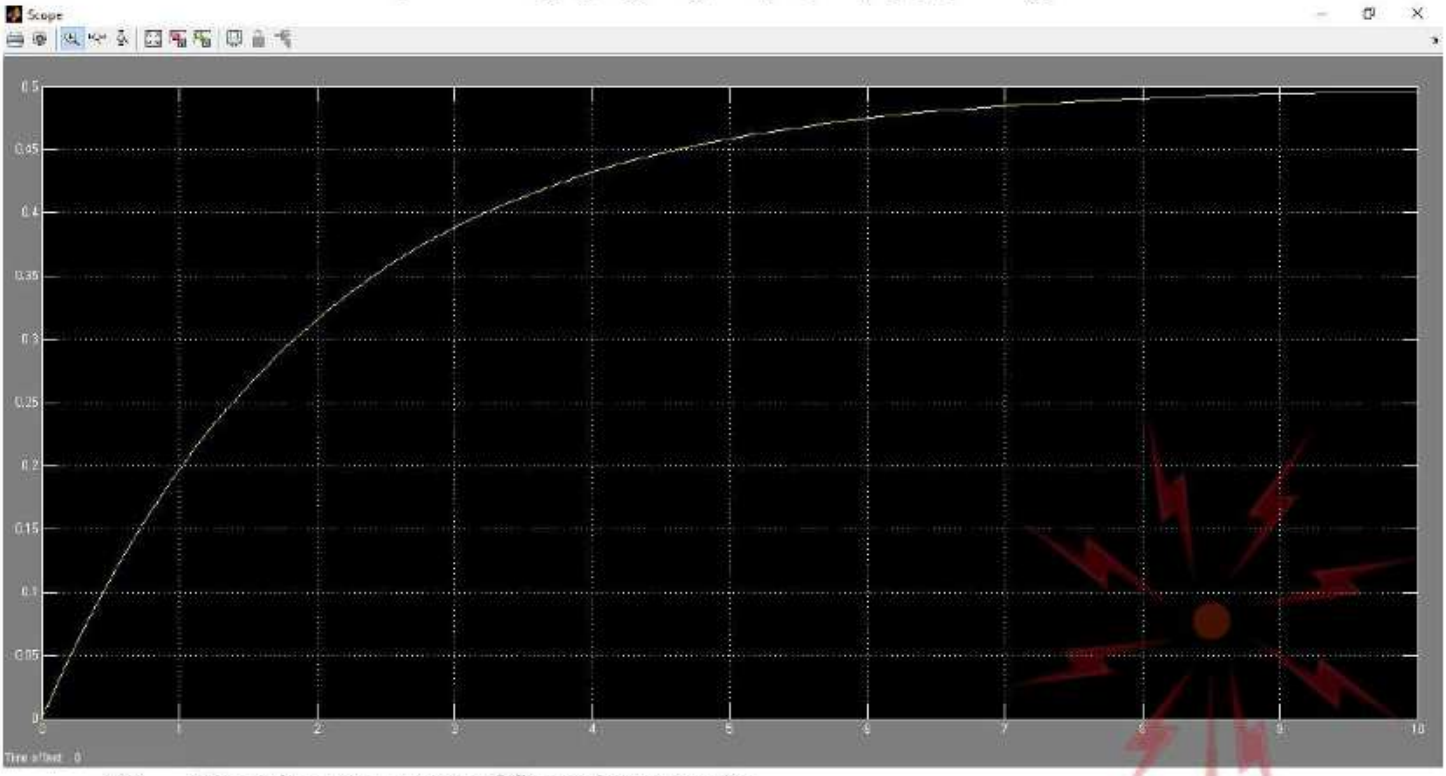
The correctness of the solution can be confirmed by using the Matlab program and calculating the time in which the output wave reaches 63.2% of the steady state value, as this represents the time constant of the wave.

The block diagram:



The output waveform on the scope:

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل



106. The unit ramp response of first order system is:

a. $c(t) = 1 - e^{-\frac{t}{4}}$

b. $c(t) = -4e^{-\frac{t}{4}}$

c. $c(t) = 2\left(t - e^{-\frac{t}{4}}\right)$

d. $c(t) = 2\left(t - 4 - 4e^{-\frac{t}{4}}\right)$

e. None of the above

Correct answer: None of the above

The general form for the unit ramp response of first order system is:

$$c(t) = t - T + Te^{-\frac{t}{T}}$$

$$c(t) = t - 4 + 4e^{-\frac{t}{4}}$$

107. The transfer function of first order system is $\frac{C(s)}{R(s)} = \frac{1}{4s+2}$. assume unit step input, the steady state output is:

a. 2
b. 0.5
c. 1
d. 4

Correct answer: 0.5

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل منشأ هذا العمل

$$\frac{C(s)}{R(s)} = \frac{1}{Ts+1}$$

Since the Laplace transform of the unit-step function is $\frac{1}{s}$, substituting R(s), we obtain:

$$C(s) = \frac{1}{Ts+1} * \frac{1}{s} = \frac{1}{s} - \frac{T}{Ts+1} = \frac{1}{s} - \frac{1}{s+\frac{1}{T}} = 1 - e^{-\frac{t}{T}}, \quad for\ t \geq 0$$

$$C(s) = \frac{1}{2}\frac{1}{2s+1} * \frac{1}{s} = \frac{1}{2}\left(\frac{1}{s} - \frac{2}{2s+1}\right) = \frac{1}{2}\left(\frac{1}{s} - \frac{1}{s+\frac{1}{2}}\right) = \frac{1}{2}\left(1 - e^{-\frac{t}{2}}\right), \quad for\ t \geq 0$$

At steady state, $t \to \infty$, so:

$$C(s) = \frac{1}{2}\left(1 - e^{-\frac{\infty}{2}}\right)$$

And $e^{-\frac{\infty}{2}} = 0$, so:

$$C(s) = \frac{1}{2}(1-0) = \frac{1}{2}$$

The correctness of the solution can be confirmed by using the Matlab program and calculating the time in which the output wave reaches 63.2% of the steady state value, as this represents the time constant of the wave.

The block diagram:

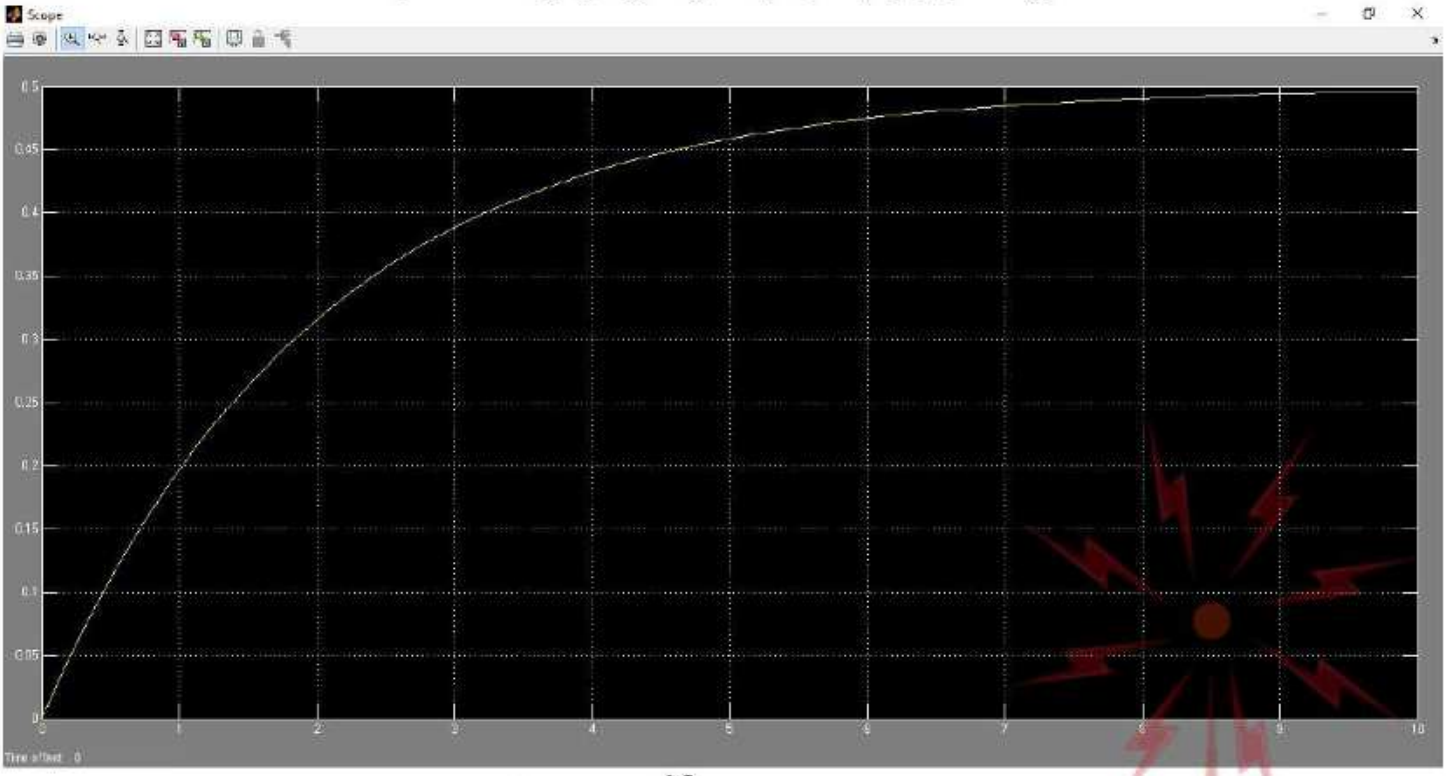


The output waveform on the scope:

POLYTECHNIC

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

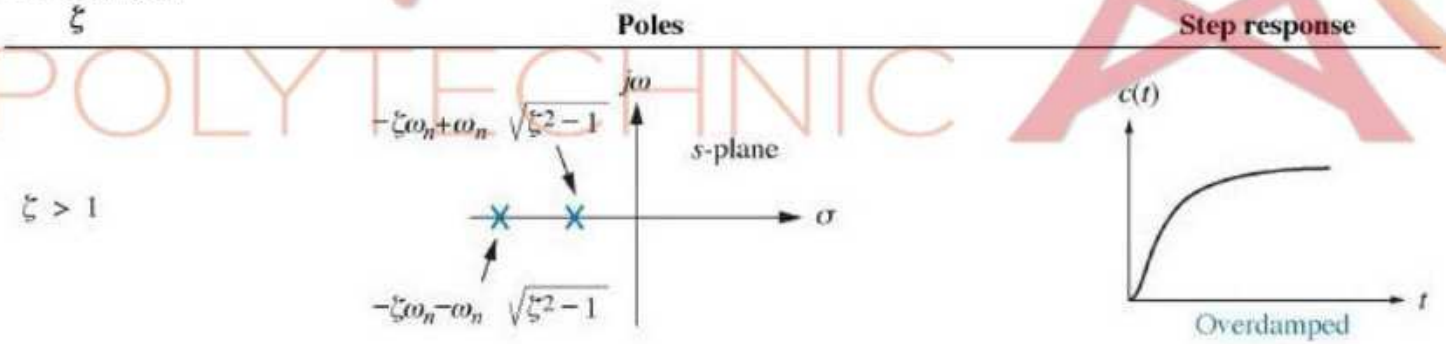


108. The following system $T(s) = \frac{10}{(s+7)(s+3)}$ is:

a. Underdamped
b. Undamped
c. Overdamped
d. Critical damped

Correct answer: Overdamped

The poles are negative real and unequal (distinct poles/ all non-repeated simple roots), then $\zeta > 1$, so the time response is overdamped.



109. Rise time t_r is the time required for the transient's damped oscillations to reach and stay within 2% of the steady-state value.

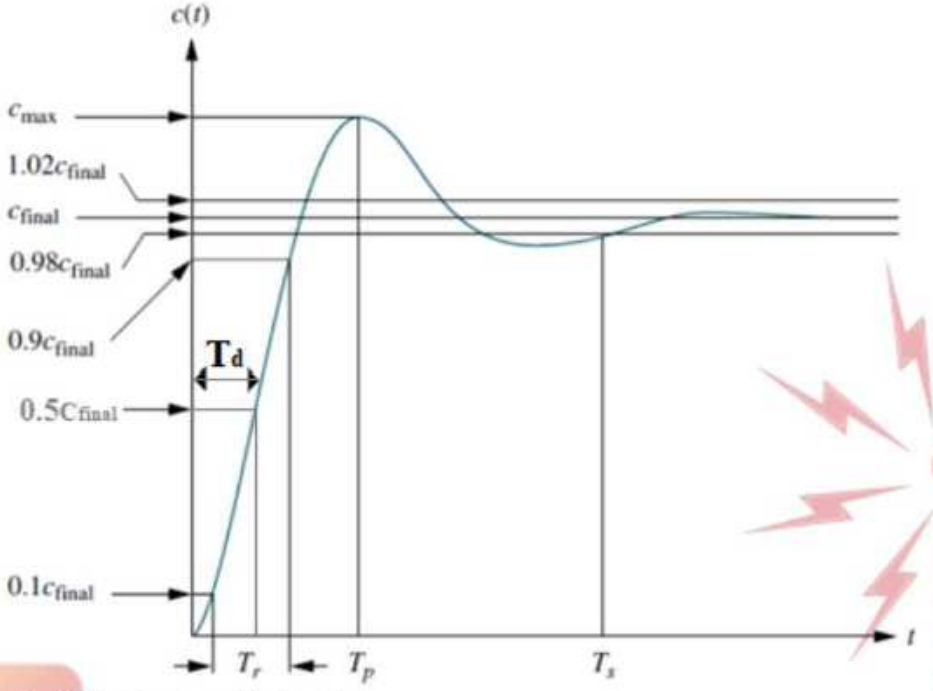
a. True
b. False

Correct answer: False

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

Rise time, t_r The time required for the waveform to go from 0.1 of the final value to 0.9 of the final value.

Settling time, t_s The time required for the transient's damped oscillations to reach and stay within prescribed percentage error (usually, 2% of the steady-state value).



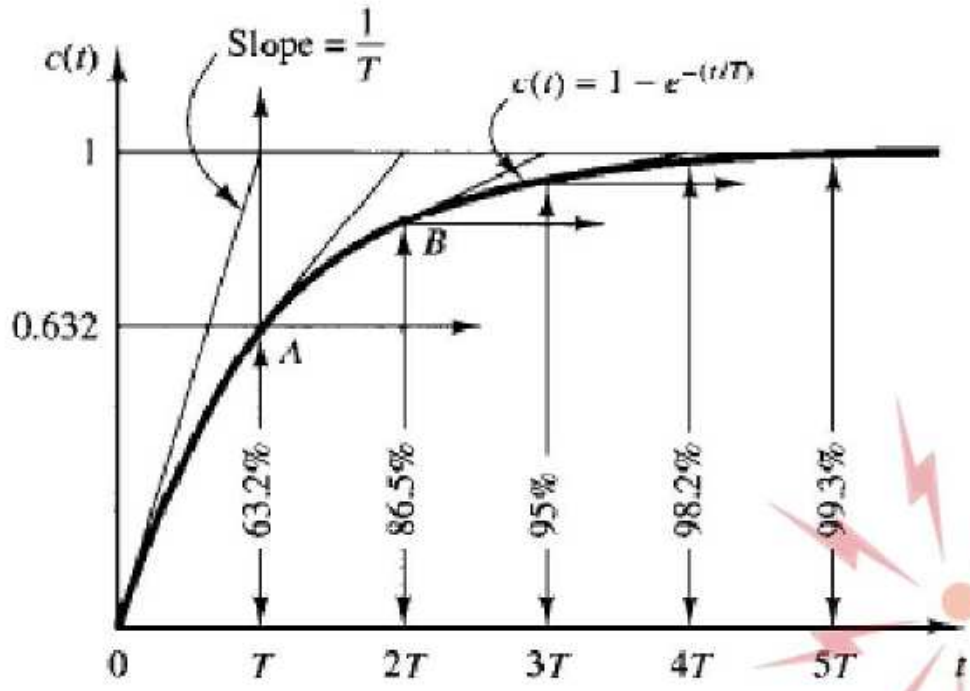
110. Given the following transfer function

$$G(S) = \frac{81}{s^2 + 9s + 81}$$

a. 0.77
b. 0.9
c. 0.99
d. 1.2

Correct answer: 0.9

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل منشأ هذا العمل



$$\frac{C(s)}{R(s)} = \frac{{\omega^2}_n}{s^2 + 2\zeta\omega_n s + {\omega^2}_n}$$

Where:

ω_n: Damped natural frequency

ζ: The damping ratio

σ: Damping constant

ω_d: Un-damped natural frequency

VIN: The coefficient of s^2 must be one

$$G(S) = \frac{81}{s^2 + 9s + 81}$$

$${\omega^2}_n = 81 \rightarrow \omega_n = 9$$

$$2\zeta\omega_n = 9$$

$$\zeta\omega_n = 4.5$$

if the question does not mention the criterion, 2% criterion is considered the permissible error). (Round to only one decimal place

$$t_s = 4T = \frac{4}{\sigma} = \frac{4}{\zeta\omega_n}$$

$$t_s = \frac{4}{4.5} = 0.88889 \cong 0.9$$

بسواعدنا نبنيها

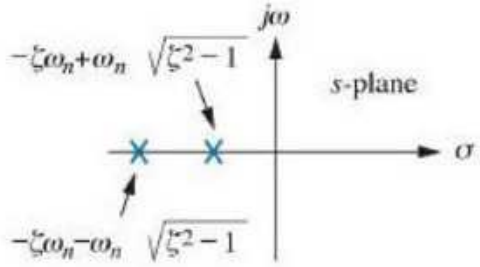
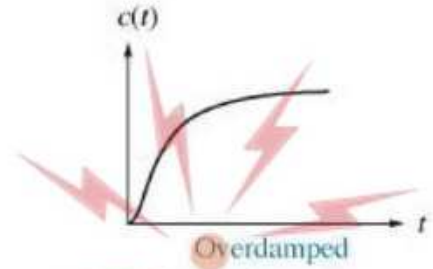
ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

111. For $\zeta > 1$, the transfer function has two imaginary poles and the response has an overshoot.

a. True
b. False

Correct answer: False

The poles are negative real and unequal (distinct poles/ all non-repeated simple roots), then $\zeta > 1$, so the time response is overdamped and hasn't an overshoot.

ζ	Poles	Step response
$\zeta > 1$		

112. The transfer function of the system is

$$G(s) = \frac{100}{(s+1)(s+100)}$$

For (a unit step/ a unit ramp/ sinusoidal/ impulse/ an) input signal to the system the approximate settling time (in sec) for 2% criterion (if the question does not mention the criterion, 2% criterion is considered the permissible error) is:

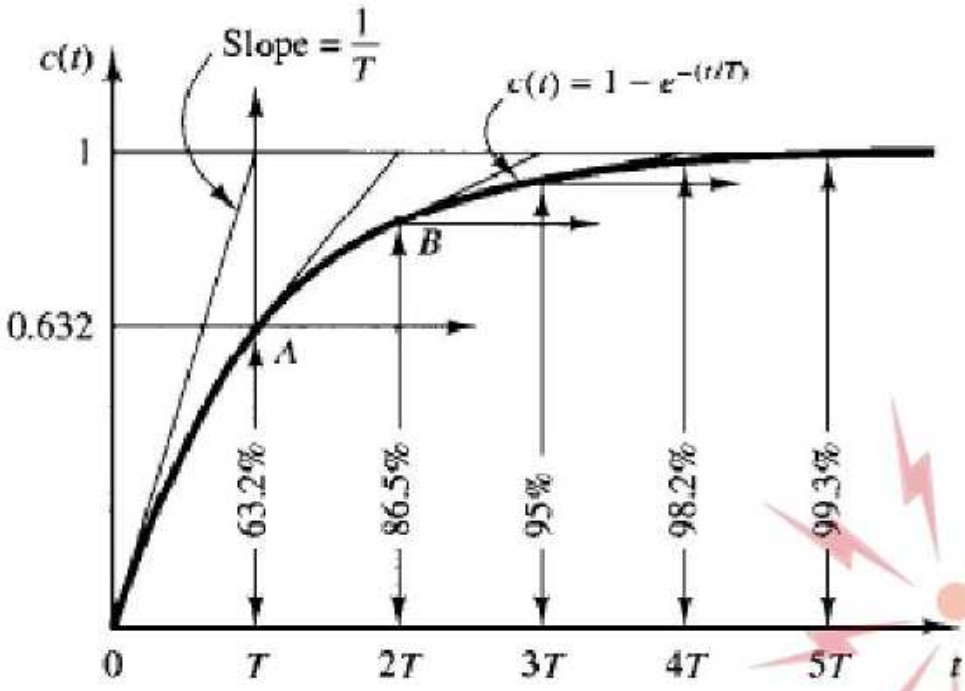
a. 0.04
b. 100
c. 1
d. 4
e. None of the above

Correct answer: None of the above ($t_s = 0.0792079207920792$)

POLYTECHNIC

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل



For 2% criterion (if the question does not mention the criterion, 2% criterion is considered the permissible error), the approximate setting time (in sec) is:

$$t_s = 4T = \frac{4}{\sigma} = \frac{4}{\zeta\omega_n}$$

$$\frac{C(s)}{R(s)} = \frac{{\omega^2}_n}{s^2 + 2\zeta\omega_n s + {\omega^2}_n}$$

Where:

ω_n: Damped natural frequency

ζ: The damping ratio

σ: Damping constant

ω_d: Un-damped natural frequency

VIN: The coefficient of s^2 must be one

$$G(S) = \frac{100}{s^2 + 101s + 100}$$

$${\omega^2}_n = 100 \rightarrow \omega_n = 10$$

$$2\zeta\omega_n = 101$$

$$\zeta\omega_n = 50.5$$

$$t_s = \frac{4}{50.5} = 0.0792079207920792$$

بسواعدنا نبنيها

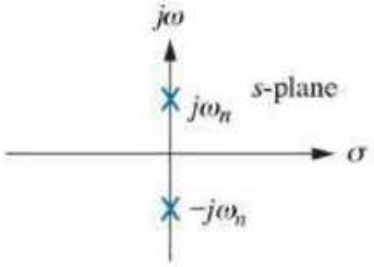
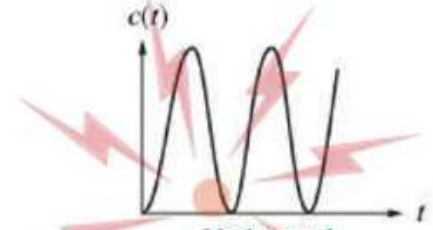
ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

113. The following system $T(s) = \frac{2s+1}{s^2+25}$ is:

a. Overdamped
b. Undamped
c. Underdamped
d. Critical damped

Correct answer: Undamped

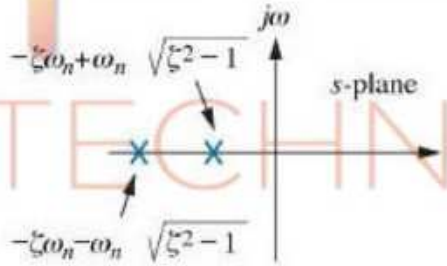
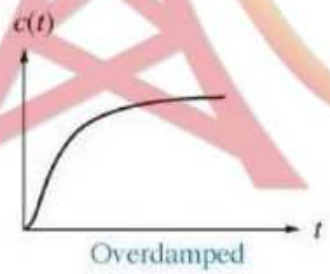
The poles of the system are imaginary conjugate, then $\zeta = 0$, so the time response is undamped.

ζ	Poles	Step response
0		

114. The response has no overshoot when $\zeta > 1$.
Or
If the damping ratio of second order system is greater than one, then time response has no overshoot.
Or
For damping factor greater than one, the system response has no overshoot.

a. True
b. False

Correct answer: True

ζ	Poles	Step response
$\zeta > 1$		

POLYTECHNIC

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

Control Action: Applying Different Types of Controllers (P, I, D, PI, PD and PID)

115. We used Ziegler-Nichols tuning method to obtain:

a. 0% Maximum overshoot
b. 10% Maximum overshoot
c. 60% Maximum overshoot
d. 25% Maximum overshoot
e. 50% Maximum overshoot
f. 20% Maximum overshoot

Correct answer: 25% Maximum overshoot

116. Second t method of Ziegler Nichols rule used when the time response of the open loop system is:

a. Undamped
b. Over damped
c. Under damped
d. None of the above

Correct answer: Under damped

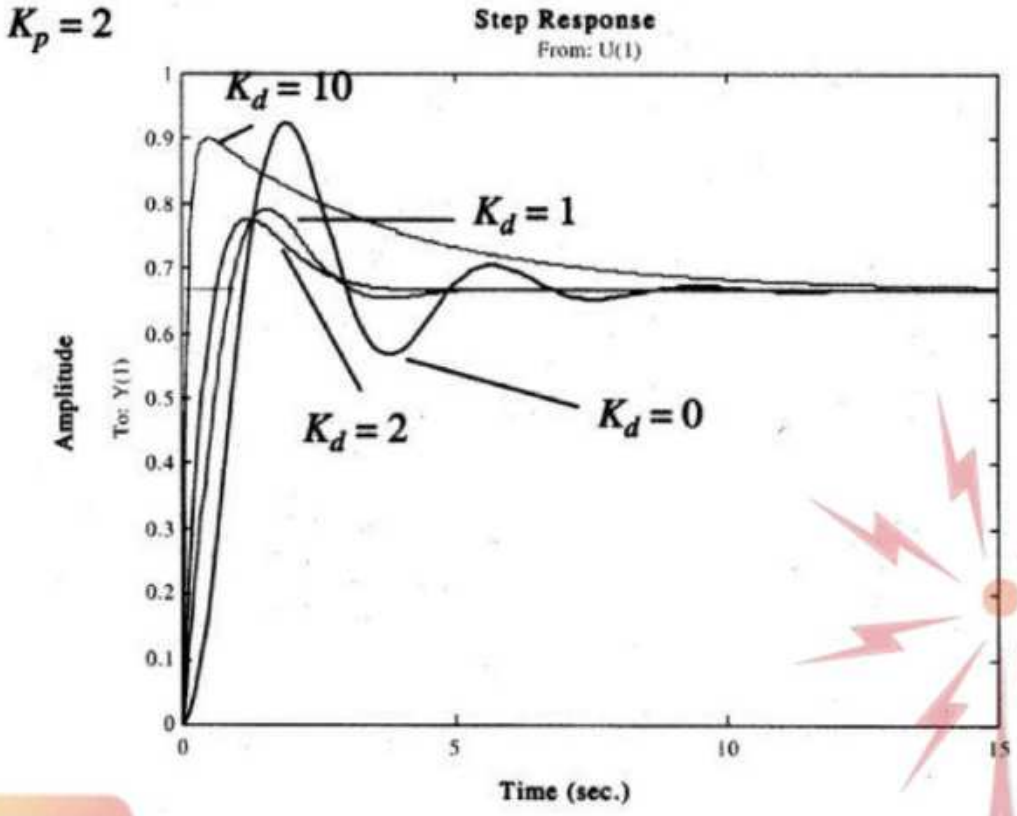
If the system doesn't have feedback path, then the time response is the time response of the open loop system which is under damped response.

117. The PD controller is used to:

a. Increase the time constant
b. Decrease the time constant
c. Decrease the maximum overshoot
d. Reduce the steady state error

Correct answer: Decrease the maximum overshoot

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل



The PD controller is mainly used to:

1. Improve the transient response of a control system.
2. Reduce the max. peak and settling time

118. Adding PID controller to any system means:

a. Adding pole and two zeros
b. Adding distinct pole
c. Adding two zeros
d. Adding zero

Correct answer: Adding pole and two zeros

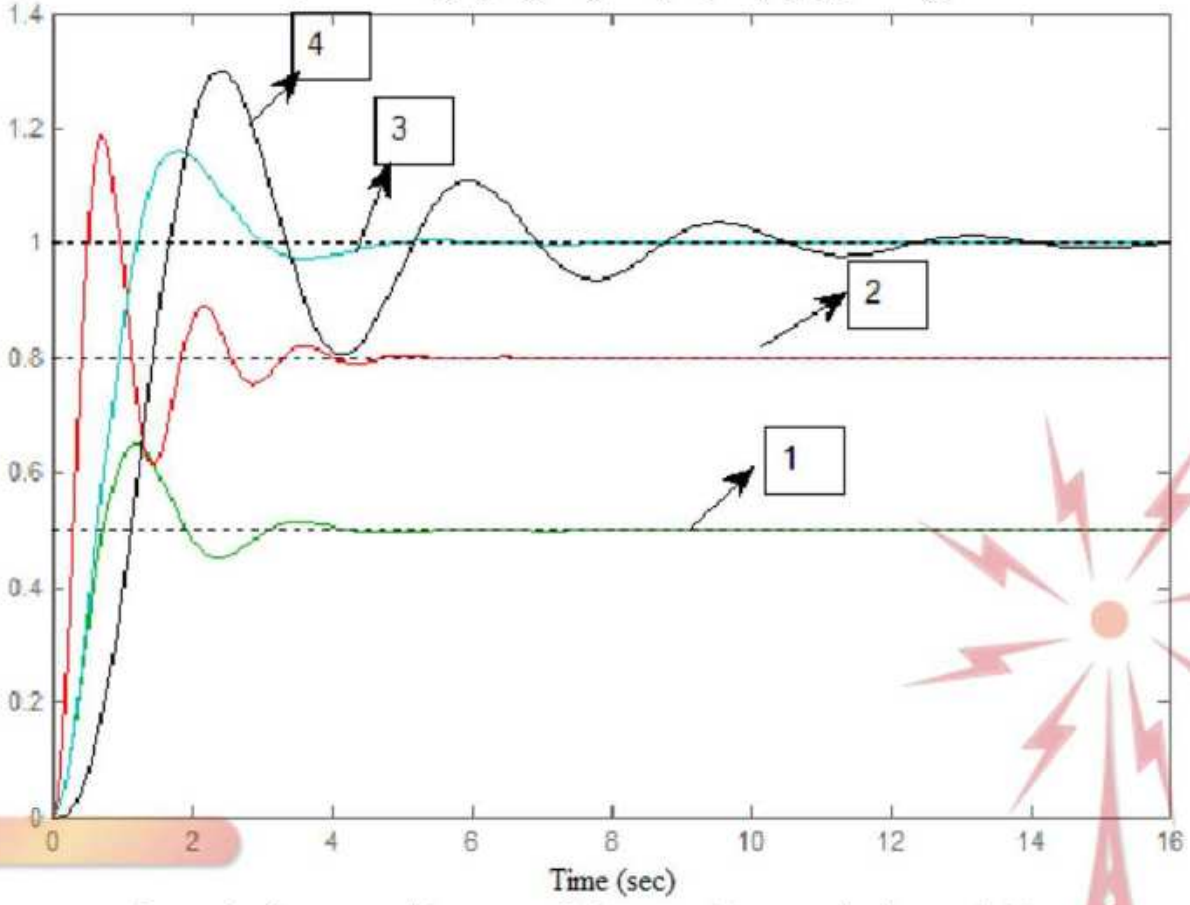
119. The result of proportional control of second order system is:

a. The system always unstable
b. Increase the time constant
c. Increase the maximum overshoot
d. Decrease the time constant

Correct answer: Increase the time constant

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل



Time response of second order system: (1) Uncontrolled system, (2) Proportional control, (3) Open loop system, (4) Integral control

Proportional controller reduces the steady state error and the damping ratio.

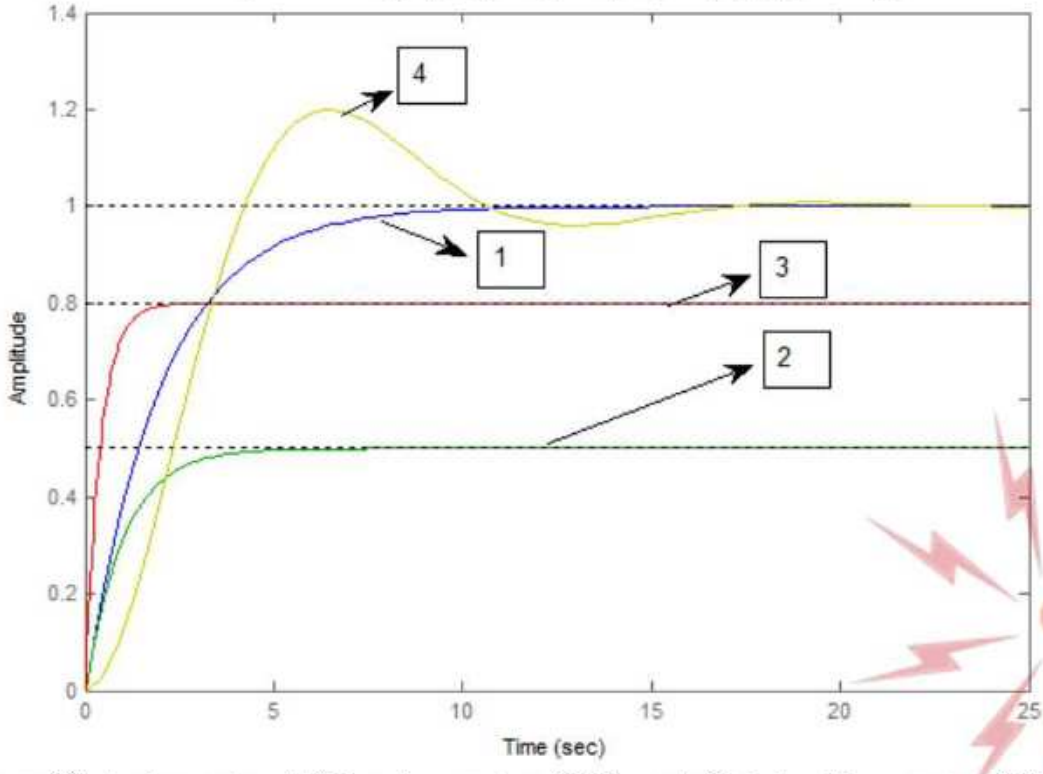
120. The result of proportional control of first order system is:

a. The system always unstable
b. Increase the maximum overshoot
c. Decrease the time constant & decrease steady state output
d. Increase the time constant
e. Increase steady state output

Correct answer: Decrease the time constant & decrease steady state output

Proportional controller reduces the steady state error and the time constant.

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

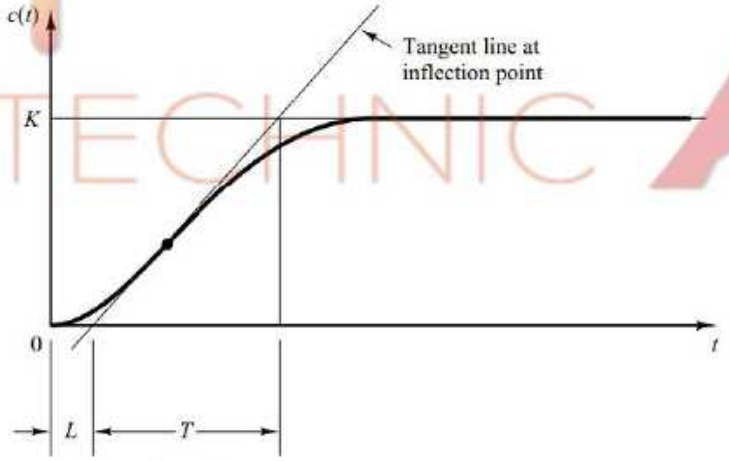


Time response of first order system: (1) Open loop system, (2) Uncontrolled closed loop system, (3) Proportional control, (4) Integral control

121. First method of Ziegler Nichols rule used when the time response of the closed loop system is:

a. Undamped
b. Overdamped
c. Underdamped
d. None of the above

Correct answer: Overdamped



122. Integral control of first order system:

a. Eliminate steady state error
b. Decrease maximum overshoot
c. Reduce settling time
d. Reduce the steady state error

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

Correct answer: Eliminates the steady state error

Integral controller:

1. Eliminates the steady state error
2. Rise the order of the system by one
3. Increase the oscillations

123. The proportional controller is mainly used to:

a. Increase maximum overshoot
b. Reduce the steady state error
c. Destabilize the open loop system
d. Reduce settling time

Correct answer: Reduce the steady state error

124. The transfer function of I controller is:

a. K
b. $\frac{1}{s}$
c. Ks
d. $\frac{K}{s}$
e. s

Correct answer: $\frac{K}{s}$ $(Specifically, \frac{K_i}{s})$

125. The transfer function of P controller is:

a. K
b. $\frac{1}{s}$
c. s
d. Ks
e. $\frac{K}{s}$

Correct answer: K $(Specifically, K_p)$

126. Sequence in tuning PID controller is:

a. K_p, T_i, T_d
b. T_d, T_i, K_p
c. T_i, T_d, T_{ip}
d. K_p, T_d, T_i

Correct answer: K_p, T_i, T_d

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

Analyzing System Stability Using Routh's and Nyquist Criteria

127. The characteristic equation of a closed loop system is $s^3 + s^2 + 2s + 1 = 0$, this system is:

a. Unstable
b. Stable
c. Critically stable
d. Conditionally stable

Correct answer: Stable

Very Important Notes (VINs):

1. A necessary (but not sufficient) condition for stability is that all the coefficients of the polynomial characteristic equation be positive.
2. A system is stable if and only if all the elements of the first column of the Routh array are positive.

See Appendix II

s^3:	1.0000	2.0000
s^2:	1.0000	1.0000
s^1:	1.0000	0
s^0:	1.0000	0

It is a stable system

Number of right-hand side poles = 0

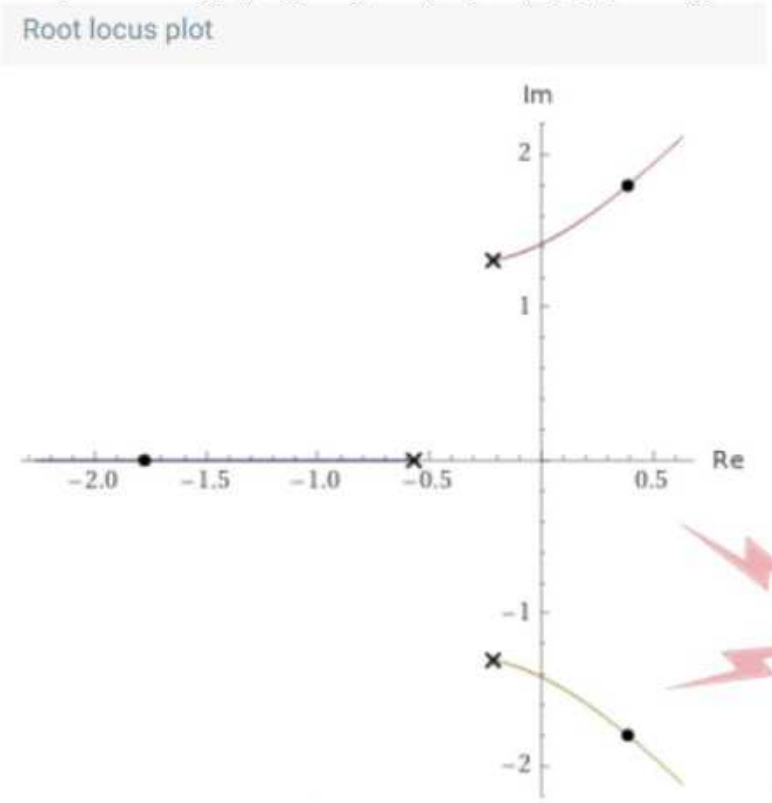
Polynomial coefficients roots:

$$s_1 = -0.2151 + j1.3071$$
$$s_2 = -0.2151 - j1.3071$$
$$s_3 = -0.5698 + j0.0000$$

spark POLYTECHNIC

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل



128. Roots on the imaginary axis indicates that:

a. Row of zeros in the first column in routh-hurwitz table
b. Row of zeros in the in routh-hurwitz table
c. One zero in the first column in routh-hurwitz table
d. Two zeros in the first column in routh-hurwitz table

Correct answer: Row of zeros in the in routh-hurwitz table

129. If we assume that s-contour encircled one zero and one pole of the function F(s), then F(s)-contour maps the origin point .... times:

a. -1
b. 0
c. 2
d. +1

Correct answer: 0

130. The characteristic equation of a closed loop system is $s^3 + s^2 + s + K = 0$, this system is:

a. Unstable
b. Stable
c. Conditionally stable
d. Critically stable

Correct answer: Conditionally stable

Very Important Notes (VINs):

3. A necessary (but not sufficient) condition for stability is that all the coefficients of the polynomial characteristic equation be positive.
4. A system is stable if and only if all the elements of the first column of the Routh array are positive.

See Appendix II

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

s^3:	1.0000	1.0000
s^2:	1.0000	K
s^1:	1-K	0
s^0:	K	0

It is a conditionally stable system if $K > 0$ and $1 - K > 0$

$$1 - K > 0$$

$$1 > K$$

So, the values of K for which the system is stable is:

$$1 > K > 0$$

131. Consider the loop transfer function. $\frac{K(s+6)}{[(s+3)(s+5)]}$, In the root locus diagram centroid will be located at t:

a. -2
b. -1
c. -4
d. -3

Correct answer: -2

$$\alpha = \frac{\sum Real\ part\ of\ finite\ open\ loop\ poles - \sum Real\ part\ of\ finite\ open\ loop\ zeros}{P - Z}$$

$$\alpha = \frac{(-3-5)-(-6)}{2-1} = -2$$

132. The order of higher order system can be reduced if it is poles are:

a. Multiple
b. Complex conjugate
c. Dominant
d. Distinct

Correct answer: Dominant

133. The poles of the control system are dominant if:

a. Positive
b. Distinct
c. Complex conjugate
d. Multiple

Correct answer: Complex conjugate

The order of higher system can be reduced to a second order system if it has dominant closed-loop poles.

134. The characteristic equation of a closed loop system is $s^3 + s^2 - s + 1 = 0$, this system is:

a. Critically stable
b. Conditionally stable
c. Stable
d. Unstable

Correct answer: Unstable

Very Important Notes (VINs):

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

1. A necessary (but not sufficient) condition for stability is that all the coefficients of the polynomial characteristic equation be positive.
2. A system is stable if and only if all the elements of the first column of the Routh array are positive.

See Appendix II

s^3:	1.0000	-1.0000
s^2:	1.0000	1.0000
s^1:	-2.0000	0
s^0:	1.0000	0

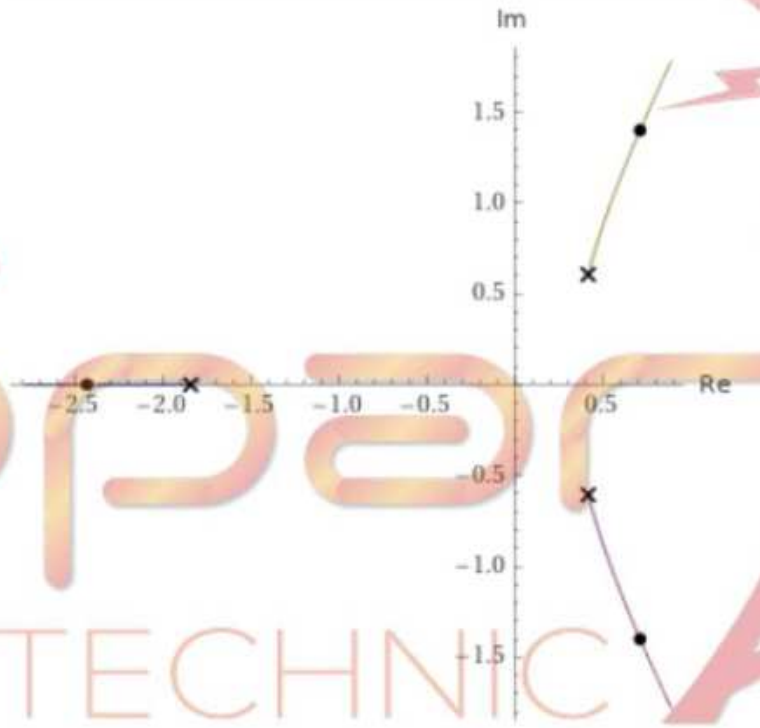
It is an unstable system

Number of right-hand side poles = 2

Polynomial coefficients roots:

$$s_1 = -1.8393 + j0.0000$$
$$s_2 = 0.4196 + j0.6063$$
$$s_3 = 0.4196 - j0.6063$$

Root locus plot



Spark POLYTECHNIC

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

Drawing Bode Plot and Nyquist Plot for a Given Transfer Function

135. The slope of logarithm of magnitude vs. frequency of lead first order (factor/ elements) is:
Or
The slope of an approximate logarithm of magnitude vs. frequency plot for lead first order (factor/ elements) is:
Or
The slope of logarithm of magnitude vs. frequency of transfer function $G(s)H(s) = Ts + 1$ is:
Or
The slope of an approximate logarithm of magnitude vs. frequency transfer function $G(s)H(s) = Ts + 1$ is:

a. Zero dB/dec at very low frequencies and +20dB/ dec at very high frequencies
b. Zero dB at very low frequencies and -20 dB at very high frequencies
c. Zero dB at all frequencies
d. Zero dB at very low frequencies and +20 dB at very high frequencies
e. Zero dB/dec at very low frequencies and -20dB/ dec at very high frequencies
f. -20dB/ dec at all frequencies
g. +20dB/dec at all frequencies

Correct answer: Zero dB/dec at very low frequencies and +20dB/ dec at very high frequencies

136. Which unit is adopted for magnitude measurement in Bode plots?

a. Decibel
b. Degree
c. Decimal
d. Deviation

Correct answer: Decibel

137. The corner frequency of lag first order (factor/ elements) is:

a. T
b. inf
c. 0
d. $\frac{1}{T}$

Correct answer: $\frac{1}{T}$

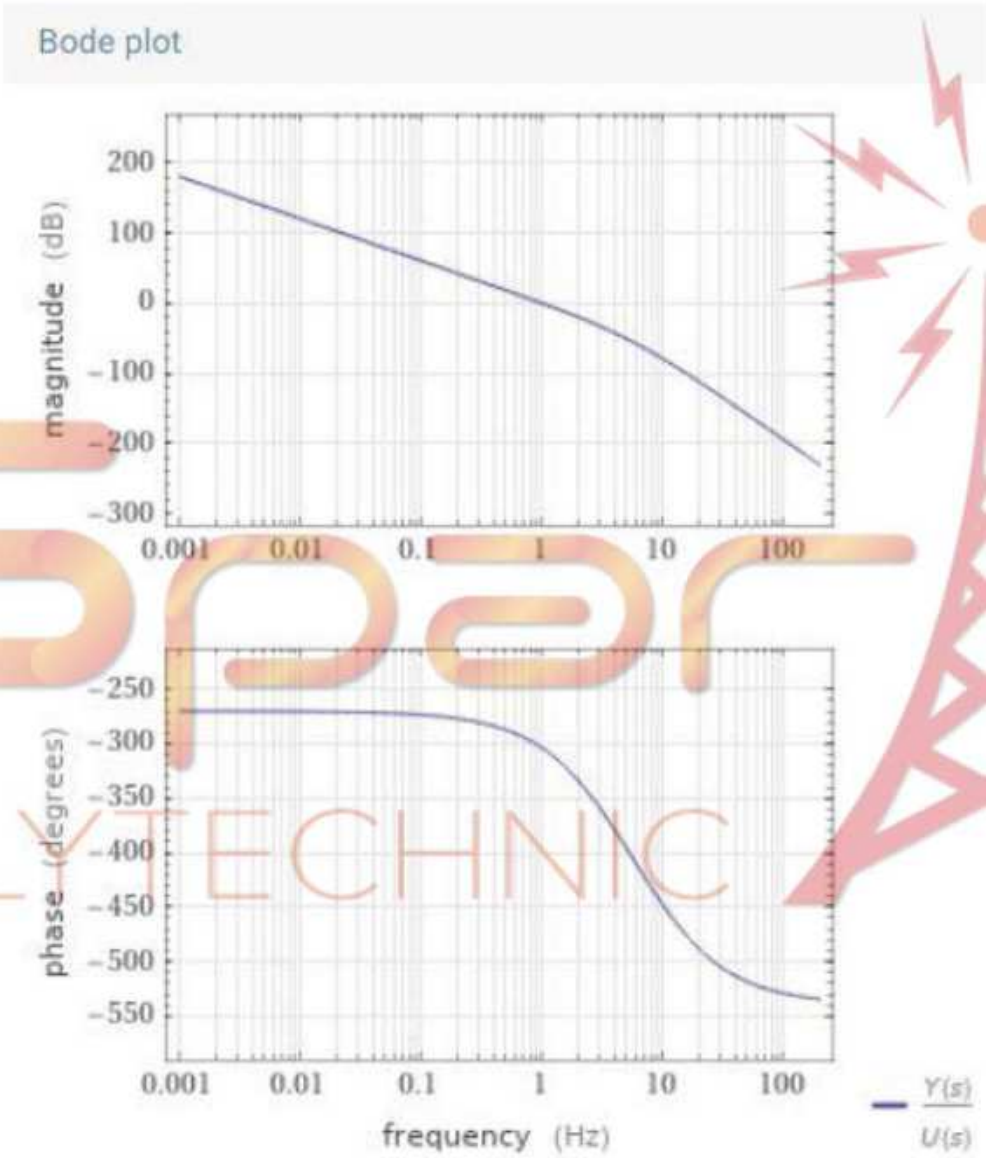
ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

138. Consider the system represented by the equation given below. What would be the total phase value at $w = 0$?

$$G(s)H(s) = \frac{200}{s^3\,(s\,+\,3)(s\,+\,6)(s\,+\,10)}$$

a. -270
b. -90
c. -180
d. Zero

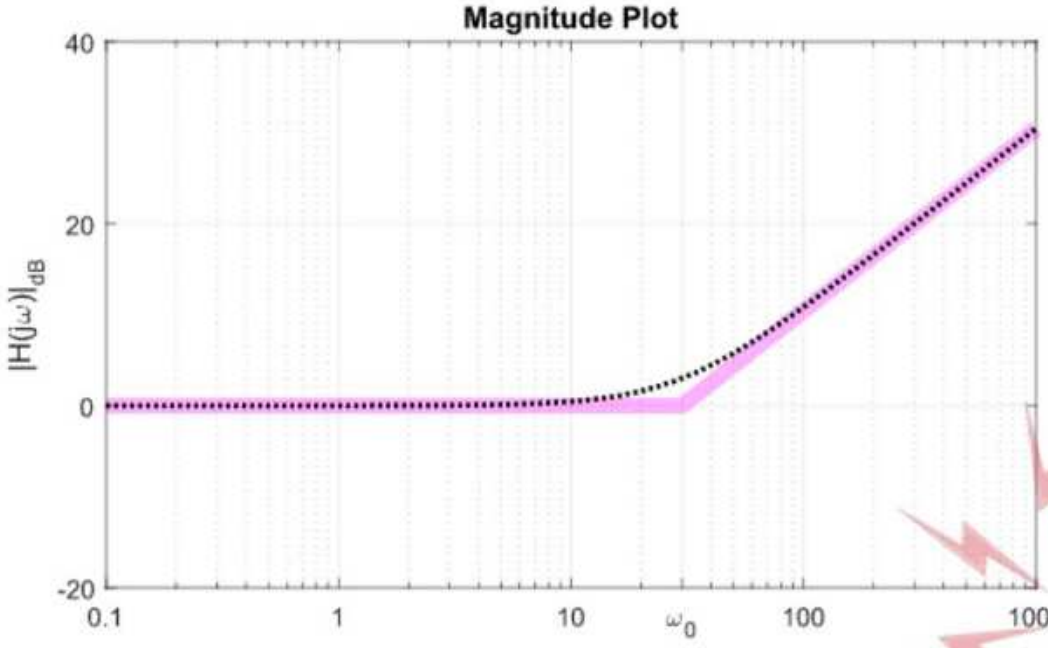
Correct answer: -270



POLYTECHNIC

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل



139. At which frequency does the magnitude of the system becomes zero dB:

a. Phase crossover frequency
b. Resonant frequency
c. Cut-off frequency
d. Gain crossover frequency

Correct answer: Gain crossover frequency

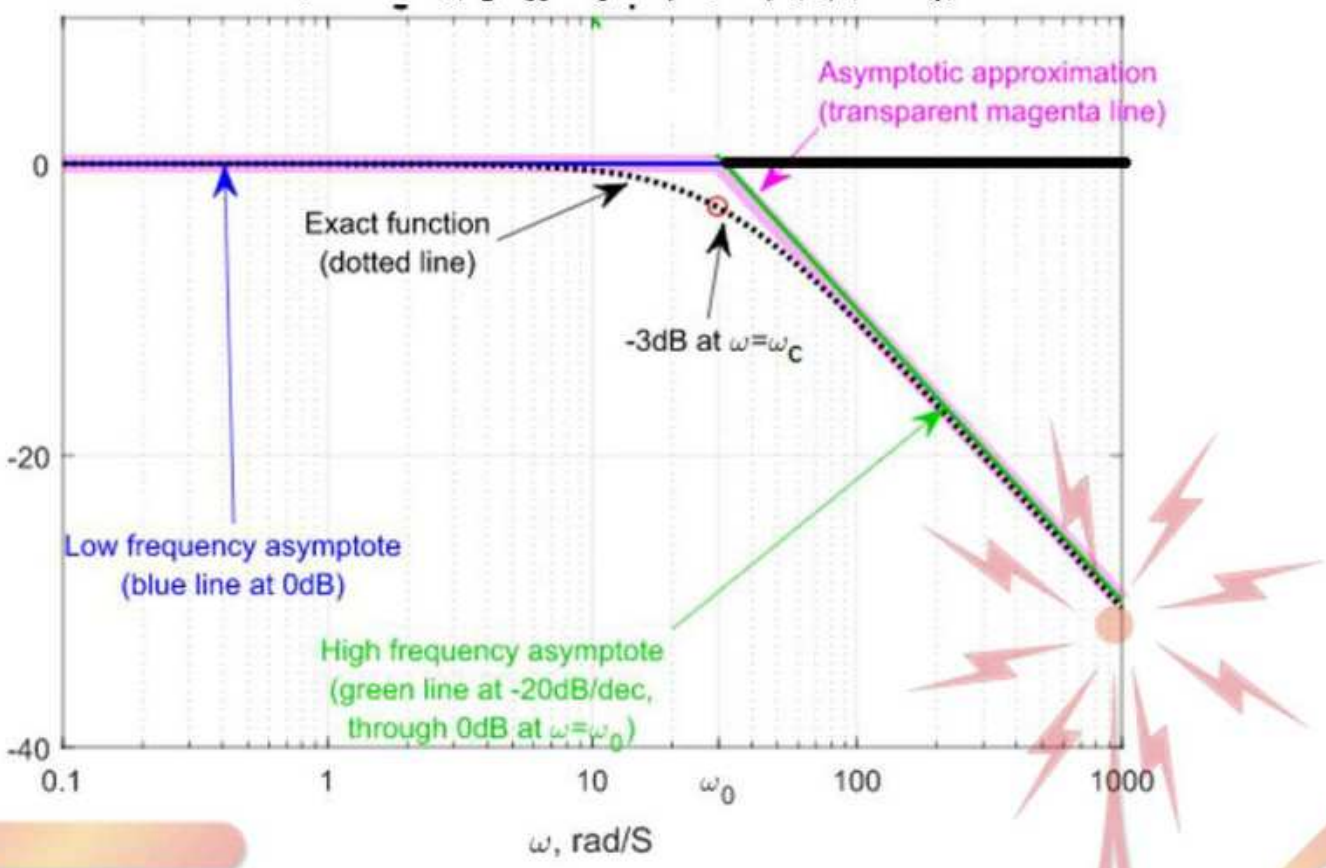
140. The slope of logarithm of magnitude vs. frequency of lag first order (factor/ elements) is:
Or
The slope of an approximate logarithm of magnitude vs. frequency plot for lag first order (factor/ elements) is:
Or
The slope of logarithm of magnitude vs. frequency of transfer function $G(s)H(s) = \frac{1}{Ts+1}$ is:
Or
The slope of an approximate logarithm of magnitude vs. frequency transfer function $G(s)H(s) = \frac{1}{Ts+1}$ is:

a. Zero dB/dec at very low frequencies and +20dB/ dec at very high frequencies
b. Zero dB at very low frequencies and +20 dB at very high frequencies
c. -20dB/ dec at all frequencies
d. Zero dB at all frequencies
e. Zero dB at very low frequencies and -20 dB at very high frequencies
f. +20dB/dec at all frequencies
g. Zero dB/dec at very low frequencies and -20dB/ dec at very high frequencies

Correct answer: Zero dB/dec at very low frequencies and -20dB/ dec at very high frequencies

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشئ هذا العمل



141. The phase shift of a second order lag (factor/ elements) when the frequency is changed from zero up to infinity is:
Or
In frequency response, the phase angle of a second order lag (factor/ elements) when the frequency is changed from zero up to infinity is:

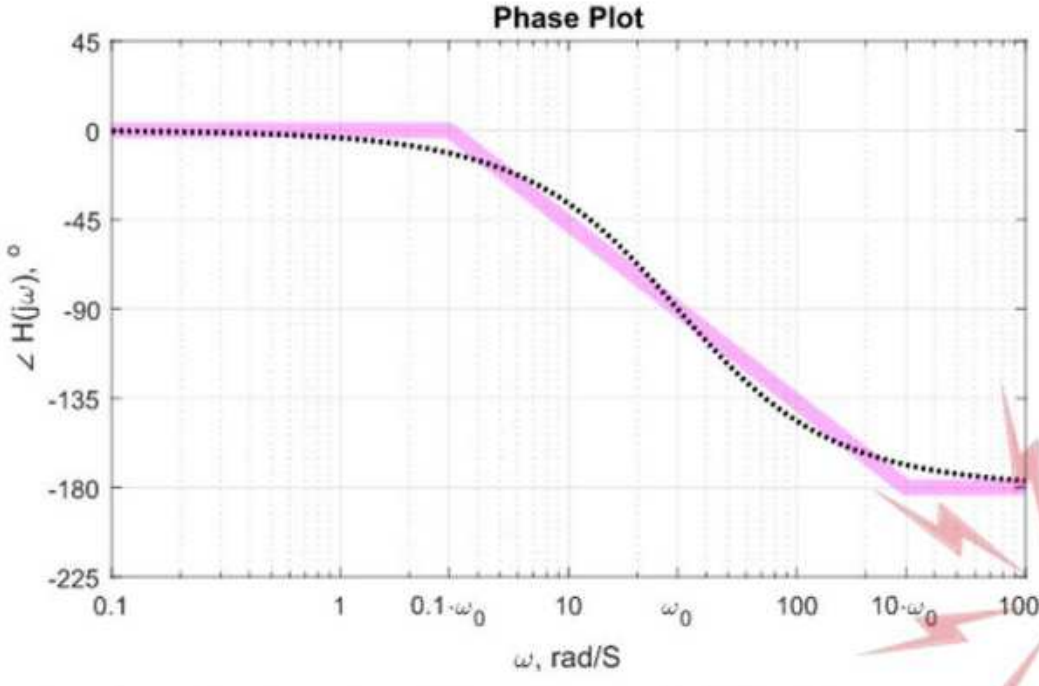
a. Changed from zero degrees up to +90 degrees
b. Changed from zero degrees up to -180 degrees
c. Changed from zero degrees up to +180 degrees
d. Changed from zero degrees up to -90 degrees

Correct answer: Changed from zero degrees up to -180 degrees

POLYTECHNIC

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل



142. The slope of logarithm of magnitude vs. frequency of a differentiator is:
Or
The slope of logarithm of magnitude vs. frequency of a derivative (factor/ elements) is:
Or
The slope of logarithm of magnitude vs. frequency of a $j\omega$ factor is:

a. -20dB/ dec at all frequencies
b. Zero dB at very low frequencies and +20 dB at very high frequencies
c. Zero dB at all frequencies
d. Zero dB/dec at very low frequencies and -20dB/ dec at very high frequencies
e. +20dB/dec at all frequencies
f. Zero dB at very low frequencies and -20 dB at very high frequencies
g. Zero dB/dec at very low frequencies and +20dB/ dec at very high frequencies

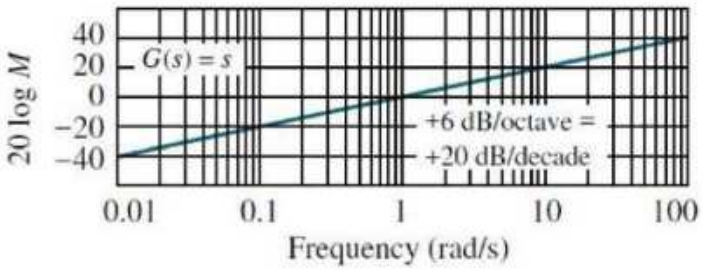
Correct answer: +20dB/dec at all frequencies

POLYTECHNIC

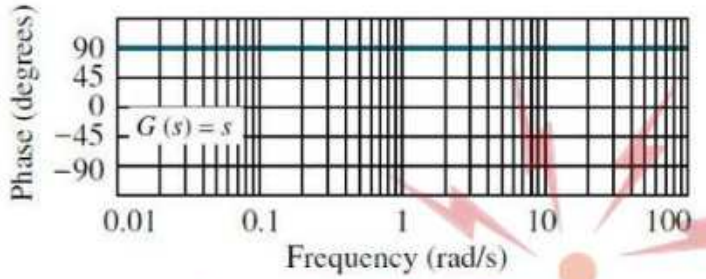
بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

The Bode magnitude plot is a straight line with a +20 dB/decade slope passing through 0 dB at ω = 1.



The Bode phase plot is equal to a constant +90°.



143. The slope of logarithm of magnitude vs. frequency of lag second order (factor/ elements) is:
Or
The slope of an approximate logarithm of magnitude vs. frequency plot for lag second order (factor/ elements) is:

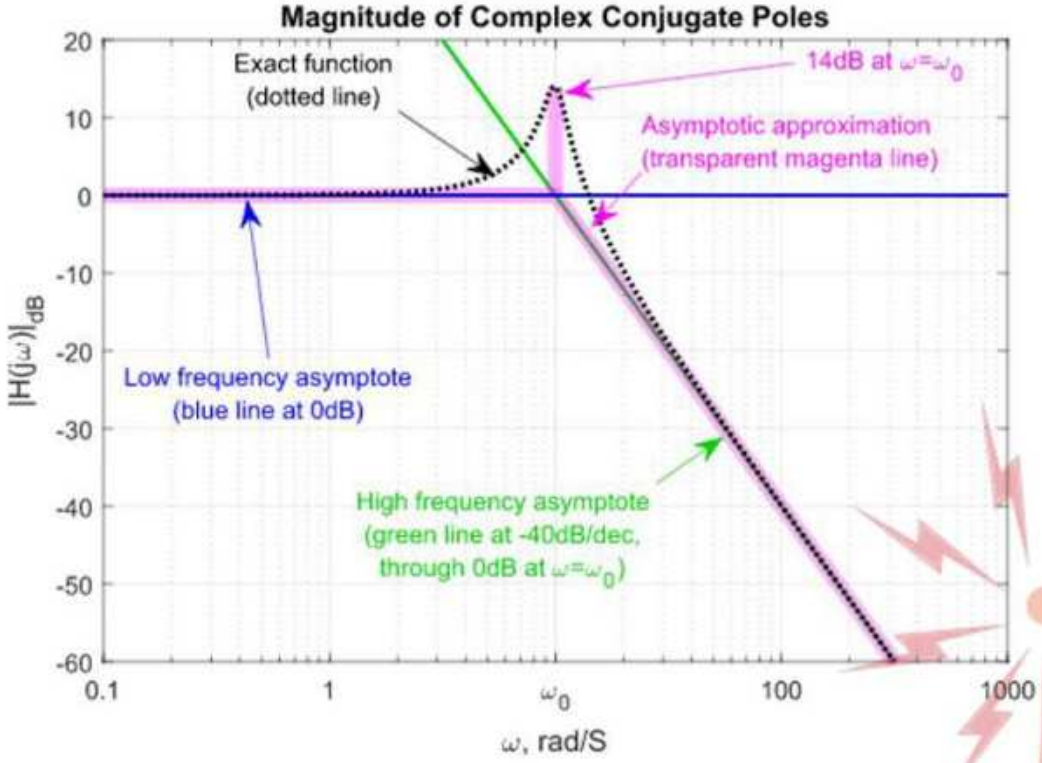
a. +40dB/dec at all frequencies
b. Zero dB at very low frequencies and -40 dB at very high frequencies
c. Zero dB at all frequencies
d. -40dB/ dec at all frequencies
e. Zero dB/dec at very low frequencies and +40dB/ dec at very high frequencies
f. Zero dB at very low frequencies and +40 dB at very high frequencies
g. Zero dB/dec at very low frequencies and -40dB/ dec at very high frequencies

Correct answer: Zero dB/dec at very low frequencies and -40dB/ dec at very high frequencies

POLYTECHNIC

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

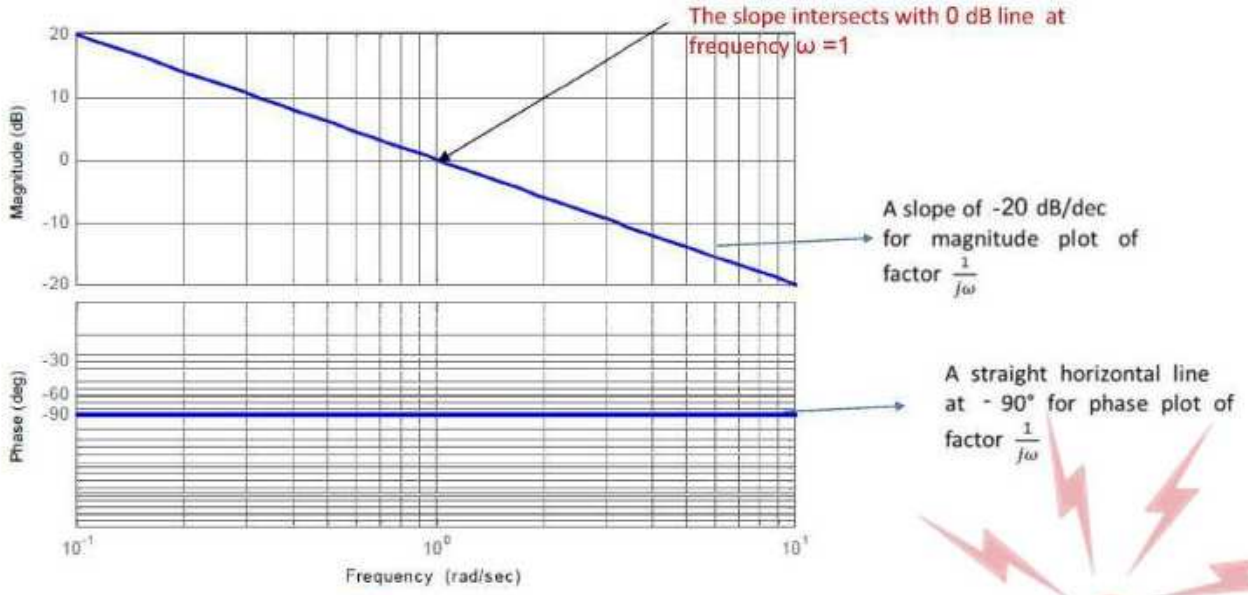


144. The slope of logarithm of magnitude vs. frequency of an integrator is:
Or
The slope of logarithm of magnitude vs. frequency of an integral (factor/ elements) is:
Or
The slope of logarithm of magnitude vs. frequency of an $(j\omega)^{-1}$ (factor/ elements) is:

a. +20dB/dec at all frequencies
b. Zero dB/dec at very low frequencies and -20dB/ dec at very high frequencies
c. Zero dB/dec at very low frequencies and +20dB/ dec at very high frequencies
d. Zero dB at all frequencies
e. -20dB/ dec at all frequencies
f. Zero dB at very low frequencies and +20 dB at very high frequencies
g. Zero dB at very low frequencies and -20 dB at very high frequencies

Correct answer: -20dB/ dec at all frequencies

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل



145. The phase shift of a differentiator when the frequency is changed from zero up to infinity is:
Or
The phase shift of a derivative (factor/ elements) when the frequency is changed from zero up to infinity is:
Or
The phase shift of a $j\omega$ (factor/ elements) when the frequency is changed from zero up to infinity is:
Or
In frequency response, the phase angle of differentiator is:
Or
In frequency response, the phase angle of derivative (factor/ elements) is:
Or
In frequency response, the phase angle of $j\omega$ (factor/ elements) is:

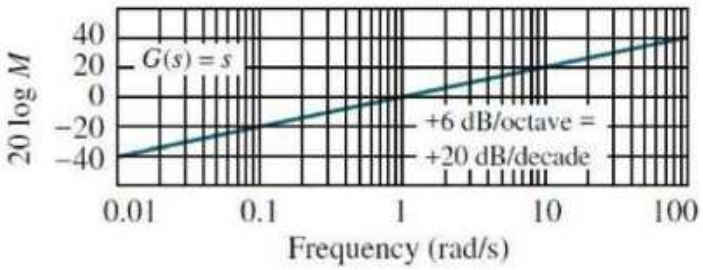
a. Changed from zero degrees up to -90 degrees
b. -90 degrees
c. -180 degrees
d. Changed from zero degrees up to +90 degrees
e. +90 degree
f. +180 degrees
g. Zero degree

Correct answer: +90 degree

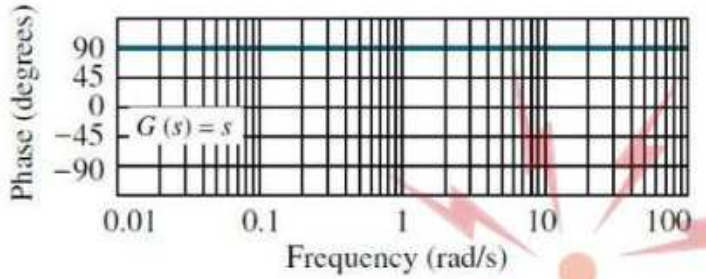
بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

The Bode magnitude plot is a straight line with a +20 dB/decade slope passing through 0 dB at ω = 1.



The Bode phase plot is equal to a constant +90°.

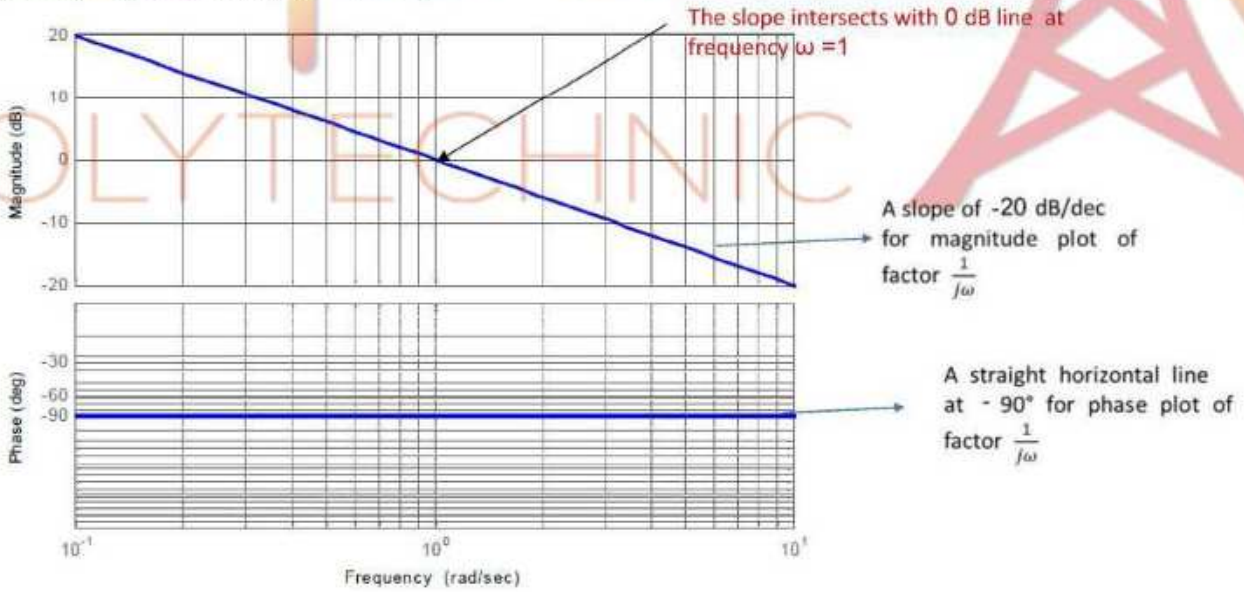


146. If a pole is located at origin, how does it get represented on the magnitude plot?

a. $-10\ \log(w)\ dB$
b. $-60\ \log(w)\ dB$
c. $-40\ \log(w)\ dB$
d. $-20\ \log(w)\ dB$

Correct answer: $-20\ \log(w)\ dB$

Since it's a pole, then this means that it's in the denominator because poles are the roots of the denominator, and as it is located at the origin, then $s = 0$, in other words, it is s in the denominator of the transfer function, which is equivalent to the integrator (integral factor/ $(j\omega)^{-1}$ factor).



ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

147. The open loop transfer function is $G(s)H(s) = \frac{s+1}{s(s+2)(s+3)}$, The Nyquist plot of this system starts at:

a. Infinity magnitude and -90 degrees phase angle
b. -90 degrees phase angle
c. 1/6 magnitude and -90 degrees phase angle
d. Infinity magnitude

Correct answer: Infinity magnitude and -90 degrees phase angle

The magnitude of $G(j\omega)H(j\omega)$, i.e., $|G(j\omega)H(j\omega)|$, is obtained as:

$$G(j\omega)H(j\omega) = \frac{j\omega + 1}{j\omega(j\omega + 2)(j\omega + 3)}$$

The start of plot where $\omega = 0$.

The magnitude of the open loop transfer function is:

$$|G(j\omega)H(j\omega)| = \frac{\sqrt{\omega^2 + 1}}{\omega\sqrt{\omega^2 + 4}\sqrt{\omega^2 + 9}}$$

$$|G(0)H(0)| = \frac{\sqrt{0^2 + 1}}{0\sqrt{0^2 + 4}\sqrt{0^2 + 9}} = \frac{1}{0} = \infty$$

The phase angle of the open loop transfer function is:

$$\angle G(j\omega)H(j\omega) = \tan^{-1}\left(\frac{\omega}{1}\right) - \frac{\pi}{2} - \tan^{-1}\left(\frac{\omega}{2}\right) - \tan^{-1}\left(\frac{\omega}{3}\right)$$

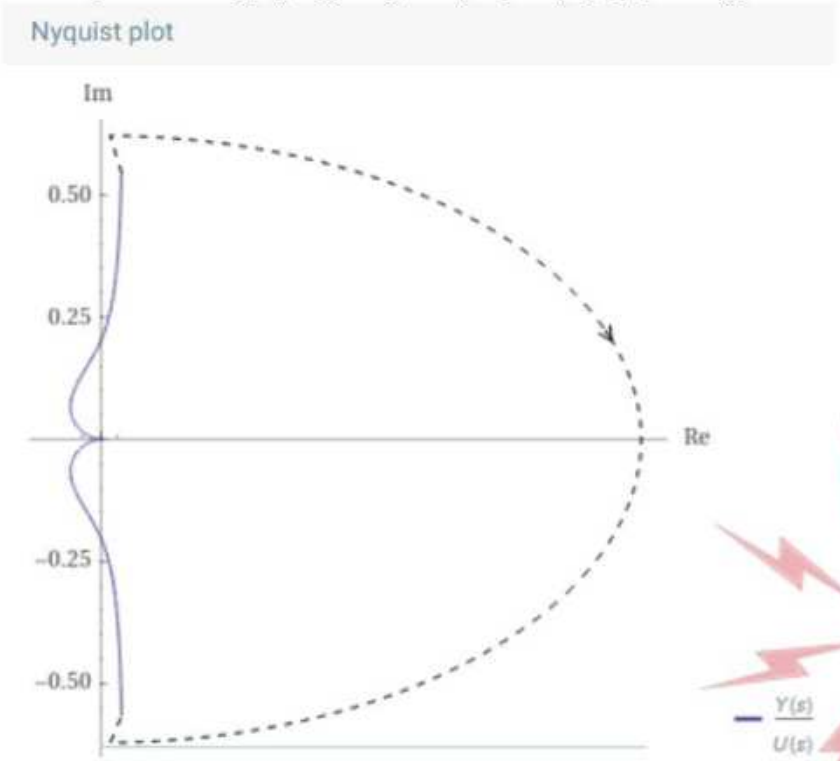
$$\angle G(0)H(0) = \tan^{-1}(0) - \frac{\pi}{2} - \tan^{-1}(0) - \tan^{-1}(0) = 0 - \frac{\pi}{2} - 0 - 0 = -\frac{\pi}{2}$$

$$\tan^{-1}(0) = -\frac{\pi}{2}$$

POLYTECHNIC

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل



148. The open loop transfer function is $G(s)H(s) = \frac{s+1}{s(s+2)(s+3)}$, The Nyquist plot of this system terminates at:

a. -180 degrees phase angle
b. Zero magnitude and +180 degrees phase angle
c. Infinity magnitude and -180 degrees phase angle
d. Zero magnitude and -180 degrees phase angle

Correct answer: Zero magnitude and -180 degrees phase angle

The magnitude of $G(j\omega)H(j\omega)$, i.e., $|G(j\omega)H(j\omega)|$, is obtained as:

$$G(j\omega)H(j\omega) = \frac{j\omega + 1}{j\omega(j\omega + 2)(j\omega + 3)}$$

The end of plot where $\omega = \infty$.

The magnitude of the open loop transfer function is:

$$|G(j\omega)H(j\omega)| = \frac{\sqrt{\omega^2 + 1}}{\omega\sqrt{\omega^2 + 4}\sqrt{\omega^2 + 9}}$$

$$|G(\infty)H(\infty)| = \frac{\sqrt{\infty^2 + 1}}{\infty\sqrt{\infty^2 + 4}\sqrt{\infty^2 + 9}} = \frac{\infty}{\infty}$$

Using L'Hôpital's rule:

$$|G(\infty)H(\infty)| = 0$$

The phase angle of the open loop transfer function is:

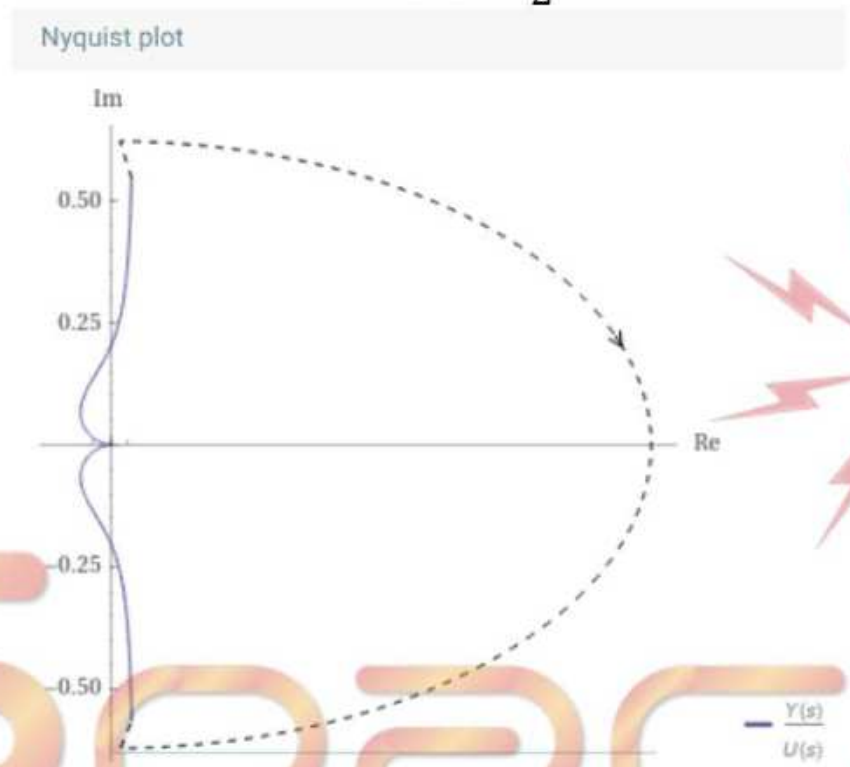
بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

$$\angle G(j\omega)H(j\omega) = \tan^{-1}\left(\frac{\omega}{1}\right) - \frac{\pi}{2} - \tan^{-1}\left(\frac{\omega}{2}\right) - \tan^{-1}\left(\frac{\omega}{3}\right)$$

$$\angle G(\infty)H(\infty) = \tan^{-1}(\infty) - \frac{\pi}{2} - \tan^{-1}(\infty) - \tan^{-1}(\infty) = \frac{\pi}{2} - \frac{\pi}{2} - \frac{\pi}{2} - \frac{\pi}{2} = \pi$$

$$\tan^{-1}(\infty) = \frac{\pi}{2}$$



149. The corner frequency of a standard lag first order (factor/ elements) is:

a. 3T
b. T
c. 1
d. 4T

Correct answer: 1

150. The corner frequency of a standard lead first order (factor/ elements) is:

a. 3T
b. 4T
c. 1
d. T

Correct answer: 1

151. The phase shift of an integrator when the frequency is changed from zero up to infinity is:
Or
The phase shift of an integral (factor/ elements) when the frequency is changed from zero up to infinity is:
Or
The phase shift of an $(j\omega)^{-1}$ (factor/ elements) when the frequency is changed from zero up to infinity is:
Or

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

In frequency response, the phase angle of integrator is:

Or

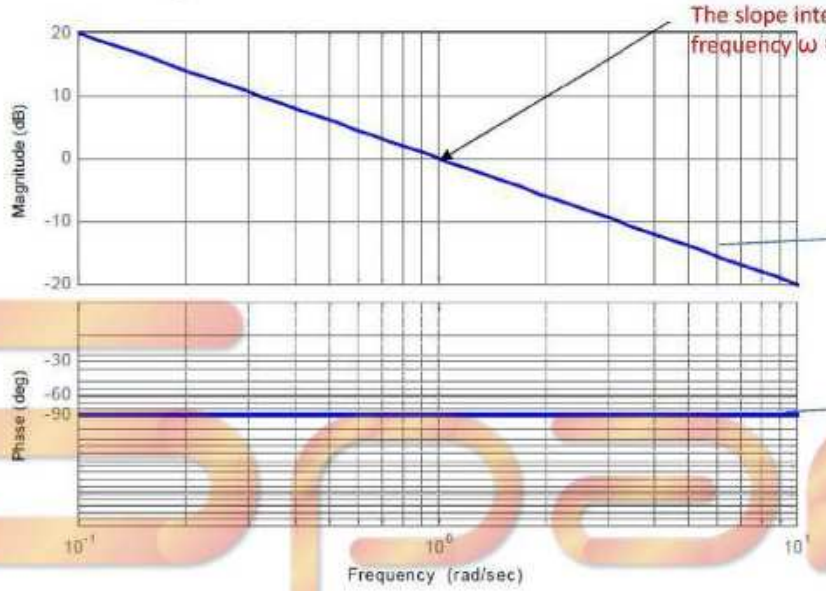
In frequency response, the phase angle of integral (factor/ elements) is:

Or

In frequency response, the phase angle of $(j\omega)^{-1}$ (factor/ elements) is:

a. Zero degree
b. Changed from zero degrees up to +90 degrees
c. +90 degree
d. -90 degrees
e. +180 degree
f. -180 degrees
g. Changed from zero degrees up to -90 degrees

Correct answer: -90 degrees



POLYTECHNIC

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

Applying Root-Locus Technique for Control System Analysis

152. For root locus which of the following are the starting points:

a. Closed loop poles
b. Open loop poles
c. Closed loop zeros
d. Open loop zeros

Correct answer: Open loop poles

153. The root locus is the part of roots of the characteristic equation traced out in the S plane. Which one of the following is correct?

a. As the sensitivity is changed
b. As the input of the system is changed
c. As a system parameter is changed
d. As the output of the system is changed

Correct answer: As a system parameter is changed

154. The root locus of a system has three asymptotes. The system can have:

a. Four poles and one zero
b. Five poles and two zeros
c. Three poles
d. All the above

Correct answer: All the above

155. The root locus is the locus of the change of the ..... traced out in the s-plane.

a. As a system parameter is changed
b. As the sensitivity is changed
c. As the input of the system is changed
d. As the output of the system is changed

Correct answer: As a system parameter is changed

The root locus is the locus of the change of the system parameters of the characteristic equation traced out in the s-plane.

156. Which one of the following is not the property of root locus?

a. The breakaway points are determined from $\frac{dK}{ds} = 0$
b. They start from the open loop poles and terminate at the open loop zeroes
c. The root locus is symmetrical about imaginary axis
d. Segments of the real axis are the part of the root locus if and only is the total number of real poles and zeroes to their right is odd.

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

Correct answer: The root locus is symmetrical about imaginary axis
The root locus is the locus traced due to the gain of the system with changing frequency and need not be symmetrical about origin.



ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

Additional Questions

Introduction to Control Systems

157. In an open loop control system
a. Only system parameters have effect on the control output
b. Output is dependent on control input
c. Output is independent of control input
d. None of the above

Correct answer: Output is independent of control input
Open-loop control systems. Those systems in which the output has no effect on the control action are called open-loop control systems.
In other words, in an open-loop control system the output is neither measured nor fed back for comparison with the input.

158. A control system in which the control action is somehow dependent on the output is known as:
a. Closed loop system
b. Semiclosed loop system
c. Open system
d. None of the above

Correct answer: Closed loop system

159. For open control system which of the following statements is incorrect?
a. Less expensive
b. Construction is simple and maintenance easy
c. Recalibration is not required for maintaining the required quality of the output
d. Errors are caused by disturbances

Correct answer: Recalibration is not required for maintaining the required quality of the output

160. The signal that drives the controller. If this signal is the difference between the input and output, it is called the error is:
a. Feedback signal
b. Input signal
c. Actuating error signal
d. None of the above

Correct answer: Actuating error signal

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل منشأ هذا العمل

Laplace Transforms

161. When the function F(s) has imaginary conjugate poles, the type of the function f(t) is:

a. (Undamped/ Sinusoidal function) (precisely, undamped sinusoidal function)
b. Exponential function
c. Damped sinusoidal
d. Critically damped sinusoidal
e. Damped ramp

Correct answer: (Undamped/ Sinusoidal function) (precisely, undamped sinusoidal function)

162. When the function F(s) has complex poles, the type of the function f(t) is:

a. (Undamped/ Sinusoidal function) (precisely, undamped sinusoidal function)
b. Damped ramp
c. Exponential function
d. Damped sinusoidal
e. Critically damped sinusoidal

Correct answer: Damped sinusoidal

163. The residues (constants) of partial fraction for $F(s) = \frac{2}{(s+1)(s+2)}$ is:

a. $2; -2$
b. $1; -1$
c. $-1; 2$
d. $1; 1$

Correct answer: $2; -2$

$$\frac{2}{(s+1)(s+2)} = \frac{A}{s+1} + \frac{B}{s+2} = \frac{As + 2A + Bs + B}{(s+1)(s+2)}$$

$$A + B = 0$$

$$B = -A$$

$$2A + B = 2$$

$$2A - A = 2$$

$$A = 2, B = -2$$

164. The final value $f(\infty)$ of the function $F(s) = \frac{s+2}{s(s+1)}$ is:

a. 0.5
b. 1
c. 2
d. 0

Correct answer:

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

$$f(\infty) = \lim_{s\to 0} sF(s) = \lim_{s\to 0} s * \frac{s+2}{s(s+1)} = \lim_{s\to 0} \frac{s+2}{s+1} = 2$$

Another way to find the solution:
Using partial fractions:

$$\frac{s+2}{s(s+1)} = \frac{A}{s} + \frac{B}{s+1} = \frac{As + A + Bs}{s(s+1)}$$

$$s + 2 = As + A + Bs$$

By using the comparison method:

$$A = 2$$
$$A + B = 1$$
$$B = -1$$
$$F(s) = \frac{2}{s} - \frac{1}{s+1}$$
$$f(t) = 2 - e^{-t}$$
$$f(t = \infty) = 2 - e^{-\infty} = 2 - 0 = 2$$

165. Laplace transform of unit ramp function u(t) is:

a. s
b. $\frac{1}{s^2}$
c. $\frac{1}{s}$
d. 1

Correct answer: $\frac{1}{s^2}$

166. Laplace transform of unit impulse function u(t) is:

a. s
b. $\frac{1}{s^2}$
c. $\frac{1}{s}$
d. 1

Correct answer: 1

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

Modeling of Mechanical, Liquid-Level and Electrical Control Systems

167. Which of the following parameters is analogous to Viscous friction coefficient in Force-voltage analogy?

a. Resistance
b. Cu
c. Lubricity
d. Capacitance

Correct answer: Resistance

168. Transfer function is used only for linear time invariant systems:

a. True
b. False

Correct answer: True

Transfer function only used for linear time invariant systems mainly single input single output (SISO)

169. Which of the following statements is correct regarding block diagram?

a. A block diagram is a representation of the interconnection of subsystems that form a system. In a linear system, the block diagram consists of blocks representing subsystems, arrows representing signals, summing junctions, and pickoff points.
b. A block diagram has the advantage of indicating more realistically the signal flows of the actual system.
c. A block diagram of a system is a pictorial representation of the functions performed by each component and of the flow of signals.
d. All the above.

Correct answer: All the above.

170. A linear time invariant system is stable if:

a. System in excited by the bounded input, the output is also bounded
b. System in excited by the bounded input, the output is not bounded
c. In the absence of input, output tends zero
d. System in excited by the bounded input, the output is also bounded and in the absence of input, output tends zero

Correct answer: System in excited by the bounded input, the output is also bounded and in the absence of input, output tends zero

A system is stable only if it is BIBO (Bounded-Input, Bounded-Output) stable and asymptotic stable.
Asymptotic stability means that solutions that start close enough not only remain close enough but also eventually converge to the equilibrium.

171. Asymptotic stability is concerned with:

a. A system not under influence of input
b. A system under influence of output
c. A system under influence of input

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

d. A system not under influence of output

Correct answer: A system not under influence of input

Asymptotic stability concerns a free system relative to its transient behavior.

172. Bounded input and Bounded output stability notion concerns with:

a. A system under influence of input
b. A system under influence of output
c. A system not under influence of input
d. A system not under influence of output

Correct answer: A system under influence of input

173. If a system is given unbounded input then the system is:

a. Linear
b. Unstable
c. Stable
d. Not defined

Correct answer: Not defined

If the system is given with the unbounded input then nothing can be clarified for the stability of the system.

174. Linear mathematical model applies to:

a. Linear systems
b. Stable systems
c. Unstable systems
d. Non-linear systems

Correct answer: Stable systems

As the output exceeds certain magnitude then the linear mathematical model no longer applies.

POLYTECHNIC

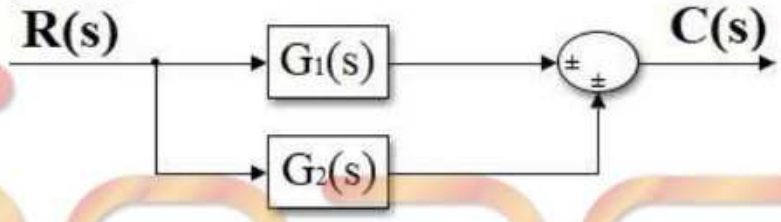
ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

Block Diagram Reduction and Obtaining Transfer Function

175. A block diagram of open loop system has two forward paths in the open loop transfer function, the number of terms in the numerator is:

a. 3
b. 1
c. 2
d. None of the above

Correct answer: 2



The equivalent transfer function of the system is the equivalent transfer function of two blocks $G_1(s)$ and $G_2(s)$ which are connected in parallel in the forward path is:

$$\frac{C(s)}{R(s)} = G(s) = G_1(s) + G_2(s)$$

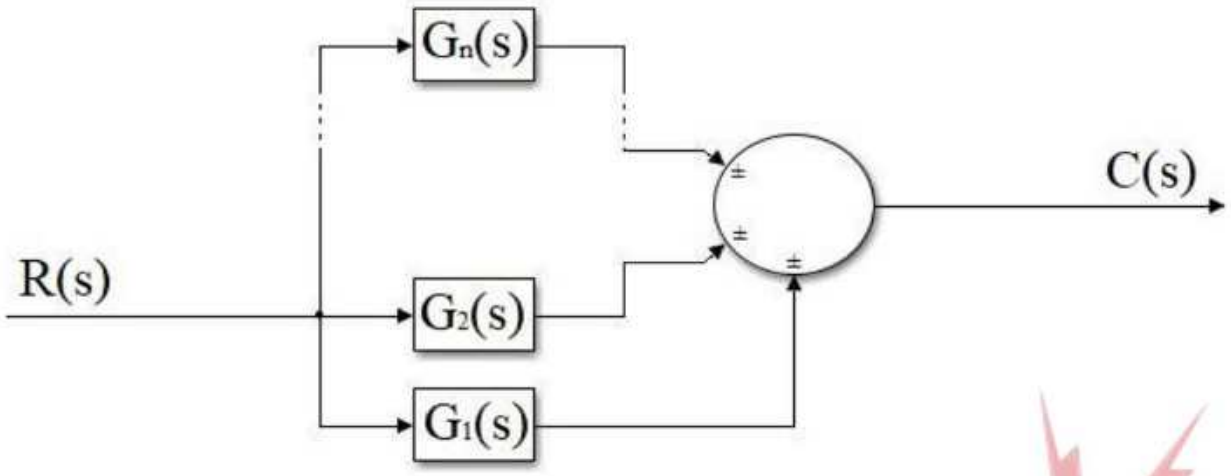
So, for a block diagram of open loop system which has two forward paths in the open loop transfer function, the number of terms in the numerator is two.

176. A block diagram of open loop system has n forward paths in the open loop transfer function, the number of terms in the numerator is:

a. n
b. $n + 1$
c. $2n$
d. $n - 1$
e. n^2

Correct answer: n

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل منشأ هذا العمل



The equivalent transfer function of the system is the equivalent transfer function of n blocks $G_1(s)$, $G_2(s)$ …… and $G_n(s)$ which are connected in parallel in the forward path is:

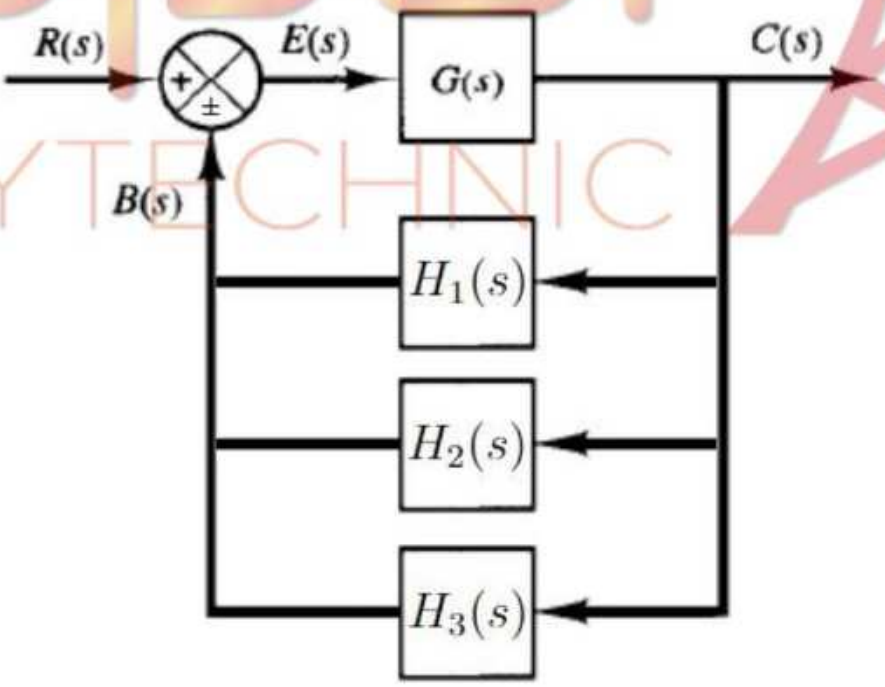
$$\frac{C(s)}{R(s)} = G(s) = G_1(s) + G_2(s) + \cdots\cdots\cdots + G_n(s)$$

So, for a block diagram of open loop system which has n forward paths in the open loop transfer function, the number of terms in the numerator is n term.

177. A block diagram of closed loop system has three feedbacks in the closed loop transfer function, the number of terms in the denominator is:

a. 2
b. 4
c. 3
d. 1

Correct answer: 4



POLYTECHNIC

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

The equivalent transfer function of three blocks $H_1(s)$, $H_2(s)$ and $H_3(s)$ which are connected in parallel in the feedback path is:

$$H(s) = H_1(s) + H_2(s) + H_3(s)$$

In closed loop system with positive/negative feedback, the forward path is G(s) and the feedback path is H(s), the transfer function:

$$\frac{C(s)}{R(s)} = \frac{G(s)}{[1 \pm G(s)H(s)]} = \frac{G(s)}{[1 \pm G(s)(H_1(s) + H_2(s) + H_3(s))]}$$

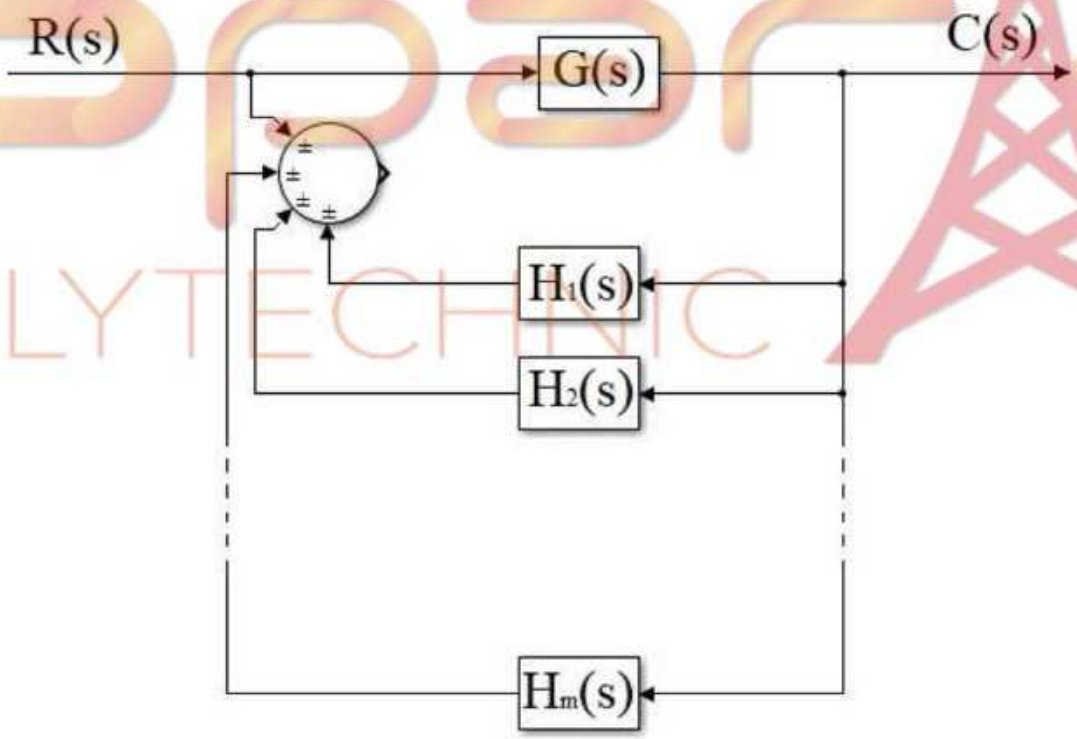
$$\frac{C(s)}{R(s)} = \frac{G(s)}{[1 \pm G(s)H_1(s) \pm G(s)H_2(s) \pm G(s)H_3(s))]}$$

So, for a block diagram of closed loop system which has three feedbacks in the closed loop transfer function, the number of terms in the denominator is four.

178. A block diagram of closed loop system has m feedbacks in the closed loop transfer function, the number of terms in the denominator is:

a. m
b. $m + 1$
c. $2m$
d. $m - 1$
e. m^2

Correct answer: $m + 1$



The equivalent transfer function of m blocks $H_1(s)$, $H_2(s) \cdots \cdots \cdots$ and $H_m(s)$ which are connected in parallel in the feedback path is:

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

$$H(s) = H_1(s) + H_2(s) + \cdots\cdots\cdots + H_m(s)$$

In closed loop system with positive/negative feedback, the forward path is G(s) and the feedback path is H(s), the transfer function:

$$\frac{C(s)}{R(s)} = \frac{G(s)}{[1 \pm G(s)H(s)]} = \frac{G(s)}{[1 \pm G(s)(H_1(s) + H_2(s) + \cdots\cdots\cdots + H_m(s))]}$$

$$\frac{C(s)}{R(s)} = \frac{G(s)}{[1 \pm G(s)H_1(s) \pm G(s)H_2(s) \pm \cdots\cdots\cdots \pm G(s)H_m(s))]}$$

So, for a block diagram of closed loop system which has m feedbacks in the closed loop transfer function, the number of terms in the denominator is m+1 term.

179. A block diagram of closed loop system has four forward paths and two feedbacks in the closed loop transfer function, answer the following questions:

I. The number of terms in the numerator:
   a. 4
   b. 1
   c. 7
   d. 9
   e. None of the above

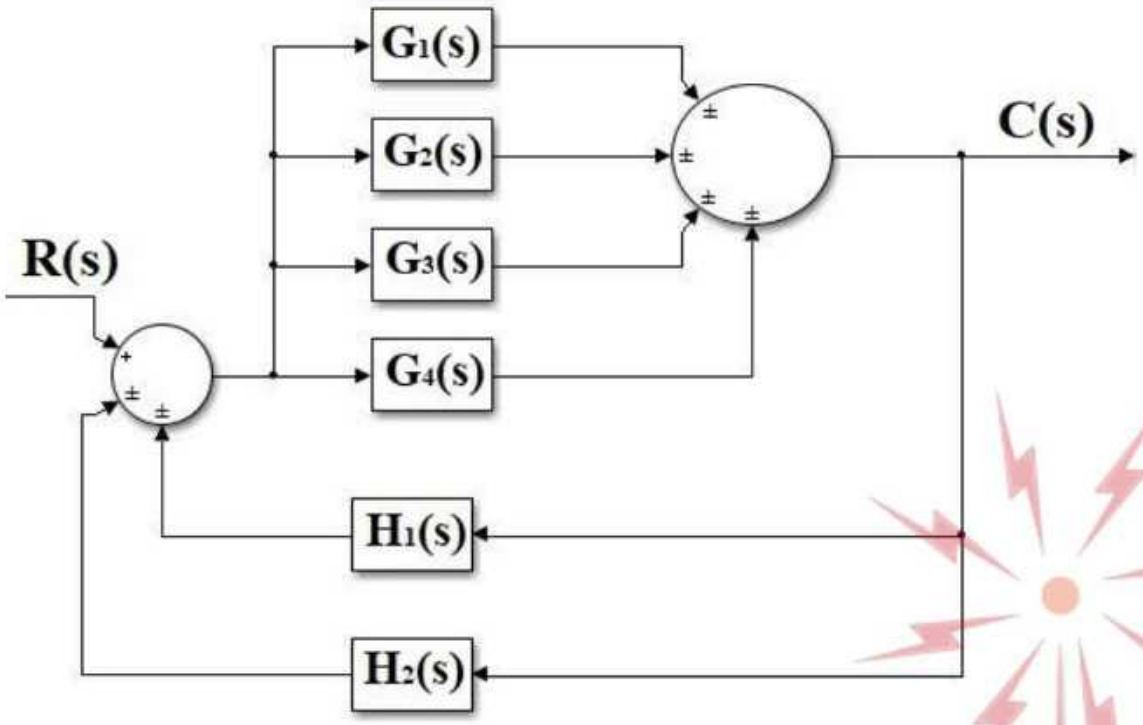
II. The number of terms in the denominator:
   a. 9
   b. 7
   c. 4
   d. 2
   e. None of the above

Correct answer:
I. 4
II. 9

Spark POLYTECHNIC

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل



I. The equivalent transfer function of forward path is the equivalent transfer function of four blocks $G_1(s)$, $G_2(s)$, $G_3(s)$ and $G_4(s)$ which are connected in parallel in the forward path is:

$$G(s) = \pm G_1(s) \pm G_2(s) \pm G_3(s) \pm G_4(s)$$

So, the number of terms in the numerator is the number of forward paths.

$$Number\ of \text{ terms in the numerator } = \text{number of forward paths} = 4$$

II. The equivalent transfer function of two blocks $H_1(s)$ and $H_2(s)$ which are connected in parallel in the feedback path is:

$$H(s) = H_1(s) + H_2(s)$$

In closed loop system with positive/negative feedback, the forward path is G(s) and the feedback path is H(s), the transfer function:

$$\frac{C(s)}{R(s)} = \frac{G(s)}{[1 \pm G(s)H(s)]}$$

$$\frac{C(s)}{R(s)} = \frac{G(s)}{[1 \pm (\pm G_1(s) \pm G_2(s) \pm G_3(s) \pm G_4(s))(H_1(s) + H_2(s))]}$$

$$\frac{C(s)}{R(s)} = \frac{G(s)}{[1 + (G_1(s) + G_2(s) + G_3(s) + G_4(s))(H_1(s) + H_2(s))]}$$

So, for a block diagram of closed loop system which has four forward paths and two feedbacks in the closed loop transfer function, the number of terms in the denominator is the number of forward paths multiplied by the number of feedbacks, then one is added to the product.

$$Number\ of \text{ terms in the denominator} = \text{number of forward paths} * \text{number of feedbacks} + 1 = 4 * 2 + 1 = 9$$

180. A block diagram of closed loop system has four forward paths and two feedbacks in the closed loop transfer function, answer the following questions:

I. The number of terms in the numerator:

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

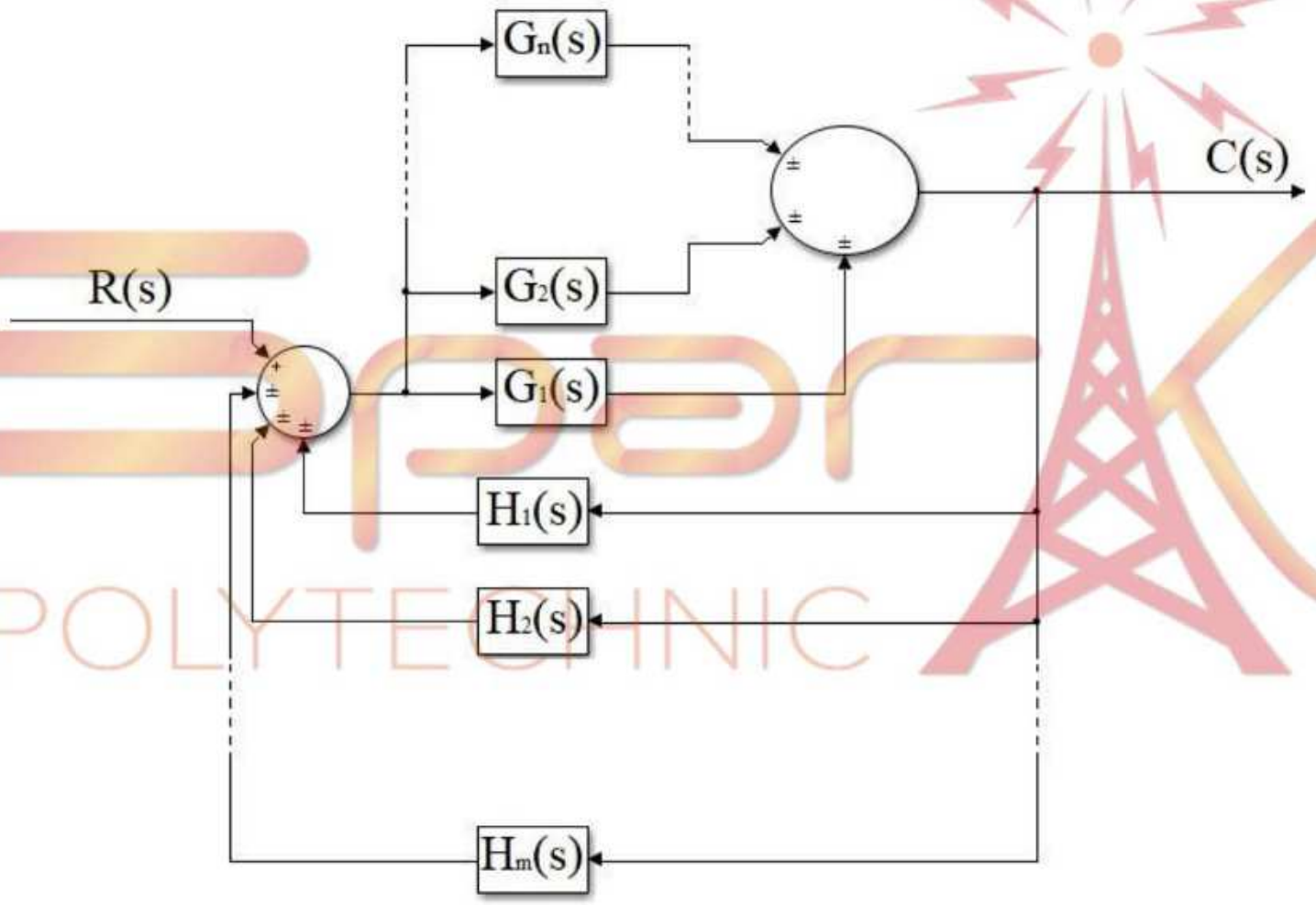
a. 2n
b. n+1
c. n-1
d. n
e. None of the above

II. The number of terms in the denominator:

a. n*m-1
b. n*m+1
c. n+m+1
d. n+m-1
e. None of the above

Correct answer:

I. n
II. n*m+1



I. The equivalent transfer function of forward path is the equivalent transfer function of n blocks $G_1(s)$, $G_2(s)\cdots\cdots$ and $G_n(s)$ which are connected in parallel in the forward path is:

$$G(s) = \pm G_1(s) \pm G_2(s) \pm \cdots \pm G_n(s)$$

So, the number of terms in the numerator is the number of forward paths.

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص او جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

$$Number\ of \text{ terms in the numerator} = \text{number of forward paths} = n$$

II. The equivalent transfer function of m blocks $H_1(s)$, $H_2(s)\cdots\cdots\cdots$ and $H_m(s)$ which are connected in parallel in the feedback path is:

$$H(s) = H_1(s) + H_2(s) + \cdots + H_m(s)$$

In closed loop system with positive/negative feedback, the forward path is G(s) and the feedback path is H(s), the transfer function:

$$\frac{C(s)}{R(s)} = \frac{G(s)}{[1 \pm G(s)H(s)]}$$

$$\frac{C(s)}{R(s)} = \frac{G(s)}{[1 \pm (\pm G_1(s) \pm G_2(s) \pm \cdots \pm G_n(s))(H_1(s) + H_2(s) + \cdots\cdots\cdots + H_m(s))]}$$

$$\frac{C(s)}{R(s)} = \frac{G(s)}{[1 + (G_1(s) + G_2(s) + \cdots + G_n(s))(H_1(s) + H_2(s) + \cdots\cdots\cdots + H_m(s))]}$$

So, for a block diagram of closed loop system which has n forward paths and m feedbacks in the closed loop transfer function, the number of terms in the denominator is the number of forward paths multiplied by the number of feedbacks, then one is added to the product.

$$Number\ of \text{ terms in the denominator} = \text{number of forward paths} * \text{number of feedbacks} + 1 = n * m + 1$$

181. In signal flow graph, the product of all _____ gains while going through a forward path is known as 'Path gain'.

a. Path
b. Loop
c. Branch
d. Node

Correct answer: Branch

182. According to signal flow graph, which among the following represents the relationship between nodes by drawing a line between them?

a. Self-loop
b. Mesh
c. Semi-node
d. Branch

Correct answer: Branch

183. Two loops are said to be non-touching only if no common _____ exists between them.

a. Loop
b. Branch
c. Node
d. Feedback path

Correct answer: Node

184. Signal flow graphs:

a. They cannot be converted back to block diagram
b. They apply to linear systems
c. The equation obtained may or may not be in the form of cause or effect
d. Arrows are not important in the graph

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

Correct answer: They apply to linear systems

Signal flow graphs are used to find the transfer function of control system by converting the block diagrams into signal flow graphs or directly but cannot be used for nonlinear systems.

185. Signal flow graphs are reliable to find transfer function than block diagram reduction technique.

a. True
b. False

Correct answer: True

As one set technique and formula is used here but in block diagram technique various methods are involved which increases complexity.

186. For which systems are the signal flow graphs applicable?

a. Linear time invariant system
b. Non-linear time invariant system
c. Non-linear time-varying system
d. Linear time-varying systems

Correct answer: Linear time invariant system

187. While solving signal flow graph using Mason's gain equation, what does the second letter in two subscript notations of 'L' stand for?

a. Parallel number of loops
b. Number of non-touching loops
c. Serial number of loops
d. Number of touching loops

Correct answer: Number of non-touching loops

188. In a signal flow graph method, how is an overall transfer function of a system obtained?

a. Block diagram reduction rules
b. Laplace's equation
c. Mason's equation
d. Poisson's equation
e. Lagrange's equation

Correct answer: Mason's equation

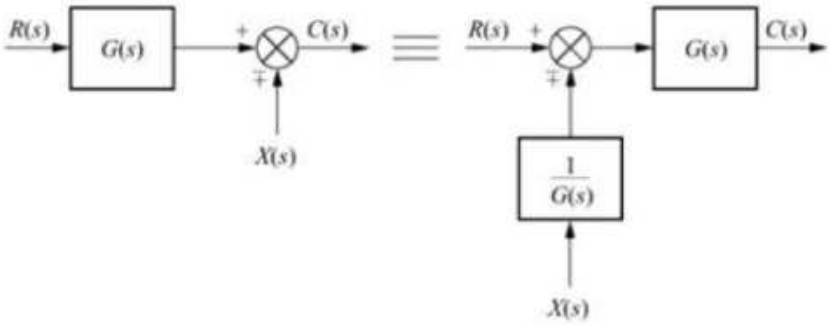
See Appendix III

189. While moving a summing point before the block, which among the following should be added?
Or
While moving a summing point ahead of the block, which among the following should be added?

a. Node
b. Block of reciprocal transfer function
c. Summing point
d. Block of the same transfer function

Correct answer: Block of reciprocal transfer function

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل



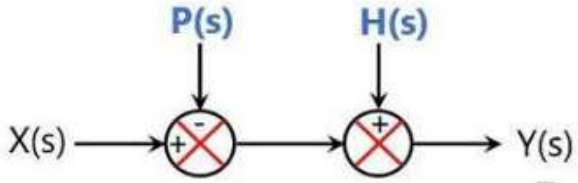
$$C = GR \pm X$$
$$= G(+R \pm X/G)$$

See Appendix III

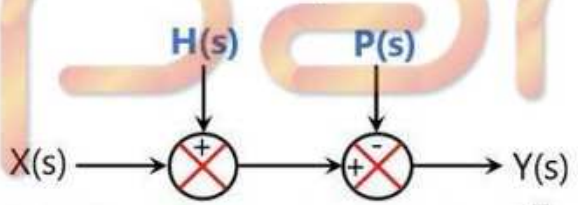
190. While replacing summing points which among the following should be done?

a. Summing points cannot be replaced
b. Inverting the signs of each summing point
c. Adding pickoff point between them
d. Nothing should be done

Correct answer: Nothing should be done

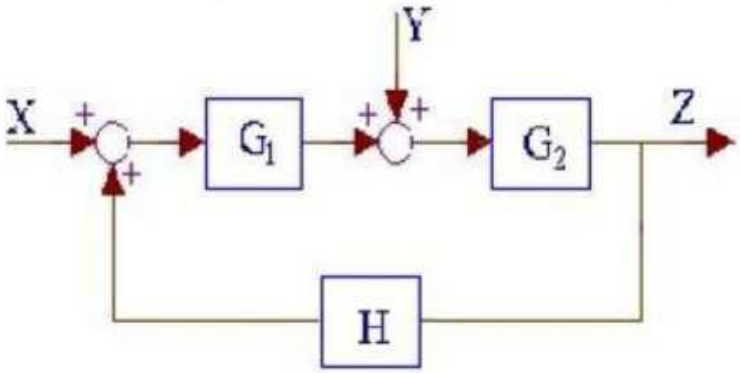


$$= X(s) - P(s) + H(s)$$



$$= X(s) - P(s) + H(s)$$

191. The transfer functions which represent the responses of the X & Y inputs individually are:

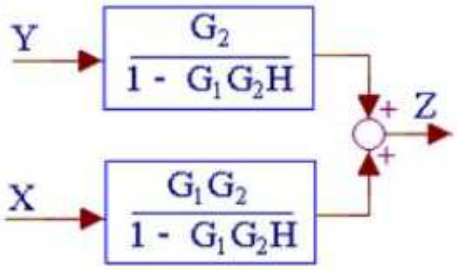


Correct answer:

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

VIN: To find the transfer function which represents the response of one input, deactivate other inputs and find the transfer function for the new single input single output system.



192. A signal flow graph is the graphical representation of the relationships between the variables of set linear algebraic equations.

a. True
b. False

Correct answer: True

193. In signal flow graph, a node having only outgoing branches.

a. Input node
b. Output node
c. Incoming node
d. Outgoing node

Correct answer: Input node

Nodes are the point by which the branches are outgoing and/or ingoing and this can be input, output or general node and input node is the node having only outgoing branches.

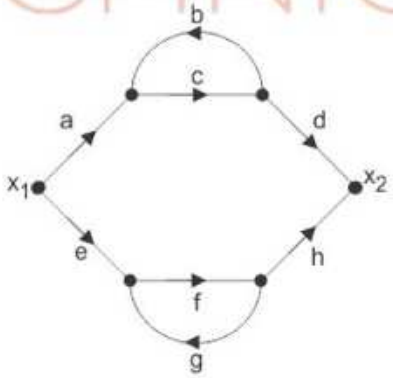
194. In signal flow graph, a node having only ingoing branches.

a. Incoming node
b. Output node
c. Outgoing node
d. Input node

Correct answer: Output node

Nodes are the point by which the branches are outgoing and/or ingoing and this can be input, output or general node and output node is the node having only ingoing branches.

195. Use mason's gain formula to find the transfer function of the following signal flow graph:



a. $\frac{acdfg+bcefh}{1-(fg+bc)+bcfg}$

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص او جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل منشأ هذا العمل

b. $\frac{acd-acdfg+efh-bcefh}{1-(fg+bc)}$

c. $\frac{acd-acdfg+efh-bcefh}{1-(fg+bc)+bcfg}$

d. $\frac{acd+efh}{1-(fg+bc)+bcfg}$

Correct answer: $\frac{acd-acdfg+efh-bcefh}{1-(fg+bc)+bcfg}$

Using mason's gain formula transfer function from signal flow graph can be calculated which relates the forward path gain to the various paths and loops.

$$T(s) = \frac{\sum_k P_k \Delta_k}{\Delta} = \frac{P_1\Delta_1 + P_2\Delta_2}{\Delta}$$

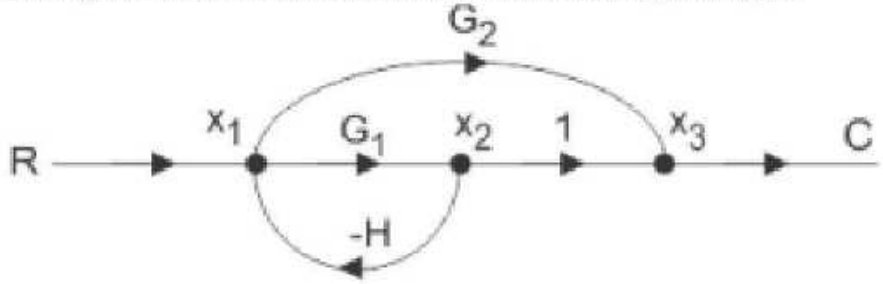
P_k is forward path transfer function for k[th] path

The cofactor Δ_k, is the determinant with the loops touching the k[th] path removed

The determinant $\Delta = 1 - (sum\ of\ all\ different\ loop\ gains) + (sum\ of\ the\ gain\ products\ of\ all\ combinations\ of\ two\ nontouching\ loops) - (sum\ of\ the\ gain\ products\ of\ all\ combinations\ of\ three\ nontouching\ loops) + \ldots$

$$P_1 = acd$$
$$P_2 = efh$$
$$\Delta_1 = 1 - fg$$
$$\Delta_2 = 1 - bc$$
$$\Delta = 1 - (fg + bc) + bcfg$$
$$T(s) = \frac{P_1\Delta_1 + P_2\Delta_2}{\Delta} = \frac{acd(1-fg) + efh(1-bc)}{1-(fg+bc)+bcfg}$$
$$= \frac{acd - acdfg + efh - bcefh}{1-(fg+bc)+bcfg}$$

196. Use mason's gain formula to calculate the transfer function of given figure:



a. $\frac{G_1}{1+G_2H}$

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص او جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل منشأ هذا العمل

b. $\frac{G_1+G_2}{1+G_1H}$

c. $\frac{G_2}{1+G_1H}$

d. None of the mentioned

Correct answer: $\frac{G_1+G_2}{1+G_1H}$

Using mason's gain formula transfer function from signal flow graph can be calculated which relates the forward path gain to the various paths and loops.

$$T(s) = \frac{\sum_k P_k \Delta_k}{\Delta} = \frac{P_1\Delta_1 + P_2\Delta_2}{\Delta}$$

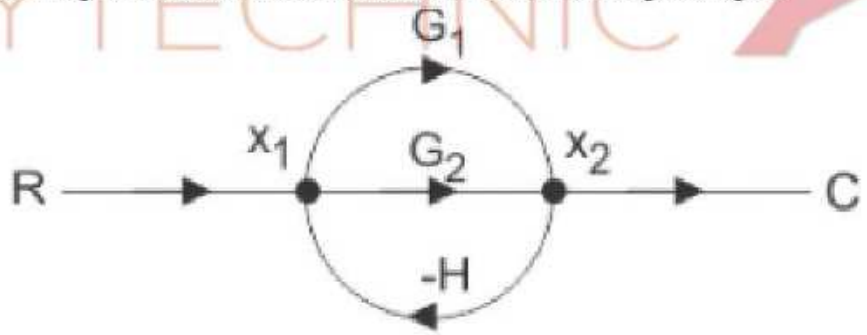
P_k is forward path transfer function for k^{th} path

The cofactor Δ_k, is the determinant with the loops touching the kth path removed

The determinant $\Delta = 1 - (sum\ of\ all\ different\ loop\ gains) + (sum\ of\ the\ gain\ products\ of\ all\ combinations\ of\ two\ nontouching\ loops) - (sum\ of\ the\ gain\ products\ of\ all\ combinations\ of\ three\ nontouching\ loops) + \ldots$

$$P_1 = G_1$$
$$P_2 = G_2$$
$$\Delta_1 = 1$$
$$\Delta_2 = 1$$
$$\Delta = 1 - (-G_1H) = 1 + G_1H$$
$$T(s) = \frac{P_1\Delta_1 + P_2\Delta_2}{\Delta} = \frac{G_1 * 1 + G_2 * 1}{1 + G_1H} = \frac{G_1 + G_2}{1 + G_1H}$$

197. Use mason's gain formula to find the transfer function of the given figure:



a. $\frac{G_1+G_2}{1-G_1H+G_2H}$

b. $G_1 - G_2$

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص او جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

c. $\frac{G_1+G_2}{1+G_1H+G_2H}$

d. $G_1 + G_2$

Correct answer: $\frac{G_1+G_2}{1+G_1H+G_2H}$

Using mason's gain formula transfer function from signal flow graph can be calculated which relates the forward path gain to the various paths and loops.

$$T(s) = \frac{\sum_k P_k \Delta_k}{\Delta} = \frac{P_1\Delta_1 + P_2\Delta_2}{\Delta}$$

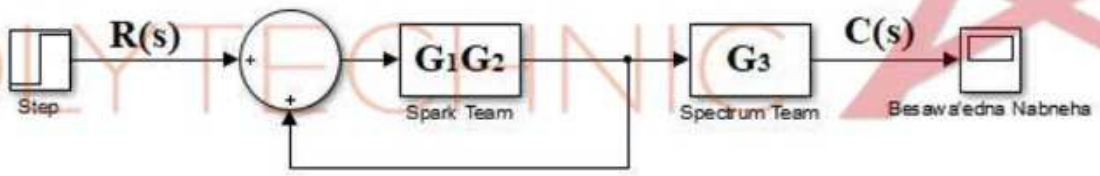
P_k is forward path transfer function for k[th] path

The cofactor Δ_k, is the determinant with the loops touching the k[th] path removed

The determinant $\Delta = 1 - (sum\ of\ all\ different\ loop\ gains) + (sum\ of\ the\ gain\ products\ of\ all\ combinations\ of\ two\ nontouching\ loops) - (sum\ of\ the\ gain\ products\ of\ all\ combinations\ of\ three\ nontouching\ loops) + \ldots$

$$P_1 = G_1$$
$$P_2 = G_2$$
$$\Delta_1 = 1$$
$$\Delta_2 = 1$$
$$\Delta = 1 - (-G_1H) = 1 + G_1H$$
$$T(s) = \frac{P_1\Delta_1 + P_2\Delta_2}{\Delta} = \frac{G_1 * 1 + G_2 * 1}{1 - (G_1H + G_2H)} = \frac{G_1 + G_2}{1 + G_1H + G_2H}$$

198. For the block diagram given in the following figure, the total gain is:



a. $\frac{G_1G_2G_3}{1-G_1G_2}$

b. $G_1G_2G_3$

c. $\frac{G_1G_2}{1-G_1G_2G_3}$

d. None of the above

Correct answer: $\frac{G_1G_2G_3}{1-G_1G_2}$

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

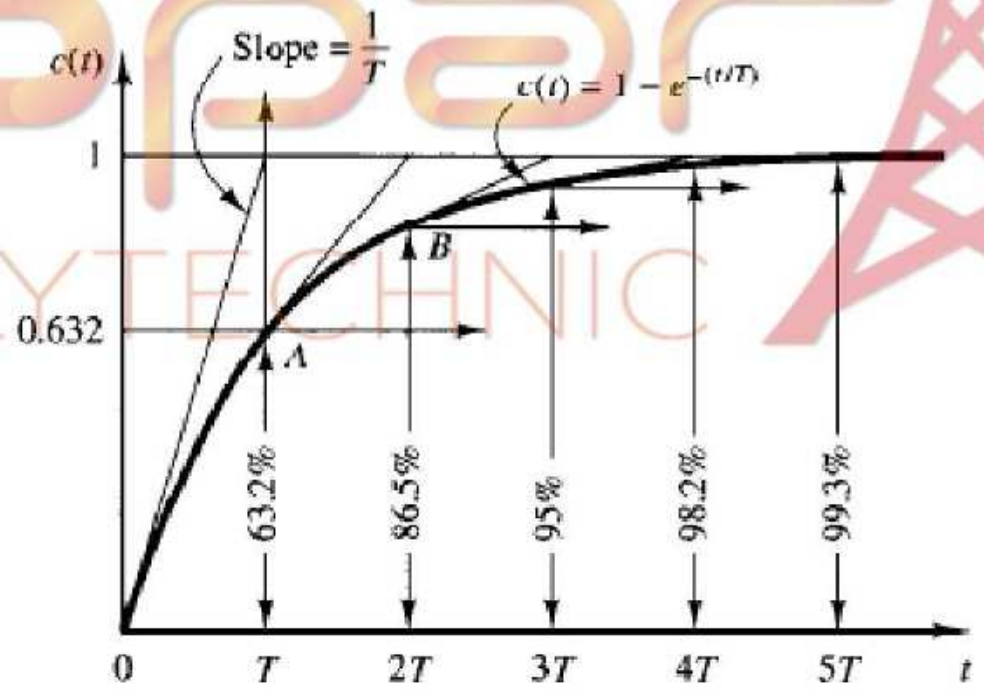
Calculation of Time Response of First-Order & Second-Order

199. The unit step response of standard first order system is:

a. $1 - e^{-\frac{t}{T}}$
b. 1
c. $-e^{-\frac{t}{T}}$
d. $e^{-\frac{t}{T}}$
e. None of the above

Correct answer: $c(t) = 1 - e^{-\frac{t}{T}}$

The following figure is an illustration for questions 124-137:



200. The error of unit step response of standard first order system is:

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص او جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

a. $e^{-\frac{t}{T}}$

b. $1 - e^{-\frac{t}{T}}$

c. 0

d. $-e^{-\frac{t}{T}}$

e. None of the above

Correct answer: $e^{-\frac{t}{T}}$

201. The slope of the tangent line of unit step response of standard first order system is:

a. $-\frac{e^{-\frac{t}{T}}}{T}$

b. $\frac{e^{-\frac{t}{T}}}{T}$

c. 1

d. $\frac{1}{T}$

e. None of the above

Correct answer: $\frac{e^{-\frac{t}{T}}}{T}$

The slope of the tangent line of unit step response of standard first order system is the derivative of the output function signal

$$\frac{dc}{dt} = \frac{e^{-\frac{t}{T}}}{T}$$

202. The slope of the tangent line of unit step response of standard first order system at $t = 0$ is:

a. $e^{-\frac{t}{T}}$

b. 1

c. $-e^{-\frac{t}{T}}$

d. $\frac{1}{T}$

e. None of the above

Correct answer: $\frac{1}{T}$

The slope of the tangent line of unit step response of standard first order system is the derivative of the output function signal

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

$$\frac{dc}{dt} = \frac{e^{-\frac{t}{T}}}{T}$$

To determine the slope of the tangent line of unit step response of standard first order system at $t = 0$, the initiating time $(t = 0)$ is substituted

203. For the unit step response of standard first order system, the output would reach the final value at t = T if it ..... its initial speed of response.

a. Maintained
b. Halved
c. Doubled
d. None of the above

Correct answer: Maintained

204. For the unit step response of standard first order system, whenever the time constant T decreased, the system response is .....

a. No effect
b. Depends on the system's parameters values
c. Slower
d. Faster

Correct answer: faster

205. For the unit step response of standard first order system, the steady state is reached mathematically after ....., and practically after the length of time the response curve needs to reach the 2% line of the final value, or .....

a. An infinite time, three-time constants
b. An infinite time, five-time constants
c. Five-time constants, four-time constants
d. An infinite time, four-time constants

Correct answer: An infinite time, four-time constants

206. Practically, the transient response time needs to reach the 36.8% line of the final value of first order system is:
Or
The time required for the output of first order system to reach 63.2% its steady state value is:

a. T
b. 2T
c. 3T
d. 4T
e. 5T
f. Infinity

Correct answer: T

207. Practically, the transient response time needs to reach the 13.5% line of the final value of first order system is:
Or
The time required for the output of first order system to reach 86.5% its steady state value is:

a. T
b. 2T

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

c. 3T
d. 4T
e. 5T
f. Infinity

Correct answer: 2T

208. Practically, the transient response time needs to reach the 2% line of the final value of first order system is:
Or
The time required for the output of first order system to reach 98.2% its steady state value is:

a. T
b. 2T
c. 3T
d. 4T
e. 5T
f. Infinity

Correct answer: 4T

209. Practically, the transient response time needs to reach the 1% line of the final value of first order system is:
Or
The time required for the output of first order system to reach 99.3% its steady state value is:

a. T
b. 2T
c. 3T
d. 4T
e. 5T
f. Infinity

Correct answer: 5T

210. The transfer function of first order system is $\frac{C(s)}{R(s)} = \frac{6}{8s+2}$. The time constant of this system is:

a. 3
b. 4
c. 8
d. 6
e. 2

Correct answer:
To determine the D.C gain or time constant of the first order function by its transfer function, the system must be in the following form:

$$\frac{C(s)}{R(s)} = \frac{K}{Ts+1}$$

$$\frac{C(s)}{R(s)} = \frac{6}{8s+2} = \frac{3}{4s+1}$$

Then, the time constant of this system is 4

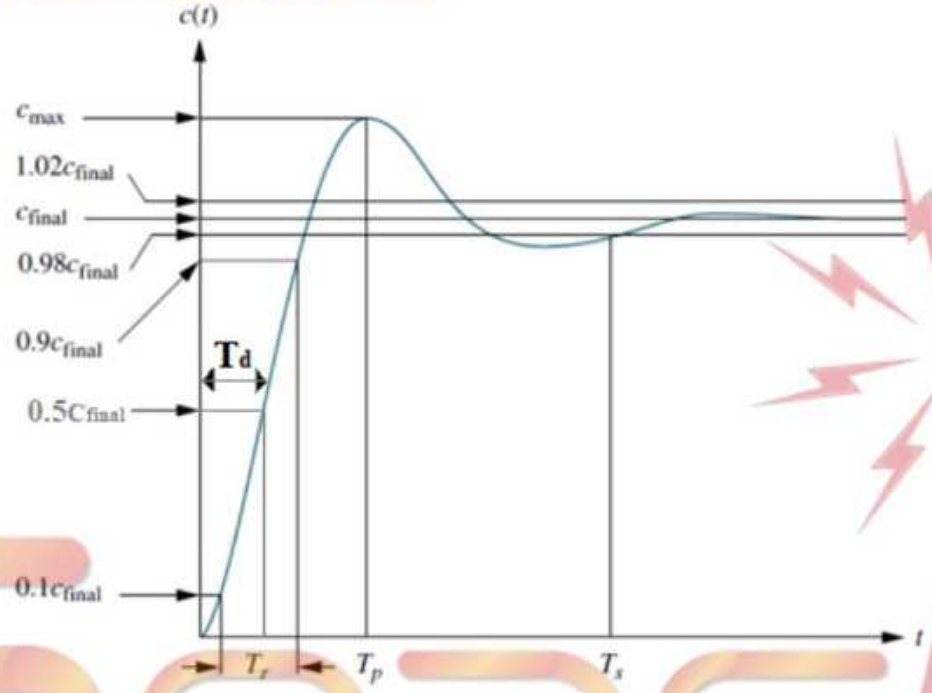
ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

211. Rise time, t_r is the time required for the waveform to go from 0.1 of the final value to 0.9 of the final value.

a. True
b. False

Correct answer: True

The following figure is an illustration for questions 15-20:



212. The peak time t_p is the time required for the response to reach the first peak of the time response or first peak overshoot.

a. True
b. False

Correct answer: True

213. The delay t_d time is the time required for the response to reach 50% of the final value in first time.

a. True
b. False

Correct answer: True

214. Settling time, t_s is the time required for the transient's damped oscillations to reach and stay within prescribed percentage error (usually, 2% of the steady-state value).

a. True
b. False

Correct answer: True

215. Steady state error e_{ss} is difference between actual output and desired output as time 't' tends to infinity.

$$\lim_{t\to\infty}\big(r(t)-c(t)\big)$$

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

a. True
b. False

Correct answer: True

216. Percent overshoot, $\%OS$ is the amount that the response exceeds the final value at t_p. In other words, it's the normalized difference between the time response peak and the steady state output and is defined as

$$\%M_p = \frac{c(t_p) - c(\infty)}{c(\infty)} * 100\%$$

a. True
b. False

Correct answer: True

217. The transfer function of second order system is $\frac{C(s)}{R(s)} = \frac{1}{s^2+2s+4}$. assume unit step input, answer the following questions:

I. Damped natural frequency.
a. 1
b. 4
c. 2
d. 0.5
e. 3

II. The damping ratio.
a. 0.5
b. 4
c. 2
d. 3
e. 1

III. Damping constant.
a. 0.5
b. 1
c. 3
d. 2
e. 4

IV. Un-damped natural frequency.
a. 2
b. 0.25
c. 3
d. 1
e. 4
f. $\sqrt{3}$

V. Time response.
a. Critically damped
b. Underdamped
c. Overdamped
d. Undamped (The transient response does not die out)

VI. The setting time for 5% error.
a. 4

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

b. 1
c. 3
d. 2

VII. The setting time for 2% error.

a. 2
b. 1
c. 4
d. 3

Correct answer:

I. 2
II. 0.5
III. 1
IV. $\sqrt{3}$
V. Underdamped
VI. 3
VII. 4

The transfer function for second order system is:

$$\frac{C(s)}{R(s)} = \frac{{\omega^2}_n}{s^2 + 2\zeta\omega_n s + {\omega^2}_n}$$

Where:

ω_n: Damped natural frequency

ζ: The damping ratio

σ: Damping constant

ω_d: Un-damped natural frequency

VIN: The coefficient of S^2 must be one

I. So, from the general form of the transfer function for second order system, damped natural frequency ω_n is:

$${\omega^2}_n = 4 \rightarrow \omega_n = 2$$

II. And from the middle term of the quadratic equation in the denominator, the damping ratio is calculated:

$$2\zeta\omega_n = 2$$
$$2\zeta * 2 = 2$$
$$\zeta = \frac{1}{2}$$

III. Now, damping constant is calculated from the following equation:

$$\sigma = \zeta\omega_n = \frac{1}{2} * 2 = 1$$

IV. Also, the un-damped natural frequency is calculated from the following equation:

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص او جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

$$\omega_d = \omega_n\sqrt{1-\zeta^2} = 2\sqrt{1-\left(\frac{1}{2}\right)^2} = \sqrt{3}$$

V. As the damping ratio is between zero and one, then the time response is underdamped.

VI. The setting time for 5% error for second order system is:

$$t_s = 3T = \frac{3}{\sigma} = \frac{3}{\zeta\omega_n} = \frac{3}{1} = 3 \quad (5\%\ criterion)$$

VII. The setting time for 2% error for second order system is:

$$t_s = 4T = \frac{4}{\sigma} = \frac{4}{\zeta\omega_n} = \frac{4}{1} = 4 \quad (2\%\ criterion)$$

218. The transfer function of second order system is $\frac{C(s)}{R(s)} = \frac{1}{s^2+8s+16}$. assume unit step input, answer the following questions:

I. Damped natural frequency.
   a. 1
   b. 0
   c. 4
   d. 2
   e. None of the above

II. The damping ratio.
   a. 4
   b. 2
   c. 0
   d. 1
   e. None of the above

III. Damping constant.
   a. 0
   b. 2
   c. 4
   d. 1
   e. None of the above

IV. Un-damped natural frequency.
   a. 0
   b. 2
   c. 4
   d. 1
   e. None of the above

V. Time response.
   a. Undamped (The transient response does not die out)
   b. Overdamped
   c. Underdamped
   d. Critically damped

VI. The setting time for 5% error.
   a. 0.25
   b. 1.25

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

c. 1
d. 0.75
e. None of the above

VII. The setting time for 2% error.

a. 1.25
b. 0.75
c. 1
d. 0.25
e. None of the above

Correct answer:

I. 4
II. 1
III. 4
IV. 0
V. Critically damped
VI. 0.75
VII. 1

The transfer function for second order system is:

$$\frac{C(s)}{R(s)} = \frac{{\omega^2}_n}{s^2 + 2\zeta\omega_n s + {\omega^2}_n}$$

Where:

ω_n: Damped natural frequency
ζ: The damping ratio
σ: Damping constant
ω_d: Un-damped natural frequency

VIN: The coefficient of S^2 must be one

I. So, from the general form of the transfer function for second order system, damped natural frequency ω_n is:

$${\omega^2}_n = 16 \rightarrow \omega_n = 4$$

II. And from the middle term of the quadratic equation in the denominator, the damping ratio is calculated:

$$2\zeta\omega_n = 8$$
$$2\zeta * 4 = 8$$
$$\zeta = 1$$

III. Now, damping constant is calculated from the following equation:

$$\sigma = \zeta\omega_n = 1 * 4 = 4$$

IV. Also, the un-damped natural frequency is calculated from the following equation:

$$\omega_d = \omega_n\sqrt{1 - \zeta^2} = 4\sqrt{1 - 1^2} = 0$$

V. As the damping ratio is equal to one, then the time response is critically damped.
VI. The setting time for 5% error for second order system is:

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

$$t_s = 3T = \frac{3}{\sigma} = \frac{3}{\zeta\omega_n} = \frac{3}{4} = 0.75 \quad (5\%\ criterion)$$

VII. The setting time for 2% error for second order system is:

$$t_s = 4T = \frac{4}{\sigma} = \frac{4}{\zeta\omega_n} = \frac{4}{4} = 1 \quad (2\%\ criterion)$$

219. The transfer function of second order system is $\frac{C(s)}{R(s)} = \frac{1}{s^2+9}$. assume unit step input, answer the following questions:

I. Damped natural frequency.
   a. 1
   b. 2
   c. 0
   d. 3
   e. None of the above

II. The damping ratio.
   a. 2
   b. 0
   c. 1
   d. 3
   e. None of the above

III. Damping constant.
   a. 2
   b. 3
   c. 0
   d. 1
   e. None of the above

IV. Un-damped natural frequency.
   a. 0
   b. 3
   c. 2
   d. 1
   e. None of the above

V. Time response.
   a. Undamped (The transient response does not die out)
   b. Overdamped
   c. Underdamped
   d. Critically damped

VI. The setting time for 5% error.
   a. 3
   b. 2
   c. 1
   d. 0
   e. None of the above

VII. The setting time for 2% error.
   a. 3
   b. 0

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

c. 1
d. 2
e. None of the above

Correct answer:

I. 3
II. 0
III. 0
IV. 3
V. Undamped (The transient response does not die out)
VI. None of the above (The output signal never reaches the steady state value.)
VII. None of the above (The output signal never reaches the steady state value.)

The transfer function for second order system is:

$$\frac{C(s)}{R(s)} = \frac{{\omega^2}_n}{s^2 + 2\zeta\omega_n s + {\omega^2}_n}$$

Where:

ω_n: Damped natural frequency

ζ: The damping ratio

σ: Damping constant

ω_d: Un-damped natural frequency

VIN: The coefficient of S^2 must be one

I. So, from the general form of the transfer function for second order system, damped natural frequency ω_n is:

$${\omega^2}_n = 9 \rightarrow \omega_n = 3$$

II. And from the middle term of the quadratic equation in the denominator, the damping ratio is calculated:

$$2\zeta\omega_n = 0$$
$$2\zeta * 3 = 0$$
$$\zeta = 0$$

III. Now, damping constant is calculated from the following equation:

$$\sigma = \zeta\omega_n = 0 * 3 = 0$$

IV. Also, the un-damped natural frequency is calculated from the following equation:

$$\omega_d = \omega_n\sqrt{1 - \zeta^2} = 3\sqrt{1 - 0^2} = 3$$

V. As the damping ratio is equal to zero, then the time response is undamped (The transient response does not die out).

VI. The setting time for 5% error for second order system is:

$$t_s = 3T = \frac{3}{\sigma} = \frac{3}{\zeta\omega_n} = \frac{3}{0} = \infty \quad (5\%\ criterion)$$

The output signal never reaches the steady state value.

VII. The setting time for 2% error for second order system is:

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

$$t_s = 4T = \frac{4}{\sigma} = \frac{4}{\zeta\omega_n} = \frac{4}{0} = \infty \quad (2\%\ criterion)$$

The output signal never reaches the steady state value.

220. The transfer function of second order system is $\frac{C(s)}{R(s)} = \frac{1}{s^2+s+4}$. assume unit step input, answer the following questions:

I. Damped natural frequency.
   a. $\frac{1}{2}$
   b. 2
   c. $\frac{1}{4}$
   d. None of the above

II. The damping ratio.
   a. 2
   b. $\frac{1}{4}$
   c. $\frac{1}{2}$
   d. None of the above

III. Damping constant.
   a. $\frac{1}{2}$
   b. $\frac{\sqrt{15}}{2}$
   c. $\frac{1}{4}$
   d. None of the above

IV. Un-damped natural frequency.
   a. $\frac{\sqrt{15}}{2}$
   b. $\frac{1}{2}$
   c. $\frac{1}{4}$
   d. None of the above

V. Time response.
   a. Underdamped
   b. Critically damped
   c. Overdamped
   d. Undamped (The transient response does not die out)

VI. The setting time for 5% error.
   a. 4

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

b. 8
c. 6
d. None of the above

VII. The setting time for 2% error.

a. 8
b. 6
c. 4
d. None of the above

Correct answer:

I. 2
II. $\frac{1}{4}$
III. $\frac{1}{2}$
IV. $\frac{\sqrt{15}}{2}$
V. Underdamped
VI. 6
VII. 8

The transfer function for second order system is:

$$\frac{C(s)}{R(s)} = \frac{{\omega^2}_n}{s^2 + 2\zeta\omega_n s + {\omega^2}_n}$$

Where:

ω_n: Damped natural frequency

ζ: The damping ratio

σ: Damping constant

ω_d: Un-damped natural frequency

VIN: The coefficient of S^2 must be one

I. So, from the general form of the transfer function for second order system, damped natural frequency ω_n is:

$${\omega^2}_n = 4 \rightarrow \omega_n = 2$$

II. And from the middle term of the quadratic equation in the denominator, the damping ratio is calculated:

$$2\zeta\omega_n = 1$$
$$2\zeta * 2 = 1$$
$$\zeta = \frac{1}{4}$$

III. Now, damping constant is calculated from the following equation:

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

$$\sigma = \zeta\omega_n = \frac{1}{4} * 2 = \frac{1}{2}$$

IV. Also, the un-damped natural frequency is calculated from the following equation:

$$\omega_d = \omega_n\sqrt{1-\zeta^2} = 2\sqrt{1-\left(\frac{1}{4}\right)^2} = \frac{\sqrt{15}}{2}$$

V. As the damping ratio is between zero and one, then the time response is underdamped.

VI. The setting time for 5% error for second order system is:

$$t_s = 3T = \frac{3}{\sigma} = \frac{3}{\zeta\omega_n} = \frac{3}{\frac{1}{2}} = 6 \quad (5\%\ criterion)$$

VII. The setting time for 2% error for second order system is:

$$t_s = 4T = \frac{4}{\sigma} = \frac{4}{\zeta\omega_n} = \frac{4}{\frac{1}{2}} = 8 \quad (2\%\ criterion)$$

221. The transfer function of second order system is $\frac{C(s)}{R(s)} = \frac{1}{\frac{s^2}{2}+\frac{1}{\sqrt{2}}s+\frac{1}{8}}$. assume unit step input, answer the following questions:

I. Damped natural frequency.

a. $\sqrt{2}$
b. $\frac{1}{2}$
c. $\frac{j}{2}$
d. $\frac{1}{\sqrt{2}}$
e. None of the above

II. The damping ratio.

a. $\frac{1}{2}$
b. $\frac{1}{\sqrt{2}}$
c. $\sqrt{2}$
d. $\frac{j}{2}$
e. None of the above

III. Damping constant.

a. $\sqrt{2}$
b. $\frac{j}{2}$

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

c. $\frac{1}{2}$

d. $\frac{1}{\sqrt{2}}$

e. None of the above

IV. Un-damped natural frequency.

a. $\frac{j}{2}$

b. $\frac{1}{2}$

c. $\frac{1}{\sqrt{2}}$

d. $\sqrt{2}$

e. None of the above

V. Time response.

a. Overdamped

b. Undamped (The transient response does not die out)

c. Critically damped

d. Underdamped

VI. The setting time for 5% error.

a. $4\sqrt{2}$

b. $\sqrt{2}$

c. $3\sqrt{2}$

d. $2\sqrt{2}$

VII. The setting time for 2% error.

a. $4\sqrt{2}$

b. $\sqrt{2}$

c. $3\sqrt{2}$

d. $2\sqrt{2}$

Spark POLYTECHNIC

Correct answer:

I. $\frac{1}{2}$

II. $\sqrt{2}$

III. $\frac{1}{\sqrt{2}}$

IV. $\frac{j}{2}$

V. Overdamped

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص او جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

VI. $3\sqrt{2}$

VII. $4\sqrt{2}$

The transfer function for second order system is:

$$\frac{C(s)}{R(s)} = \frac{{\omega^2}_n}{s^2 + 2\zeta\omega_n s + {\omega^2}_n}$$

Where:

ω_n: Damped natural frequency

ζ: The damping ratio

σ: Damping constant

ω_d: Un-damped natural frequency

VIN: The coefficient of S^2 must be one

$$\frac{C(s)}{R(s)} = 2\frac{1}{s^2 + \sqrt{2}s + \frac{1}{4}}$$

I. So, from the general form of the transfer function for second order system, damped natural frequency ω_n is:

$${\omega^2}_n = \frac{1}{4} \rightarrow \omega_n = \frac{1}{2}$$

II. And from the middle term of the quadratic equation in the denominator, the damping ratio is calculated:

$$2\zeta\omega_n = \sqrt{2}$$

$$2\zeta * \frac{1}{2} = \sqrt{2}$$

$$\zeta = \sqrt{2}$$

III. Now, damping constant is calculated from the following equation:

$$\sigma = \zeta\omega_n = \sqrt{2} * \frac{1}{2} = \frac{1}{\sqrt{2}}$$

IV. Also, the un-damped natural frequency is calculated from the following equation:

$$\omega_d = \omega_n\sqrt{1-\zeta^2} = \frac{1}{2}\sqrt{1-\left(\sqrt{2}\right)^2} = \frac{1}{2}\sqrt{-1} = \frac{j}{2}$$

V. As the damping ratio is greater than one, then the time response is overdamped.

VI. The setting time for 5% error for second order system is:

$$t_s = 3T = \frac{3}{\sigma} = \frac{3}{\zeta\omega_n} = \frac{3}{\frac{1}{\sqrt{2}}} = 3\sqrt{2} \quad (5\%\ criterion)$$

VII. The setting time for 2% error for second order system is:

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص او جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

$$t_s = 4T = \frac{4}{\sigma} = \frac{4}{\zeta\omega_n} = \frac{4}{\frac{1}{\sqrt{2}}} = 4\sqrt{2} \quad (2\%\ criterion)$$

222. The transfer function of second order system is $\frac{C(s)}{R(s)} = \frac{1}{s^2+8s+4}$. assume unit step input, answer the following questions:

I. Damped natural frequency.

a. 2
b. 4
c. $\frac{\sqrt{3}}{2}j$
d. $2\sqrt{3}j$
e. None of the above

II. The damping ratio.

a. $2\sqrt{3}j$
b. 2
c. 4
d. $\frac{\sqrt{3}}{2}j$
e. None of the above

III. Damping constant.

a. 2
b. $2\sqrt{3}j$
c. 4
d. $\frac{\sqrt{3}}{2}j$
e. None of the above

IV. Un-damped natural frequency.

a. $\frac{\sqrt{3}}{2}j$
b. 2
c. 4
d. $2\sqrt{3}j$
e. None of the above

V. Time response.

a. Underdamped
b. Critically damped

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

c. Overdamped
d. Undamped (The transient response does not die out)

VI. The setting time for 5% error.
a. 0.75
b. 1.5
c. 1
d. 0.5
e. None of the above

VII. The setting time for 2% error.
a. 1.5
b. 0.5
c. 1
d. 0.75
e. None of the above

Correct answer:

I. 2
II. 2
III. 4
IV. $2\sqrt{3}j$
V. Overdamped
VI. 0.75
VII. 1

The transfer function for second order system is:

$$\frac{C(s)}{R(s)} = \frac{{\omega^2}_n}{s^2 + 2\zeta\omega_n s + {\omega^2}_n}$$

Where:

ω_n: Damped natural frequency
ζ: The damping ratio
σ: Damping constant
ω_d: Un-damped natural frequency

VIN: The coefficient of S^2 must be one

I. So, from the general form of the transfer function for second order system, damped natural frequency ω_n is:

$${\omega^2}_n = 4 \rightarrow \omega_n = 2$$

II. And from the middle term of the quadratic equation in the denominator, the damping ratio is calculated:

$$2\zeta\omega_n = 8$$
$$2\zeta * 2 = 8$$
$$\zeta = 2$$

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص او جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

III. Now, damping constant is calculated from the following equation:

$$\sigma = \zeta\omega_n = 2 * 2 = 4$$

IV. Also, the un-damped natural frequency is calculated from the following equation:

$$\omega_d = \omega_n\sqrt{1-\zeta^2} = 2\sqrt{1-2^2} = 2\sqrt{-3} = 2\sqrt{3}j$$

V. As the damping ratio is greater than one, then the time response is overdamped.

VI. The setting time for 5% error for second order system is:

$$t_s = 3T = \frac{3}{\sigma} = \frac{3}{\zeta\omega_n} = \frac{3}{4} = 0.75 \quad (5\%\ criterion)$$

VII. The setting time for 2% error for second order system is:

$$t_s = 4T = \frac{4}{\sigma} = \frac{4}{\zeta\omega_n} = \frac{4}{4} = 1 \quad (2\%\ criterion)$$

223. The transfer function of second order system is $\frac{C(s)}{R(s)} = \frac{1}{4s^2+32s+100}$. assume unit step input, answer the following questions:

I. Damped natural frequency.

a. 5
b. 3
c. 0.8
d. 4
e. None of the above

II. The damping ratio.

a. 4
b. 5
c. 0.8
d. 3
e. None of the above

III. Damping constant.

a. 5
b. 4
c. 0.8
d. 3
e. None of the above

IV. Un-damped natural frequency.

a. 3
b. 5
c. 0.8
d. 4
e. None of the above

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

V. Time response.
   a. Overdamped
   b. Undamped (The transient response does not die out)
   c. Underdamped
   d. Critically damped

VI. The setting time for 5% error.
   a. 0.75
   b. 1
   c. 1.25
   d. 1.5
   e. None of the above

VII. The setting time for 2% error.
   a. 1.5
   b. 1
   c. 1.25
   d. 0.75
   e. None of the above

Correct answer:

I. 5
II. 0.8
III. 4
IV. 3
V. Underdamped
VI. 0.75
VII. 1

The transfer function for second order system is:

$$\frac{C(s)}{R(s)} = \frac{{\omega^2}_n}{s^2 + 2\zeta\omega_n s + {\omega^2}_n}$$

Where:

ω_n: Damped natural frequency

ζ: The damping ratio

σ: Damping constant

ω_d: Un-damped natural frequency

VIN: The coefficient of S^2 must be one

$$\frac{C(s)}{R(s)} = \frac{1}{4}\frac{1}{s^2 + 8s + 25}$$

I. So, from the general form of the transfer function for second order system, damped natural frequency ω_n is:

$${\omega^2}_n = 25 \rightarrow \omega_n = 5$$

II. And from the middle term of the quadratic equation in the denominator, the damping ratio is calculated:

$$2\zeta\omega_n = 8$$

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

$$2\zeta * 5 = 8$$

$$\zeta = 0.8$$

III. Now, damping constant is calculated from the following equation:

$$\sigma = \zeta\omega_n = 0.8 * 5 = 4$$

IV. Also, the un-damped natural frequency is calculated from the following equation:

$$\omega_d = \omega_n\sqrt{1-\zeta^2} = 5\sqrt{1-\left(\frac{8}{10}\right)^2} = 5\sqrt{1-0.64} = 5\sqrt{0.36} = 5 * 0.6 = 3$$

V. As the damping ratio is between zero and one, then the time response is underdamped.

VI. The setting time for 5% error for second order system is:

$$t_s = 3T = \frac{3}{\sigma} = \frac{3}{\zeta\omega_n} = \frac{3}{4} = 0.75 \quad (5\%\ criterion)$$

VII. The setting time for 2% error for second order system is:

$$t_s = 4T = \frac{4}{\sigma} = \frac{4}{\zeta\omega_n} = \frac{4}{4} = 1 \quad (2\%\ criterion)$$

224. The transfer function of second order system is $\frac{C(s)}{R(s)} = \frac{1}{2s^2+4\pi s+2\pi^2}$. assume unit step input, answer the following questions:

I. Damped natural frequency.

a. π
b. 1
c. 0
d. $\frac{\pi}{2}$
e. None of the following

II. The damping ratio.

a. π
b. 1
c. 0
d. $\frac{\pi}{2}$
e. None of the following

III. Damping constant.

a. π
b. 1
c. 0
d. $\frac{\pi}{2}$
e. None of the following

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص او جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

IV. Un-damped natural frequency.

a. π
b. 1
c. 0
d. j
e. None of the following

V. Time response.

a. Overdamped
b. Undamped (The transient response does not die out)
c. Critically damped
d. Underdamped

VI. The setting time for 5% error.

a. $\frac{4}{\pi}$
b. $\frac{2}{\pi}$
c. $\frac{3}{\pi}$
d. $\frac{1}{\pi}$
e. None of the above

VII. The setting time for 2% error.

a. $\frac{5}{\pi}$
b. $\frac{4}{\pi}$
c. $\frac{3}{\pi}$
d. $\frac{1}{\pi}$
e. None of the above

SPARK POLYTECHNIC

Correct answer:

I. π
II. 1
III. π
IV. 0
V. Critically damped
VI. $\frac{3}{\pi}$
VII. $\frac{4}{\pi}$

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص او جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

The transfer function for second order system is:

$$\frac{C(s)}{R(s)} = \frac{{\omega^2}_n}{s^2 + 2\zeta\omega_n s + {\omega^2}_n}$$

Where:

ω_n: Damped natural frequency

ζ: The damping ratio

σ: Damping constant

ω_d: Un-damped natural frequency

VIN: The coefficient of S^2 must be one

$$\frac{C(s)}{R(s)} = \frac{1}{2}\frac{1}{s^2 + 2\pi s + \pi^2}$$

I. So, from the general form of the transfer function for second order system, damped natural frequency ω_n is:

$${\omega^2}_n = \pi^2 \rightarrow \omega_n = \pi$$

II. And from the middle term of the quadratic equation in the denominator, the damping ratio is calculated:

$$2\zeta\omega_n = 2\pi$$

$$2\zeta * \pi = 2\pi$$

$$\zeta = 1$$

III. Now, damping constant is calculated from the following equation:

$$\sigma = \zeta\omega_n = 1 * \pi = \pi$$

IV. Also, the un-damped natural frequency is calculated from the following equation:

$$\omega_d = \omega_n\sqrt{1 - \zeta^2} = 2\sqrt{1 - 1^2} = 0$$

V. As the damping ratio is equal to one, then the time response is critically damped.

VI. The setting time for 5% error for second order system is:

$$t_s = 3T = \frac{3}{\sigma} = \frac{3}{\zeta\omega_n} = \frac{3}{\pi} \quad (5\%\ criterion)$$

VII. The setting time for 2% error for second order system is:

$$t_s = 4T = \frac{4}{\sigma} = \frac{4}{\zeta\omega_n} = \frac{4}{\pi} \quad (2\%\ criterion)$$

225. The transfer function of second order system is $\frac{C(s)}{R(s)} = \frac{1}{\frac{1}{e}s^2 + e}$. assume unit step input, answer the following questions:

I. Damped natural frequency.

a. e

b. 0

c. $\sqrt{e}$

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص او جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

d. 1

e. None of the above

II. The damping ratio.

a. e

b. 0

c. $\sqrt{e}$

d. 1

e. None of the above

III. Damping constant.

a. e

b. 0

c. $\sqrt{e}$

d. 1

e. None of the above

IV. Un-damped natural frequency.

a. e

b. 0

c. $\sqrt{e}$

d. j

e. None of the above

V. Time response.

a. Critically damped

b. Undamped (The transient response does not die out)

c. Underdamped

d. Overdamped

VI. The setting time for 5% error.

a. 1

b. $3e$

c. 1.25

d. $4\sqrt{e}$

e. The output signal never reaches the steady state value.

VII. The setting time for 2% error.

a. 1

b. $3e$

c. 1.25

d. $4\sqrt{e}$

e. The output signal never reaches the steady state value.

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

Correct answer:

I. e
II. 0
III. 0
IV. e
V. Undamped (The transient response does not die out)
VI. The output signal never reaches the steady state value.
VII. The output signal never reaches the steady state value.

The transfer function for second order system is:

$$\frac{C(s)}{R(s)} = \frac{{\omega^2}_n}{s^2 + 2\zeta\omega_n s + {\omega^2}_n}$$

Where:

ω_n: Damped natural frequency
ζ: The damping ratio
σ: Damping constant
ω_d: Un-damped natural frequency

VIN: The coefficient of S^2 must be one

$$\frac{C(s)}{R(s)} = e\frac{1}{s^2 + e^2}$$

I. So, from the general form of the transfer function for second order system, damped natural frequency ω_n is:

$${\omega^2}_n = e^2 \rightarrow \omega_n = e$$

II. And from the middle term of the quadratic equation in the denominator, the damping ratio is calculated:

$$2\zeta\omega_n = 0$$
$$2\zeta * e = 0$$
$$\zeta = 0$$

III. Now, damping constant is calculated from the following equation:

$$\sigma = \zeta\omega_n = 0 * e = 0$$

IV. Also, the un-damped natural frequency is calculated from the following equation:

$$\omega_d = \omega_n\sqrt{1 - \zeta^2} = e\sqrt{1 - 0^2} = e$$

V. As the damping ratio is equal to zero, then the time response is Undamped (The transient response does not die out).

VI. The setting time for 5% error for second order system is:

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

$$t_s = 3T = \frac{3}{\sigma} = \frac{3}{\zeta\omega_n} = \frac{3}{0} = \infty \quad (5\%\ criterion)$$

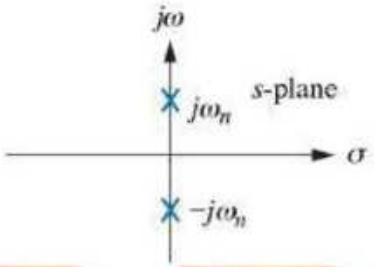
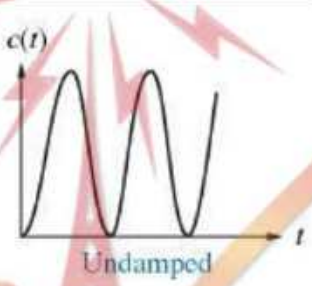
VII. The setting time for 2% error for second order system is:

$$t_s = 4T = \frac{4}{\sigma} = \frac{4}{\zeta\omega_n} = \frac{4}{0} = \infty \quad (2\%\ criterion)$$

226. The response has no overshoot when $\zeta = 0$.
Or
If the damping ratio of second order system is zero, then time response has no overshoot.
Or
For zero damping factor, the system response has no overshoot.

a. True
b. False

Correct answer: False

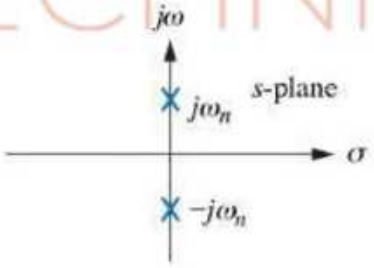
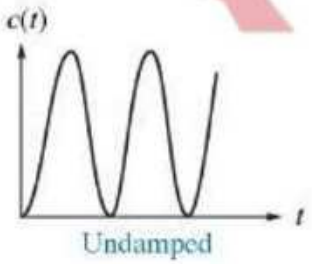
ζ	Poles	Step response
0		

227. For $\zeta = 0$, the transfer function has two imaginary conjugate poles and the response has an overshoot.

a. True
b. False

Correct answer: True

The poles of the system are imaginary conjugate, then $\zeta = 0$, so the time response is undamped.

ζ	Poles	Step response
0		

228. If the damping ratio of second order system is greater than zero and less than one, the time response is:
Or
For damping factor greater than zero and less than one, the system will be:

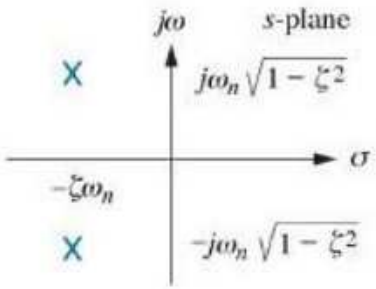
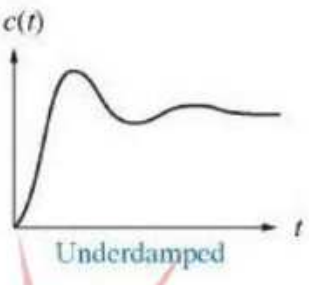
a. Underdamped
b. Overdamped
c. Critically damped

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

d. Undamped (The transient response does not die out)

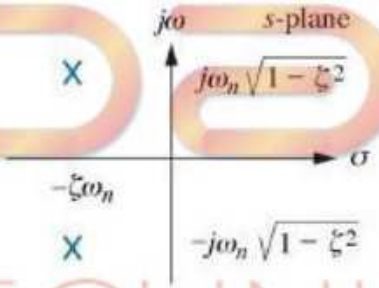
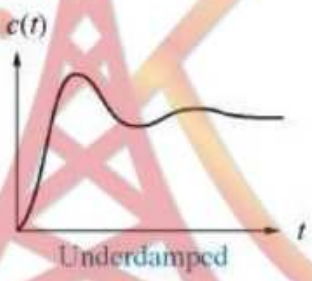
Correct answer: Underdamped

ζ	Poles	Step response
$0 < \zeta < 1$		

229. The response has no overshoot when $1 > \zeta > 0$.
Or
If the damping ratio of second order system is greater than zero and less than one, then time response has no overshoot.
Or
For damping factor greater than zero and less than one, the system response has no overshoot.

a. True
b. False

Correct answer: False

ζ	Poles	Step response
$0 < \zeta < 1$		

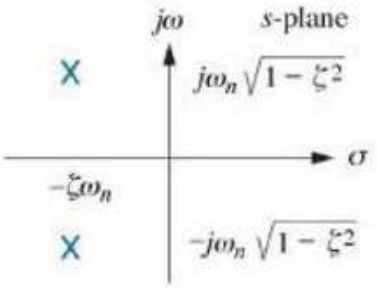
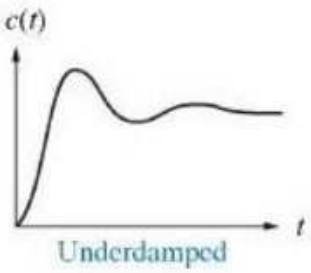
230. For $1 > \zeta > 0$, the transfer function has two complex poles and the response has an overshoot.

a. True
b. False

Correct answer: True

The poles of the system are complex poles, then $1 > \zeta > 0$, so the time response is Underdamped and has an overshoot.

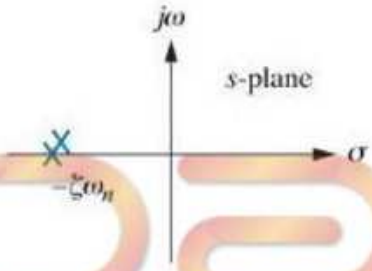
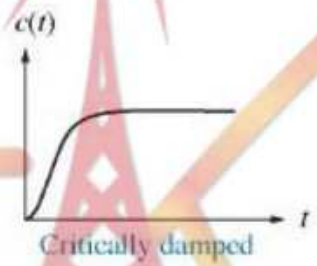
ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

ζ	Poles	Step response
$0 < \zeta < 1$		

231. The response has no overshoot when $\zeta = 1$.
Or
If the damping ratio of second order system is unity, then time response has no overshoot.
Or
For unity damping factor, the system response has no overshoot.

a. True
b. False

Correct answer: True

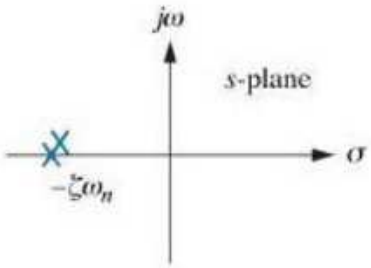
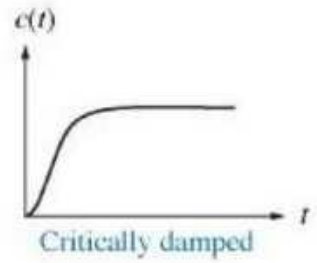
ζ	Poles	Step response
$\zeta = 1$		

232. For $\zeta = 1$, the transfer function has two repeated poles and the response has an overshoot.

a. True
b. False

Correct answer: False

The poles of the system are multiple (repeated) poles, then $\zeta = 1$, so the time response is Critically damped and hasn't an overshoot.

ζ	Poles	Step response
$\zeta = 1$		

233. If the damping ratio of second order system is greater than one, the time response is:
Or

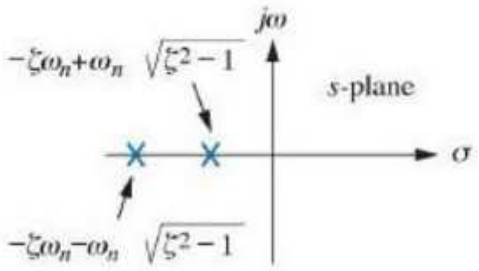
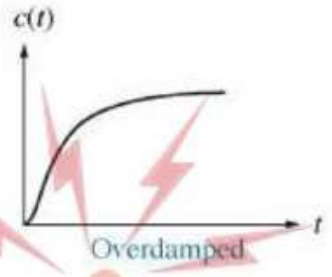
بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

For damping factor greater than one, the system will be:

a. Undamped (The transient response does not die out)
b. Overdamped
c. Critically damped
d. Underdamped

Correct answer: Overdamped

ζ	Poles	Step response
$\zeta > 1$		

234. The forced response is also called:

a. Dynamic response
b. Steady state response
c. Free response
d. All the above

Correct answer: Steady state response

235. The natural response is also called:

a. Free response
b. Transient response
c. Dynamic response
d. All the above

Correct answer: All the above

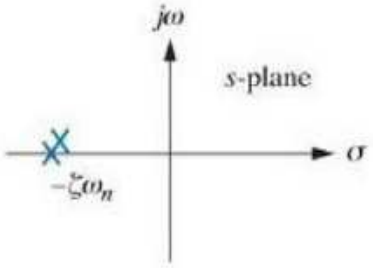
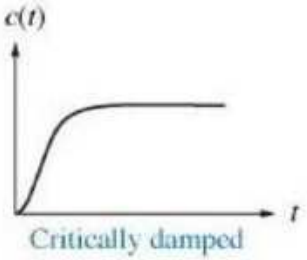
236. The following system $T(s) = \frac{s+1}{s^2+10s+25}$ is:

a. Overdamped
b. Undamped
c. Underdamped
d. Critical damped

Correct answer: Critically damped

The poles of the system are multiple (repeated) poles, then $\zeta = 1$, so the time response is Critically damped.

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

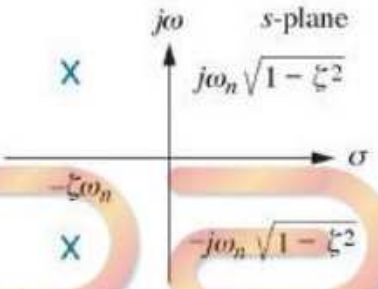
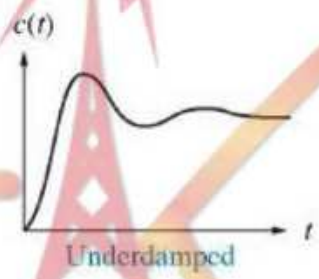
ζ	Poles	Step response
$\zeta = 1$		

237. The following system $T(s) = \frac{s+4}{s^2+5s+25}$ is:

a. Critical damped
b. Overdamped
c. Underdamped
d. Undamped

Correct answer: Underdamped

The poles of the system are complex poles, then $1 > \zeta > 0$, so the time response is Underdamped.

ζ	Poles	Step response
$0 < \zeta < 1$		

238. The transfer function of the system is

$$G(s) = \frac{100}{s^2 + 20s + 100}$$

For (a unit step/ a unit ramp/ sinusoidal/ impulse/ an) input signal to the system, find the approximate setting time (in sec) for:

I. 5% criterion.
   a. 0.3
   b. 0.4
   c. 0.5
   d. 0.33
   e. None of the above

II. 2% criterion (if the question does not mention the criterion, 2% criterion is considered the permissible error).
   a. 0.33
   b. 0.5
   c. 0.4
   d. 0.3
   e. None of the above

III. 1% criterion.

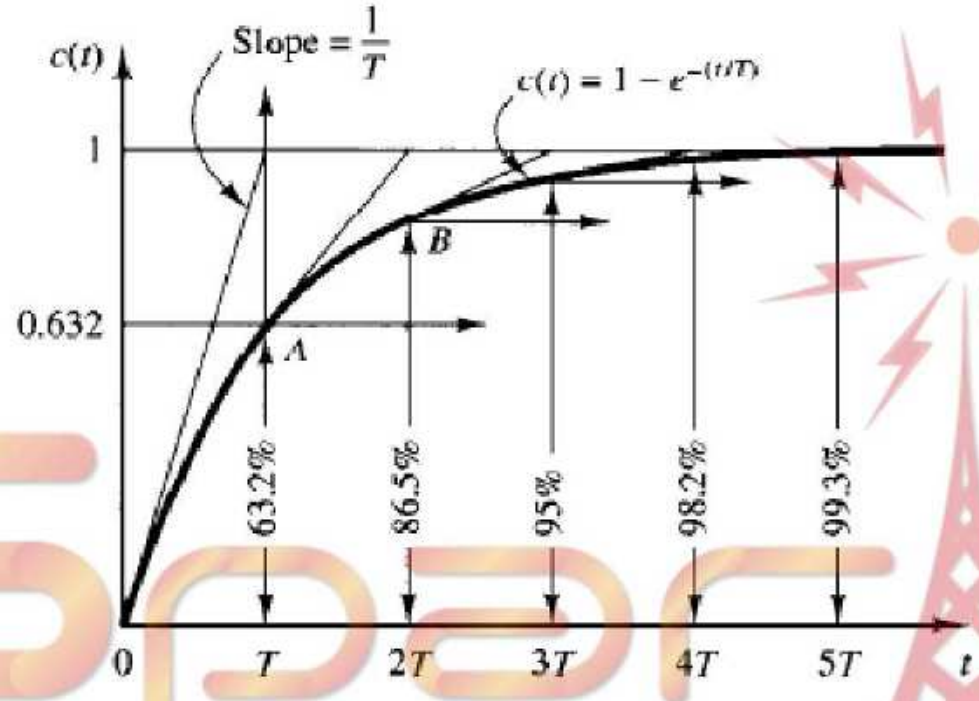
بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

a. 0.4
b. 0.3
c. 0.5
d. 0.33
e. None of the above

Correct answer:

I. 0.3
II. 0.4
III. 0.5



$$\frac{C(s)}{R(s)} = \frac{\omega^2{}_n}{s^2 + 2\zeta\omega_n s + \omega^2{}_n}$$

Where:

ω_n: Damped natural frequency

ζ: The damping ratio

σ: Damping constant

ω_d: Un-damped natural frequency

VIN: The coefficient of s^2 must be one

$$G(S) = \frac{100}{s^2 + 20s + 100}$$

$$\omega^2{}_n = 100 \rightarrow \omega_n = 10$$

$$2\zeta\omega_n = 20$$

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل منشأ هذا العمل

$$\zeta\omega_n = 10$$

I. For 5% criterion, the approximate setting time (in sec) is:

$$t_s = 3T = \frac{3}{\sigma} = \frac{3}{\zeta\omega_n}$$

$$t_s = \frac{3}{10} = 0.3$$

II. For 2% criterion (if the question does not mention the criterion, 2% criterion is considered the permissible error), the approximate setting time (in sec) is:

$$t_s = 4T = \frac{4}{\sigma} = \frac{4}{\zeta\omega_n}$$

$$t_s = \frac{4}{10} = 0.4$$

III. For 1% criterion, the approximate setting time (in sec) is:

$$t_s = 5T = \frac{5}{\sigma} = \frac{5}{\zeta\omega_n}$$

$$t_s = \frac{5}{10} = 0.5$$

239. The transfer function of the system is

$$G(s) = \frac{100}{s^2 + 29s + 100}$$

For (a unit step/ a unit ramp/ sinusoidal/ impulse/ an) input signal to the system, find the approximate setting time (in sec) for:

I. 5% criterion.
   a. 0.28
   b. 0.24
   c. 0.21
   d. 0.34
   e. None of the above

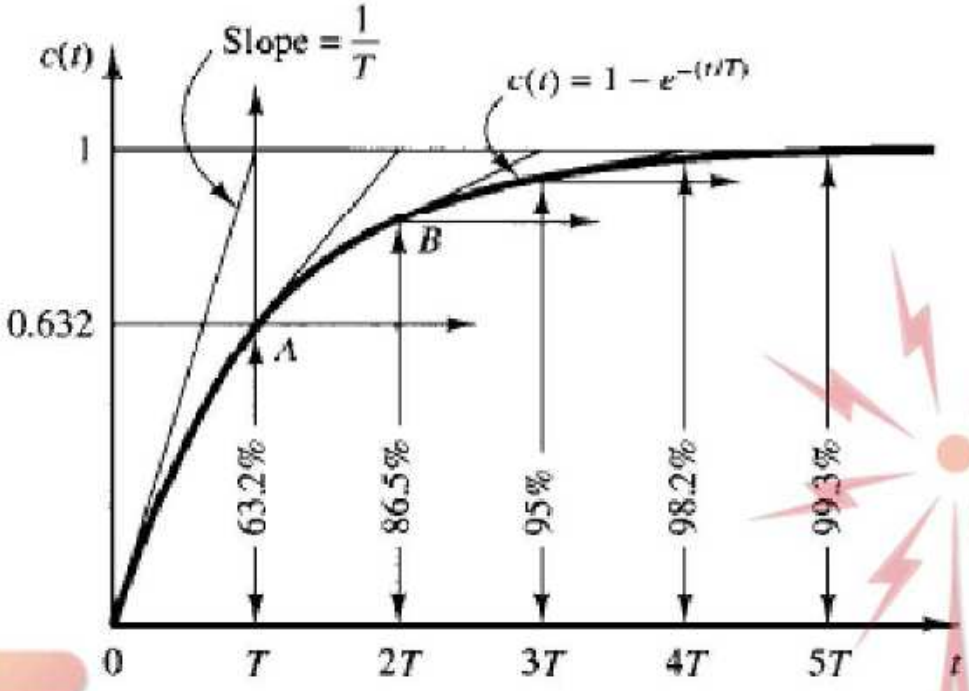
II. 2% criterion (if the question does not mention the criterion, 2% criterion is considered the permissible error).
   a. 0.34
   b. 0.28
   c. 0.21
   d. 0.24
   e. None of the above

III. 1% criterion.
   a. 0.24
   b. 0.21
   c. 0.28
   d. 0.34
   e. None of the above

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

Correct answer:

I 0.21
II 0.28
III 0.34



$$\frac{C(s)}{R(s)} = \frac{{\omega^2}_n}{s^2 + 2\zeta\omega_n s + {\omega^2}_n}$$

Where:

ω_n: Damped natural frequency

ζ: The damping ratio

σ: Damping constant

ω_d: Un-damped natural frequency

VIN: The coefficient of s^2 must be one

$$G(S) = \frac{100}{s^2 + 29s + 100}$$

$${\omega^2}_n = 100 \rightarrow \omega_n = 10$$

$$2\zeta\omega_n = 29$$

$$\zeta\omega_n = 14.5$$

I. For 5% criterion, the approximate setting time (in sec) is:

$$t_s = 3T = \frac{3}{\sigma} = \frac{3}{\zeta\omega_n}$$

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

$$t_s = \frac{3}{14.5} = 0.2068965517$$

II. For 2% criterion (if the question does not mention the criterion, 2% criterion is considered the permissible error), the approximate setting time (in sec) is:

$$t_s = 4T = \frac{4}{\sigma} = \frac{4}{\zeta\omega_n}$$

$$t_s = \frac{4}{14.5} = 0.275862069$$

III. For 1% criterion, the approximate setting time (in sec) is:

$$t_s = 5T = \frac{5}{\sigma} = \frac{5}{\zeta\omega_n}$$

$$t_s = \frac{5}{14.5} = 0.3448275862$$

240. The transfer function of the system is

$$G(s) = \frac{100}{s^2 + 25s + 100}$$

For (a unit step/ a unit ramp/ sinusoidal/ impulse/ an) input signal to the system, find the approximate setting time (in sec) for:

I. 5% criterion.
   a. 0.2
   b. 0.24
   c. 0.3
   d. 0.15
   e. None of the above

II. 2% criterion (if the question does not mention the criterion, 2% criterion is considered the permissible error).
   a. 0.4
   b. 0.3
   c. 0.32
   d. 0.24
   e. None of the above

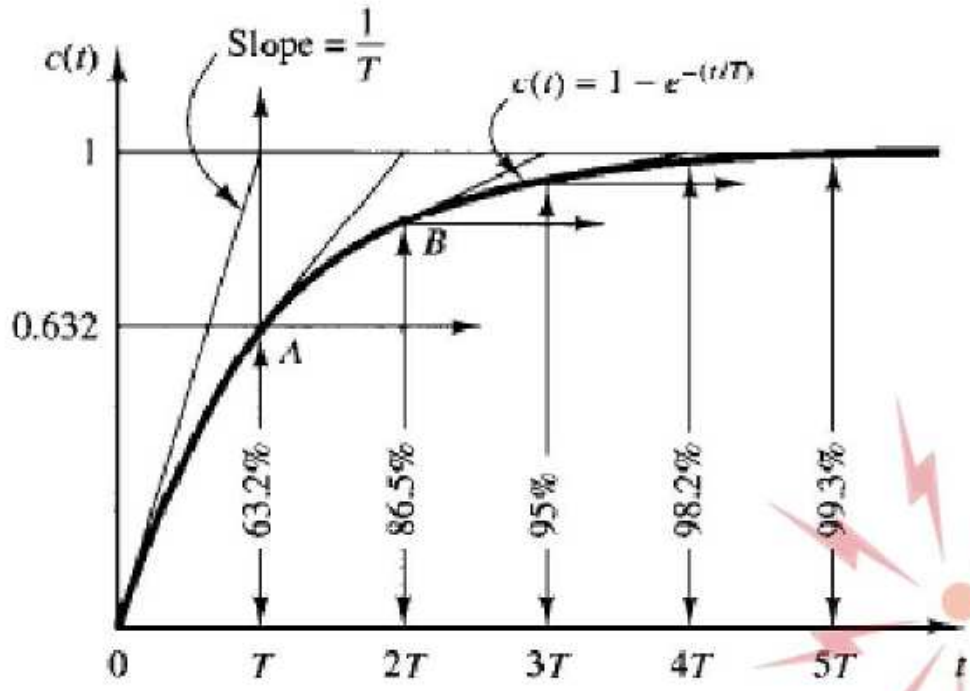
III. 1% criterion.
   a. 0.4
   b. 0.5
   c. 0.45
   d. 0.3
   e. None of the above

Correct answer:

I. 0.24
II. 0.32
III. 0.4

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل منشأ هذا العمل



$$\frac{C(s)}{R(s)} = \frac{{\omega^2}_n}{s^2 + 2\zeta\omega_n s + {\omega^2}_n}$$

Where:

ω_n: Damped natural frequency

ζ: The damping ratio

σ: Damping constant

ω_d: Un-damped natural frequency

VIN: The coefficient of s^2 must be one

$$G(S) = \frac{100}{s^2 + 25s + 100}$$

$${\omega^2}_n = 100 \rightarrow \omega_n = 10$$

$$2\zeta\omega_n = 25$$

$$\zeta\omega_n = 12.5$$

I. For 5% criterion, the approximate setting time (in sec) is:

$$t_s = 3T = \frac{3}{\sigma} = \frac{3}{\zeta\omega_n}$$

$$t_s = \frac{3}{12.5} = 0.24$$

II. For 2% criterion (if the question does not mention the criterion, 2% criterion is considered the permissible error), the approximate setting time (in sec) is:

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

$$t_s = 4T = \frac{4}{\sigma} = \frac{4}{\zeta\omega_n}$$

$$t_s = \frac{4}{12.5} = 0.32$$

III. For 1% criterion, the approximate setting time (in sec) is:

$$t_s = 5T = \frac{5}{\sigma} = \frac{5}{\zeta\omega_n}$$

$$t_s = \frac{5}{12.5} = 0.4$$

241. The transfer function of the system is

$$G(s) = \frac{100}{s^2 + 100}$$

For (a unit step/ a unit ramp/ sinusoidal/ impulse/ an) input signal to the system, find the approximate setting time (in sec) for:

I. 5% criterion.
   a. 0.27
   b. 0.21
   c. 0.24
   d. The output signal never reaches the steady state value
   e. None of the above

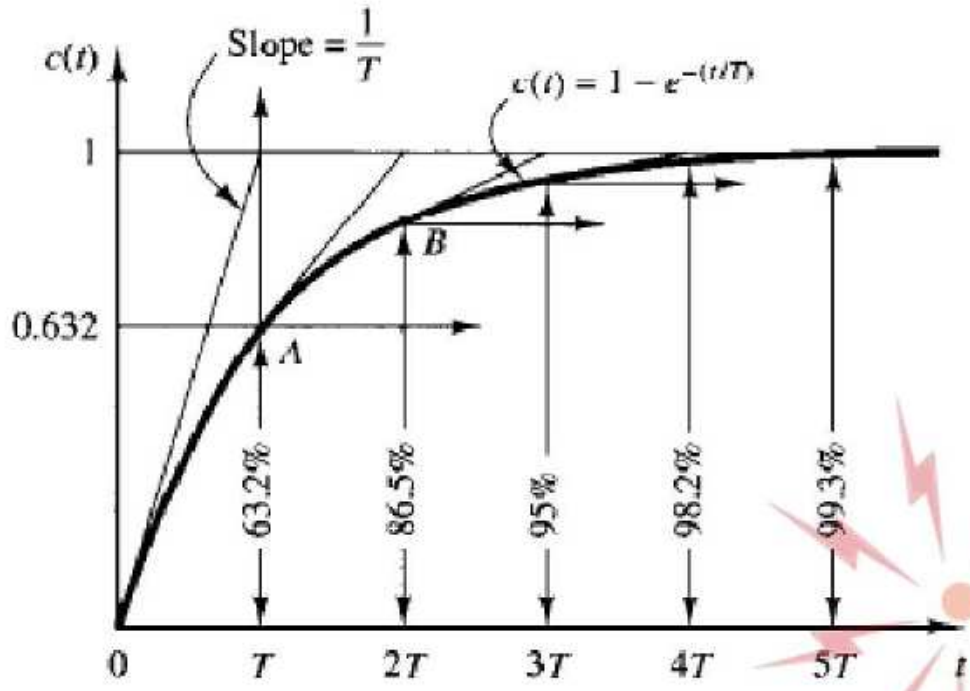
II. 2% criterion (if the question does not mention the criterion, 2% criterion is considered the permissible error).
   a. 0.25
   b. 0.4
   c. 0.3
   d. The output signal never reaches the steady state value
   e. None of the above

III. 1% criterion.
   a. 0.35
   b. The output signal never reaches the steady state value
   c. 0.3
   d. 0.45
   e. None of the above

Correct answer:

I. The output signal never reaches the steady state value
II. The output signal never reaches the steady state value
III. The output signal never reaches the steady state value

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل



$$\frac{C(s)}{R(s)} = \frac{{\omega^2}_n}{s^2 + 2\zeta\omega_n s + {\omega^2}_n}$$

Where:

ω_n: Damped natural frequency

ζ: The damping ratio

σ: Damping constant

ω_d: Un-damped natural frequency

VIN: The coefficient of s^2 must be one

$$G(S) = \frac{100}{s^2 + 100}$$

$${\omega^2}_n = 100 \rightarrow \omega_n = 10$$

$$2\zeta\omega_n = 0$$

$$\zeta\omega_n = 0$$

So, the output signal never reaches the steady state value.

I. For 5% criterion, the approximate setting time (in sec) is:

$$t_s = 3T = \frac{3}{\sigma} = \frac{3}{\zeta\omega_n}$$

$$t_s = \frac{3}{0} = \infty$$

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

II. For 2% criterion (if the question does not mention the criterion, 2% criterion is considered the permissible error), the approximate setting time (in sec) is:

$$t_s = 4T = \frac{4}{\sigma} = \frac{4}{\zeta\omega_n}$$

$$t_s = \frac{4}{0} = \infty$$

III. For 1% criterion, the approximate setting time (in sec) is:

$$t_s = 5T = \frac{5}{\sigma} = \frac{5}{\zeta\omega_n}$$

$$t_s = \frac{5}{0} = \infty$$

242. The transfer function of the system is

$$G(s) = \frac{32}{2s^2 + 16s + 32}$$

For (a unit step/ a unit ramp/ sinusoidal/ impulse/ an) input signal to the system, find the approximate setting time (in sec) for:

I 5% criterion.

a. 0.33
b. 0.75
c. 0.8
d. 0.5
e. None of the above

II 2% criterion (if the question does not mention the criterion, 2% criterion is considered the permissible error).

a. 0.75
b. 1.0
c. 1.2
d. 0.8
e. None of the above

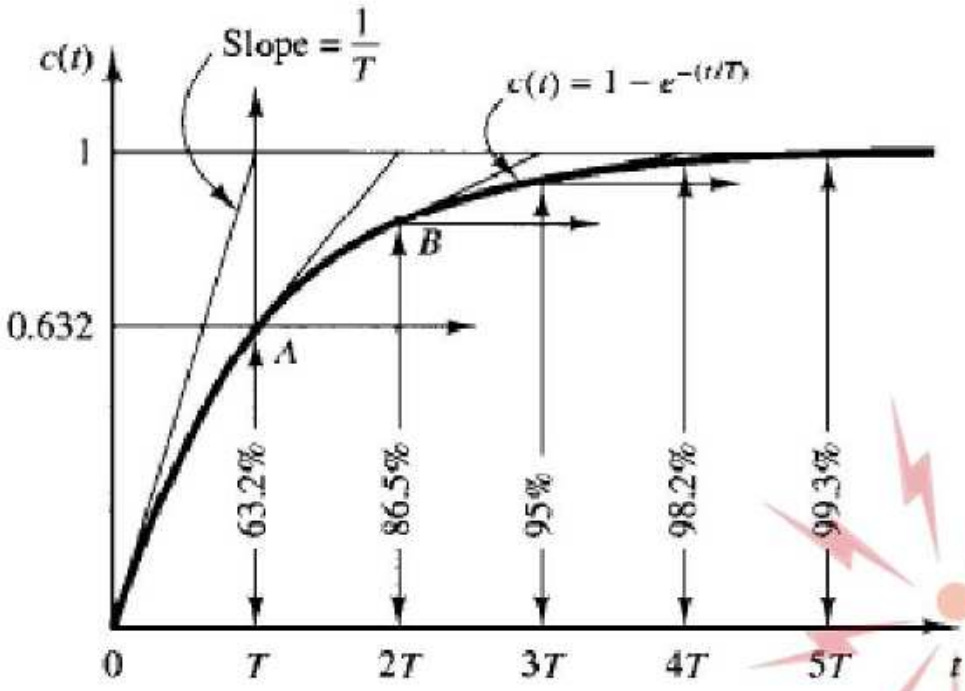
III 1% criterion.

a. 1.5
b. 0.5
c. 1.0
d. 1.3
e. None of the above

Correct answer:

I 0.75
II 1.0
III None of the above (1.25)

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل



$$\frac{C(s)}{R(s)} = \frac{{\omega^2}_n}{s^2 + 2\zeta\omega_n s + {\omega^2}_n}$$

Where:

ω_n: Damped natural frequency

ζ: The damping ratio

σ: Damping constant

ω_d: Un-damped natural frequency

VIN: The coefficient of s^2 must be one

$$G(S) = \frac{32}{s^2 + 16s + 32} = \frac{16}{s^2 + 8s + 16}$$

$${\omega^2}_n = 16 \rightarrow \omega_n = 4$$

$$2\zeta\omega_n = 8$$

$$\zeta\omega_n = 4$$

I. For 5% criterion, the approximate setting time (in sec) is:

$$t_s = 3T = \frac{3}{\sigma} = \frac{3}{\zeta\omega_n}$$

$$t_s = \frac{3}{4} = 0.75$$

II. For 2% criterion (if the question does not mention the criterion, 2% criterion is considered the permissible error), the approximate setting time (in sec) is:

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

$$t_s = 4T = \frac{4}{\sigma} = \frac{4}{\zeta\omega_n}$$

$$t_s = \frac{4}{4} = 1$$

III. For 1% criterion, the approximate setting time (in sec) is:

$$t_s = 5T = \frac{5}{\sigma} = \frac{5}{\zeta\omega_n}$$

$$t_s = \frac{5}{4} = 1.25$$

243. The transfer function of the system is

$$G(s) = \frac{8}{0.5s^2 + 3s + 4}$$

For (a unit step/ a unit ramp/ sinusoidal/ impulse/ an) input signal to the system, find the approximate setting time (in sec) for:

I. 5% criterion.
   a. 0.67
   b. 1.5
   c. 1
   d. 1.33
   e. None of the above

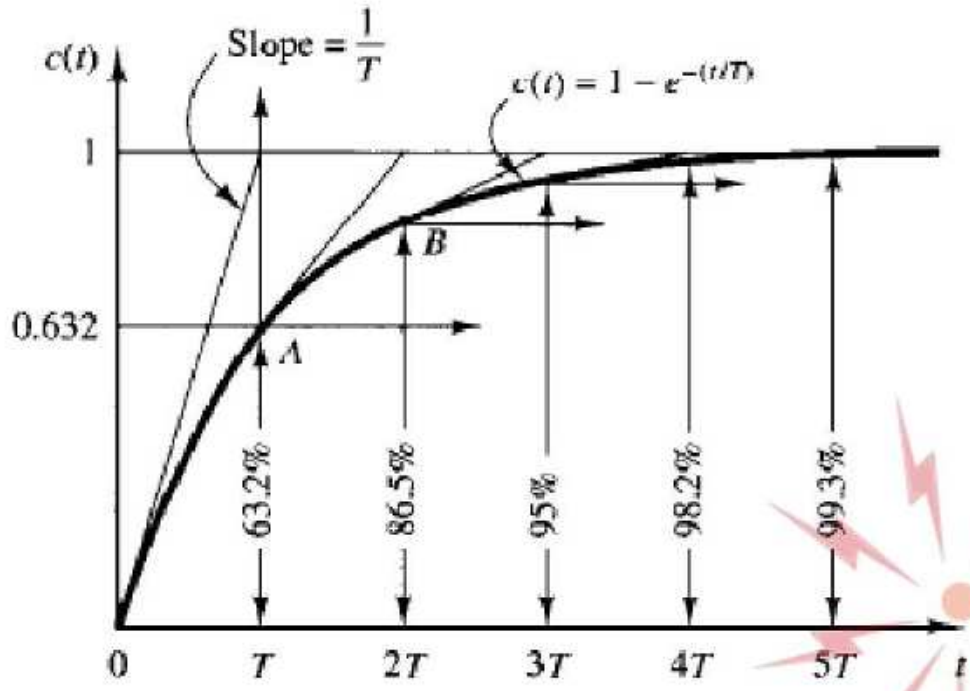
II. 2% criterion (if the question does not mention the criterion, 2% criterion is considered the permissible error).
   a. 1.67
   b. 0.5
   c. 1
   d. 2
   e. None of the above

III. 1% criterion.
   a. 1
   b. 1.67
   c. 1.33
   d. 2
   e. None of the above

Correct answer:

I. 1
II. None of the above (1.33)
III. 1.67

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل منشأ هذا العمل



$$\frac{C(s)}{R(s)} = \frac{{\omega^2}_n}{s^2 + 2\zeta\omega_n s + {\omega^2}_n}$$

Where:

ω_n: Damped natural frequency

ζ: The damping ratio

σ: Damping constant

ω_d: Un-damped natural frequency

VIN: The coefficient of s^2 must be one

$$G(S) = \frac{8}{0.5s^2 + 3s + 4} = \frac{16}{s^2 + 6s + 8}$$

$${\omega^2}_n = 8 \rightarrow \omega_n = \sqrt{8}$$

$$2\zeta\omega_n = 6$$

$$\zeta\omega_n = 3$$

I. For 5% criterion, the approximate setting time (in sec) is:

$$t_s = 3T = \frac{3}{\sigma} = \frac{3}{\zeta\omega_n}$$

$$t_s = \frac{3}{3} = 1$$

II. For 2% criterion (if the question does not mention the criterion, 2% criterion is considered the permissible error), the approximate setting time (in sec) is:

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

$$t_s = 4T = \frac{4}{\sigma} = \frac{4}{\zeta\omega_n}$$

$$t_s = \frac{4}{3} = 1.33$$

III. For 1% criterion, the approximate setting time (in sec) is:

$$t_s = 5T = \frac{5}{\sigma} = \frac{5}{\zeta\omega_n}$$

$$t_s = \frac{5}{3} = 1.67$$

244. Given the following transfer function

$$G(S) = \frac{81}{s^2 + 9s + 81}$$

For (a unit step/ a unit ramp/ sinusoidal/ impulse/ an) input signal to the system, find the approximate setting time (in sec) for:

I. 5% criterion. (Round to two decimal places)
   a. 0.67
   b. 0.77
   c. 0.6
   d. None of the above

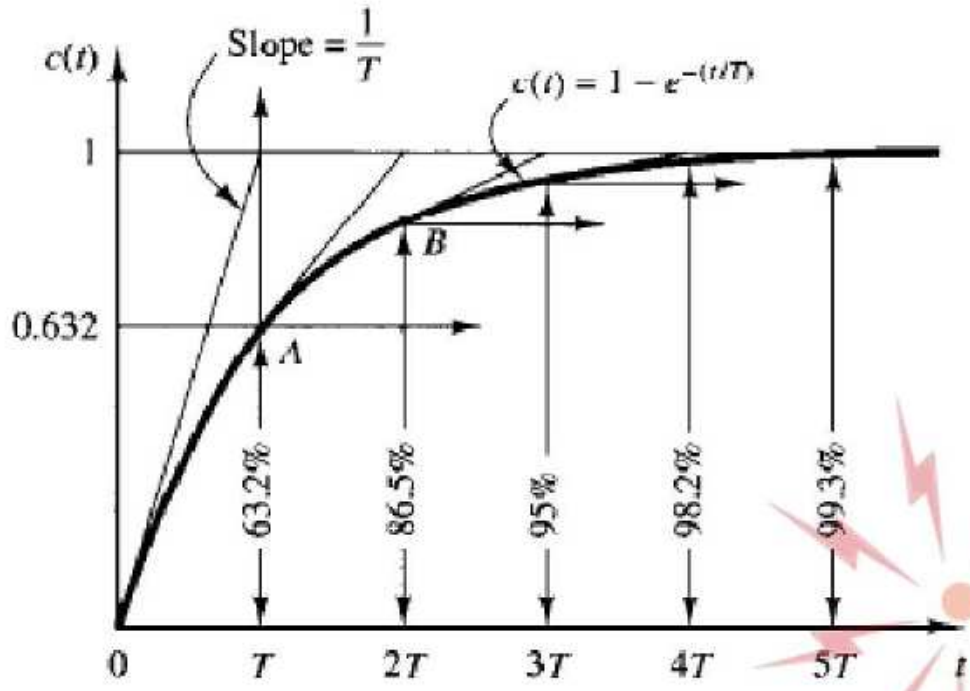
II. 2% criterion (if the question does not mention the criterion, 2% criterion is considered the permissible error). (Round to only one decimal place)
   a. 0.77
   b. 0.9
   c. 0.99
   d. 1.2

III. 1% criterion. (Round to only one decimal place)
   a. 1.2
   b. 0.9
   c. 1.1
   d. 0.67

Correct answer:

I. 0.67
II. 0.9
III. 1.1

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل منشأ هذا العمل



$$\frac{C(s)}{R(s)} = \frac{{\omega^2}_n}{s^2 + 2\zeta\omega_n s + {\omega^2}_n}$$

Where:

ω_n: Damped natural frequency

ζ: The damping ratio

σ: Damping constant

ω_d: Un-damped natural frequency

VIN: The coefficient of s^2 must be one

$$G(S) = \frac{81}{s^2 + 9s + 81}$$

$${\omega^2}_n = 81 \rightarrow \omega_n = 9$$

$$2\zeta\omega_n = 9$$

$$\zeta\omega_n = 4.5$$

I. For 5% criterion, the approximate setting time (in sec) is:

$$t_s = 3T = \frac{3}{\sigma} = \frac{3}{\zeta\omega_n}$$

$$t_s = \frac{3}{4.5} = 0.66667 \cong 0.67$$

II. For 2% criterion (if the question does not mention the criterion, 2% criterion is considered the permissible error), the approximate setting time (in sec) is:

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

$$t_s = 4T = \frac{4}{\sigma} = \frac{4}{\zeta\omega_n}$$

$$t_s = \frac{4}{4.5} = 0.88889 \cong 0.9$$

III. For 1% criterion, the approximate setting time (in sec) is:

$$t_s = 5T = \frac{5}{\sigma} = \frac{5}{\zeta\omega_n}$$

$$t_s = \frac{5}{4.5} = 1.11111 \cong 1.1$$

245. The transfer function of the system is

$$G(s) = \frac{75}{3s^2 + 75}$$

For (a unit step/ a unit ramp/ sinusoidal/ impulse/ an) input signal to the system, find the approximate setting time (in sec) for:

I. 5% criterion.
   a. 0.5
   b. 1.5
   c. 1
   d. The output signal never reaches the steady state value
   e. None of the above

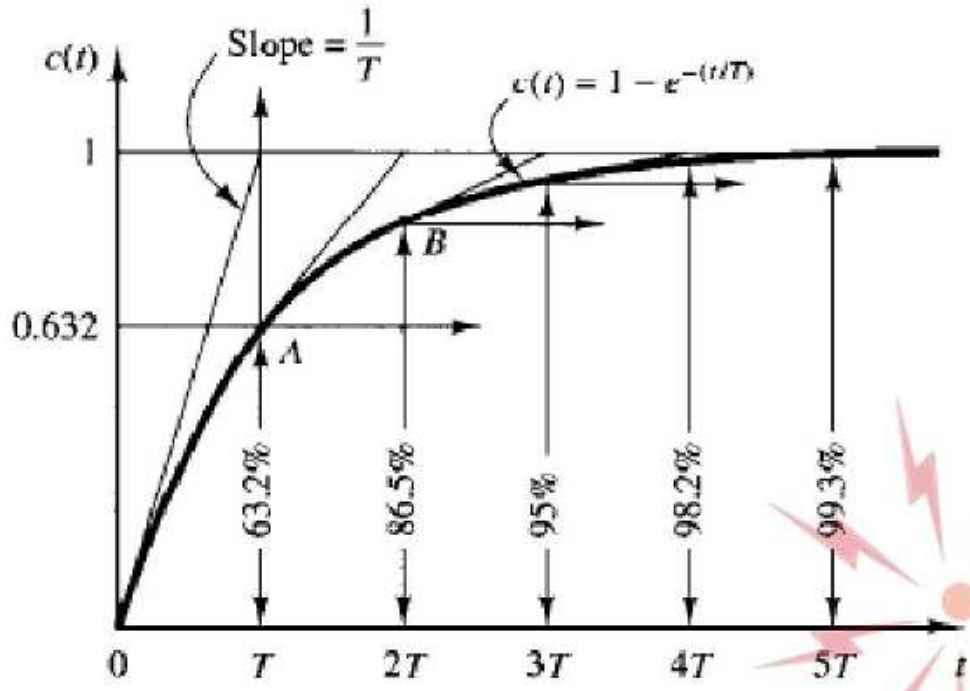
II. 2% criterion (if the question does not mention the criterion, 2% criterion is considered the permissible error).
   a. 0.5
   b. 1.5
   c. 2
   d. 1
   e. None of the above

III. 1% criterion.
   a. 1
   b. 2
   c. 2.33
   d. 1.33
   e. The output signal never reaches the steady state value

Correct answer:

I. The output signal never reaches the steady state value
II. None of the above (The output signal never reaches the steady state value)
III. The output signal never reaches the steady state value

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل



$$\frac{C(s)}{R(s)} = \frac{{\omega^2}_n}{s^2 + 2\zeta\omega_n s + {\omega^2}_n}$$

Where:

ω_n: Damped natural frequency

ζ: The damping ratio

σ: Damping constant

ω_d: Un-damped natural frequency

VIN: The coefficient of s^2 must be one

$$G(S) = \frac{75}{3s^2 + 75} = \frac{25}{s^2 + 25}$$

$${\omega^2}_n = 25 \rightarrow \omega_n = 5$$

$$2\zeta\omega_n = 0$$

$$\zeta\omega_n = 0$$

So, the output signal never reaches the steady state value

I. For 5% criterion, the approximate setting time (in sec) is:

$$t_s = 3T = \frac{3}{\sigma} = \frac{3}{\zeta\omega_n}$$

$$t_s = \frac{3}{0} = \infty$$

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

II. For 2% criterion (if the question does not mention the criterion, 2% criterion is considered the permissible error), the approximate setting time (in sec) is:

$$t_s = 4T = \frac{4}{\sigma} = \frac{4}{\zeta\omega_n}$$

$$t_s = \frac{4}{0} = \infty$$

III. For 1% criterion, the approximate setting time (in sec) is:

$$t_s = 5T = \frac{5}{\sigma} = \frac{5}{\zeta\omega_n}$$

$$t_s = \frac{5}{0} = \infty$$

246. The transfer function of the system is

$$G(s) = \frac{10}{2s^2 + 4s + 10}$$

For (a unit step/ a unit ramp/ sinusoidal/ impulse/ an) input signal to the system, find the approximate setting time (in sec) for:

I 5% criterion.

a. 2
b. 4
c. 5
d. 3
e. None of the above

II 2% criterion (if the question does not mention the criterion, 2% criterion is considered the permissible error).

a. 2
b. 3.5
c. 4
d. 3
e. None of the above

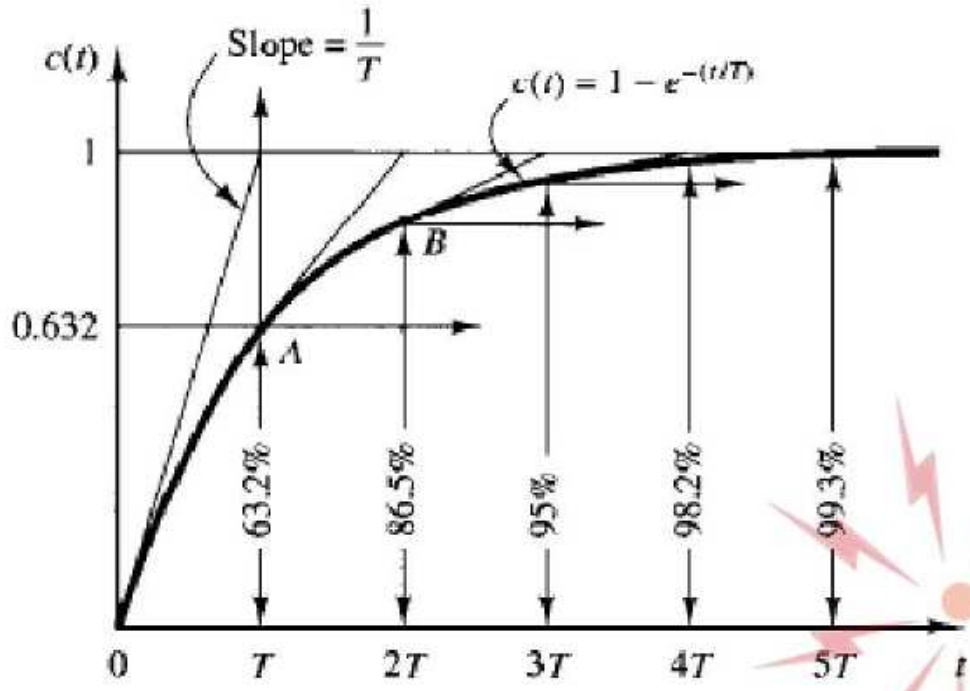
III 1% criterion.

a. 4
b. 2.5
c. 4.5
d. 6
e. None of the above

Correct answer:

I 3
II 4
III 5

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل منشأ هذا العمل



$$\frac{C(s)}{R(s)} = \frac{{\omega^2}_n}{s^2 + 2\zeta\omega_n s + {\omega^2}_n}$$

Where:

ω_n: Damped natural frequency

ζ: The damping ratio

σ: Damping constant

ω_d: Un-damped natural frequency

VIN: The coefficient of s^2 must be one

$$G(S) = \frac{10}{2s^2 + 4s + 10} = \frac{5}{s^2 + 2s + 5}$$

$${\omega^2}_n = 5 \rightarrow \omega_n = \sqrt{5}$$

$$2\zeta\omega_n = 2$$

$$\zeta\omega_n = 1$$

I. For 5% criterion, the approximate setting time (in sec) is:

$$t_s = 3T = \frac{3}{\sigma} = \frac{3}{\zeta\omega_n}$$

$$t_s = \frac{3}{1} = 3$$

II. For 2% criterion (if the question does not mention the criterion, 2% criterion is considered the permissible error), the approximate setting time (in sec) is:

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

$$t_s = 4T = \frac{4}{\sigma} = \frac{4}{\zeta\omega_n}$$

$$t_s = \frac{4}{1} = 4$$

III. For 1% criterion, the approximate setting time (in sec) is:

$$t_s = 5T = \frac{5}{\sigma} = \frac{5}{\zeta\omega_n}$$

$$t_s = \frac{5}{1} = 5$$

247. A second order system has $\frac{C(s)}{R(s)} = \frac{3}{s^2+4s+13}$ for 2% error the settling time of the system is

a. 3
b. 2
c. 4
d. 1.5

Correct answer: 2

The standard form for transfer function of second order system is:

$$\frac{C(s)}{R(s)} = \frac{{\omega^2}_n}{s^2 + 2\zeta\omega_n s + {\omega^2}_n} = \frac{3}{s^2 + 4s + 13}$$

$$2\zeta\omega_n = 4$$

$$\zeta\omega_n = 2$$

The setting time for second order system is:

$$t_s = 4T = \frac{4}{\sigma} = \frac{4}{\zeta\omega_n} = \frac{4}{2} = 2 \quad (2\%\ criterion)$$

248. A second order system has $\frac{C(s)}{R(s)} = \frac{3}{s^2+4s+13}$ for 2% error the settling time of the system is

a. 2
b. 1.5
c. 4
d. 3

Correct answer:

The standard form for transfer function of second order system is:

$$\frac{C(s)}{R(s)} = \frac{{\omega^2}_n}{s^2 + 2\zeta\omega_n s + {\omega^2}_n} = \frac{3}{s^2 + 4s + 13}$$

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

$$2\zeta\omega_n = 4$$

$$\zeta\omega_n = 2$$

The setting time for second order system is:

$$t_s = 3T = \frac{3}{\sigma} = \frac{3}{\zeta\omega_n} = \frac{3}{2} = 1.5 \quad (5\%\ criterion)$$

249. If the forward gain of a negative unity feedback control system is:

$$\frac{100}{s(s+100)}$$

Find Settling time (2%) in seconds?

a. 0.02
b. 0.06
c. 0.08
d. Not mentioned

Correct answer: 0.06

The standard form for transfer function of second order system is:

$$\frac{C(s)}{R(s)} = \frac{{\omega^2}_n}{s^2 + 2\zeta\omega_n s + {\omega^2}_n} = \frac{\frac{100}{s(s+100)}}{1 + \frac{100}{s(s+100)}} = \frac{100}{s^2 + 100s + 100}$$

$$2\zeta\omega_n = 100$$

$$\zeta\omega_n = 50$$

The setting time for second order system is:

$$t_s = 3T = \frac{3}{\sigma} = \frac{3}{\zeta\omega_n} = \frac{3}{50} = 0.06 \quad (5\%\ criterion)$$

POLYTECHNIC

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

Control Action: Applying Different Types of Controllers (P, I, D, PI, PD and PID)

250. Integral control of second order system:

a. Rise the order of the system by one
b. The system may be unstable
c. Eliminates the steady state error
d. All the above

Correct answer: All the above

Integral controller:

1. Eliminates the steady state error
2. Rise the order of the system by one
3. The system may be unstable

251. The transfer function of D controller is:

a. K
b. Ks
c. s
d. $\frac{K}{s}$
e. $\frac{1}{s}$

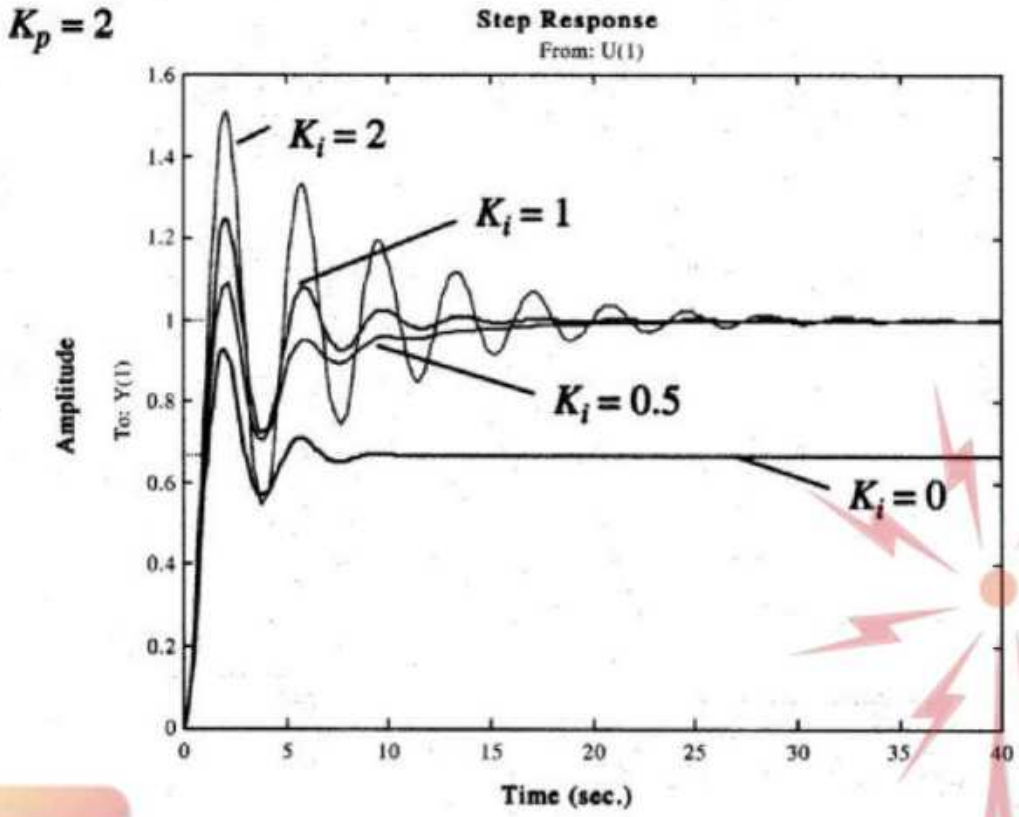
Correct answer: $Ks\ (Specifically, K_d s)$

252. The PI controller is used to:

a. Increase oscillations and overshoot greater than I controller
b. Upgrade steady state responses
c. Increase oscillations and overshoot, but less than I controller
d. None of the above

Correct answer: Increase oscillations and overshoot, but less than I controller

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل



253. Which of the following statements is true for I controller?

a. Increase the system type by 1
b. Amplify overshoot and oscillations
c. Eliminate steady-state error for a unit step input
d. Second and higher order systems may be unstable
e. All the above

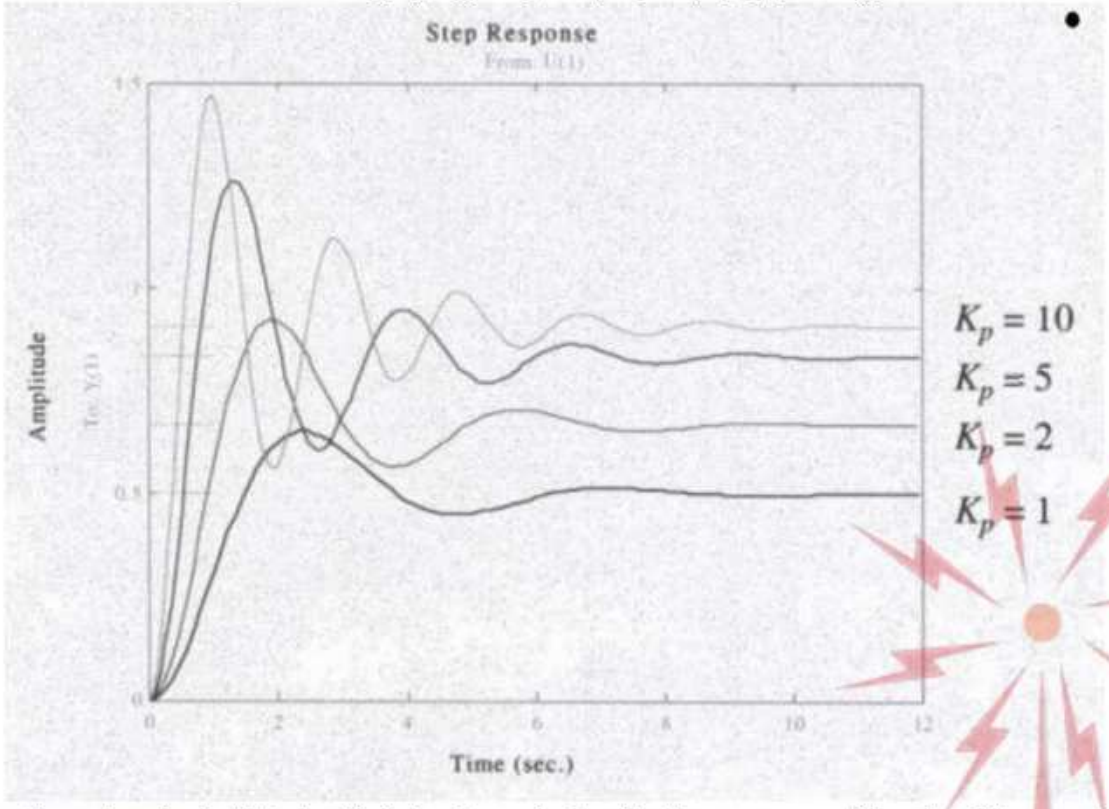
Correct answer: All the above

254. Which of the following would happen if the gain of P controller has been increased?

a. Reduce steady-state error
b. Upgrade both steady-state and transient responses
c. Reduce stability of second order system and the higher order system may be unstable
d. All the above

Correct answer: All the above

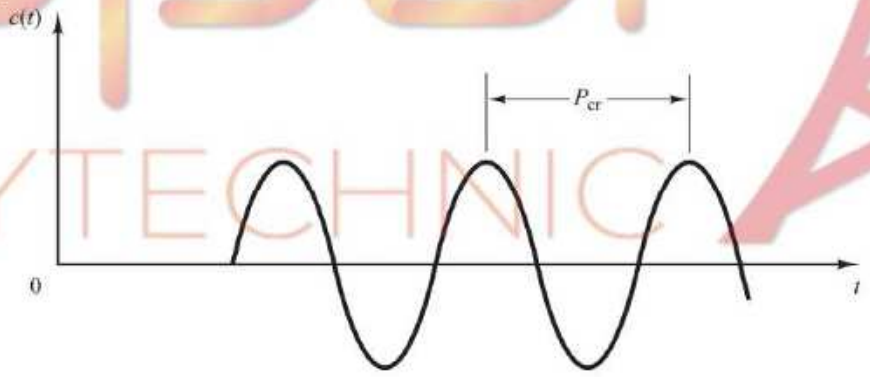
ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل



255. Second method of Ziegler Nichols rule used when the time response of the closed loop system:

a. Underdamped
b. Overdamped
c. Undamped
d. None of the above

Correct answer: Undamped



ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

Analyzing System Stability Using Routh's and Nyquist Criteria

256. The characteristic equation of a closed loop system is $4s^5 + 8s^4 + 7s^3 + 3s^2 + s + 5 = 0$, this system is:

a. Conditionally stable
b. Stable
c. Unstable
d. Critically stable

Correct answer: Unstable

Very Important Notes (VINs):

1. A necessary (but not sufficient) condition for stability is that all the coefficients of the polynomial characteristic equation be positive.
2. A system is stable if and only if all the elements of the first column of the Routh array are positive.

See Appendix II

s^5:	4.0000	7.0000	1.0000
s^4:	8.0000	3.0000	5.0000
s^3:	5.5000	-1.5000	0
s^2:	5.1818	5.0000	0
s^1:	-6.8070	0	0
s^0:	5.0000	0	0

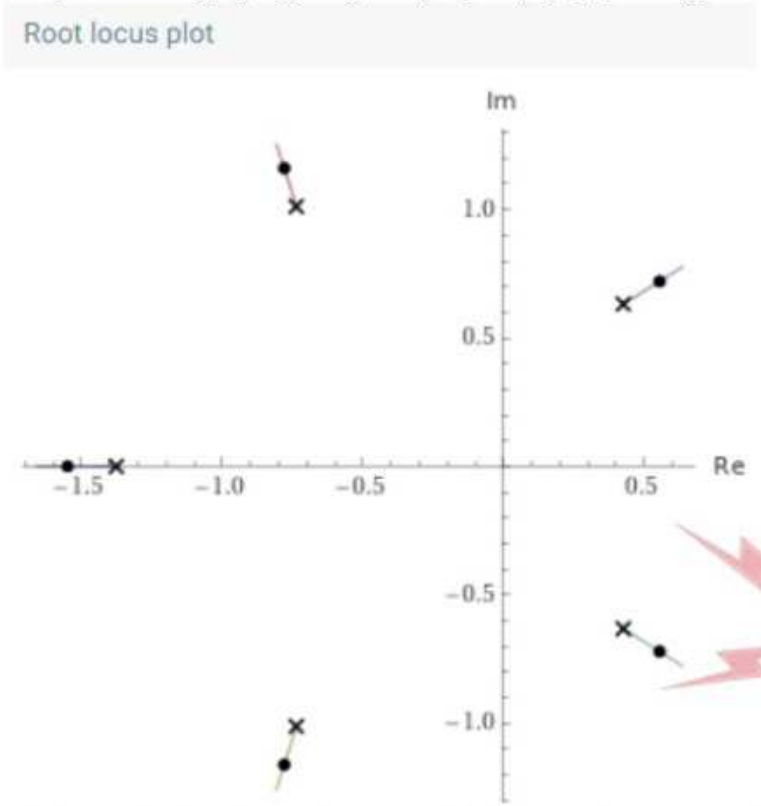
It is an unstable system

Number of right-hand side poles = 2

Polynomial coefficients roots:

$$s_1 = -1.3767 + j0.0000j$$
$$s_2 = -0.7370 + j1.0114$$
$$s_3 = -0.7370 - j1.0114$$
$$s_4 = 0.4254 + j0.6315$$
$$s_5 = 0.4254 - j0.6315$$

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل



257. The characteristic equation of a closed loop system is $s^3 + 5s^2 + 7s + 9 = 0$, this system is:

a. Unstable
b. Conditionally stable
c. Stable
d. Critically stable

Correct answer: Stable

Very Important Notes (VINs):

1. A necessary (but not sufficient) condition for stability is that all the coefficients of the polynomial characteristic equation be positive.
2. A system is stable if and only if all the elements of the first column of the Routh array are positive.

See Appendix II

s^3:	1.0000	7.0000
s^2:	5.0000	9.0000
s^1:	5.2000	0
s^0:	9.0000	0

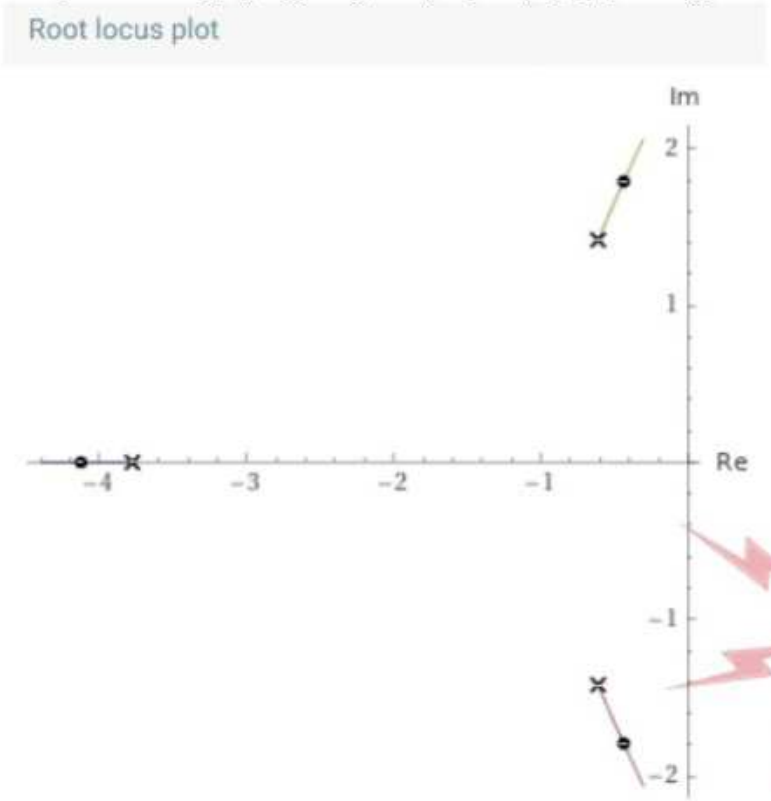
It is a stable system

Number of right-hand side poles = 0

Polynomial coefficients roots:

$$s_1 = -3.7777 + j0.0000$$
$$s_2 = -0.6112 + j1.4174$$
$$s_3 = -0.6112 - j1.4174$$

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل



258. The characteristic equation of a closed loop system is $s^3 + s^2 + s + 1 = 0$, this system is:

a. Stable
b. Critically stable
c. Conditionally stable
d. Unstable

Correct answer: Critically stable

Very Important Notes (VINs):

1. A necessary (but not sufficient) condition for stability is that all the coefficients of the polynomial characteristic equation be positive.
2. A system is stable if and only if all the elements of the first column of the Routh array are positive.

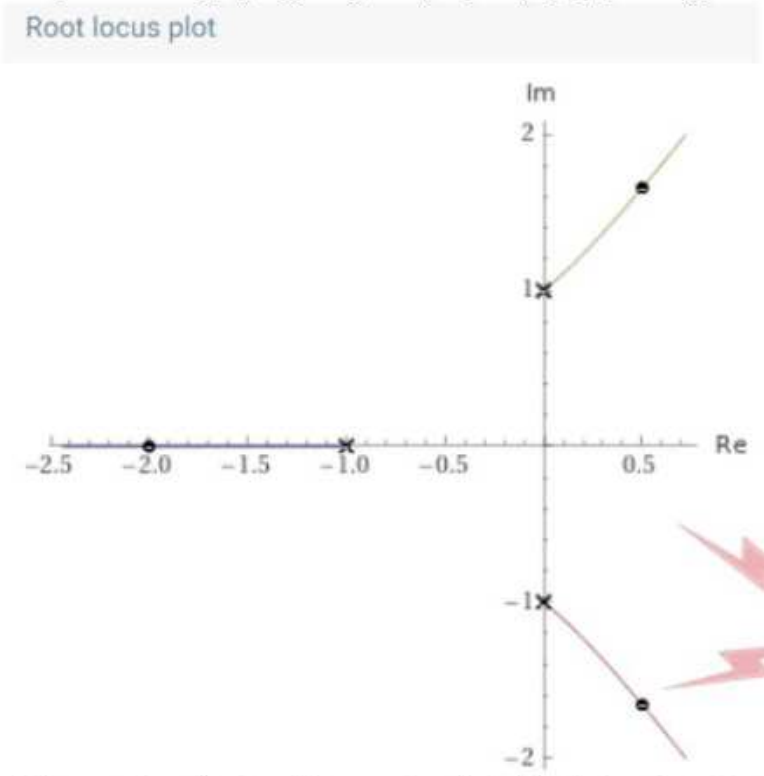
See Appendix II

s^3:	1.0000	1.0000
s^2:	1.0000	1.0000
s^1:	0.0000	0
s^0:	1.0000	0

It is a critically stable system
Number of right-hand side poles = 0
Polynomial coefficients roots:

$$s_1 = -1.0000 + j0.0000$$
$$s_2 = 0.0000 + j1.0000$$
$$s_3 = 0.0000 - j1.0000$$

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل



259. The characteristic equation of a closed loop system is $2s^3 + 4s^2 + 6s + 3 = 0$, this system is:

a. Stable
b. Critically stable
c. Unstable
d. Conditionally stable

Correct answer: Stable

Very Important Notes (VINs):

1. A necessary (but not sufficient) condition for stability is that all the coefficients of the polynomial characteristic equation be positive.
2. A system is stable if and only if all the elements of the first column of the Routh array are positive.

See Appendix II

s^3:	2.0000	6.0000
s^2:	4.0000	3.0000
s^1:	4.5000	0
s^0:	3.0000	0

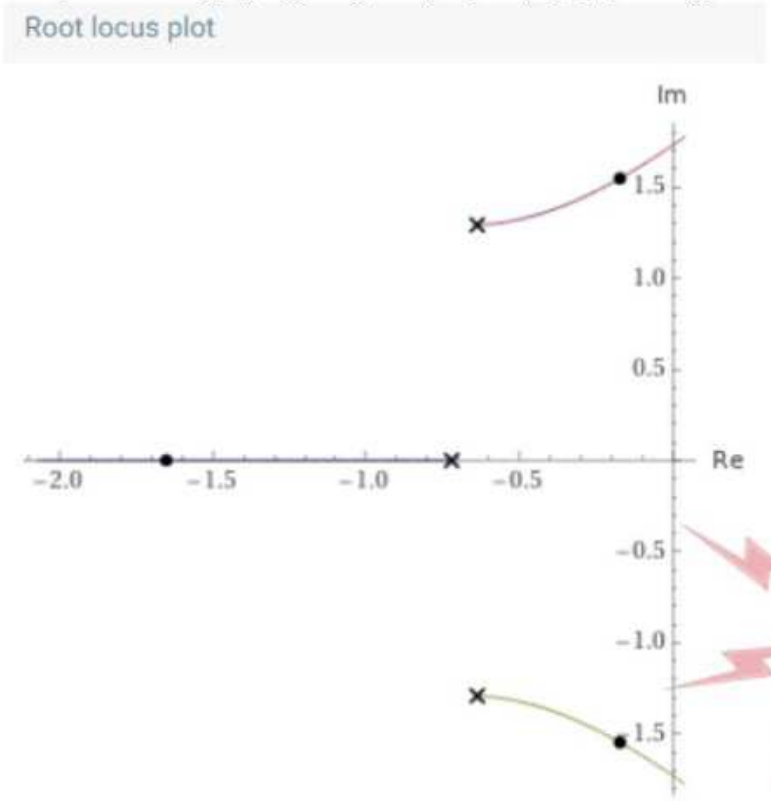
It is a stable system

Number of right-hand side poles = 0

Polynomial coefficients roots:

$$s_1 = -0.6389 + j1.2919$$
$$s_2 = -0.6389 - j1.2919$$
$$s_3 = -0.7221 + j0.0000$$

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل



260. The characteristic equation of a closed loop system is $2s^3 + 4s^2 + 3s + 6 = 0$, this system is:

a. Conditionally stable
b. Unstable
c. Critically stable
d. Stable

Correct answer: Critically stable

Very Important Notes (VINs):

1. A necessary (but not sufficient) condition for stability is that all the coefficients of the polynomial characteristic equation be positive.
2. A system is stable if and only if all the elements of the first column of the Routh array are positive.

See Appendix II

s^3:	2.0000	3.0000
s^2:	4.0000	6.0000
s^1:	0.0000	0
s^0:	6.0000	0

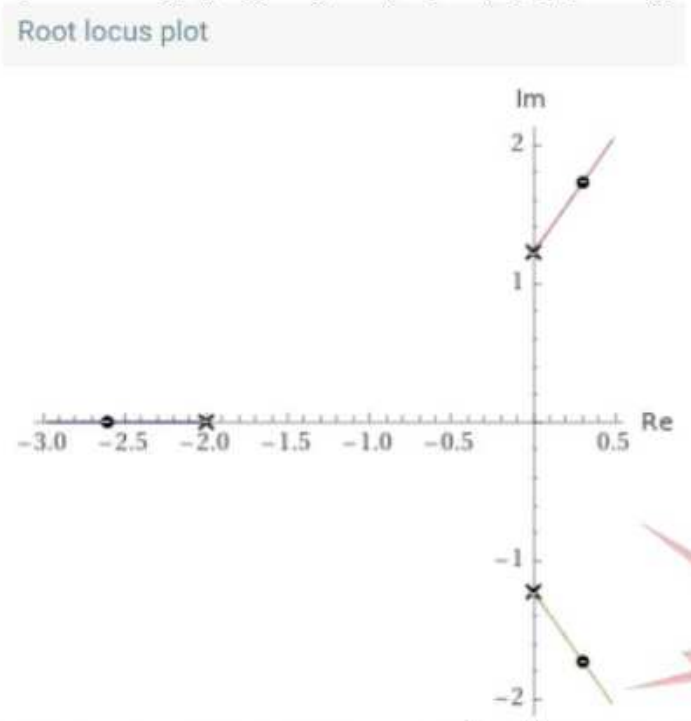
It is a critically stable system

Number of right-hand side poles = 0

Polynomial coefficients roots:

$$s_1 = -2.0000 + j0.0000$$
$$s_2 = -0.0000 + j1.2247$$
$$s_3 = -0.0000 - j1.2247$$

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل



261. The characteristic equation of a closed loop system is $s^3 - s^2 + s - 1 = 0$, this system is:

a. Unstable
b. Conditionally stable
c. Stable
d. Critically stable

Correct answer: Unstable

Very Important Notes (VINs):

1. A necessary (but not sufficient) condition for stability is that all the coefficients of the polynomial characteristic equation be positive.
2. A system is stable if and only if all the elements of the first column of the Routh array are positive.

See Appendix II

s^3:	1.0000	1.0000
s^2:	-1.0000	-1.0000
s^1:	0.0000	0
s^0:	-1.0000	0

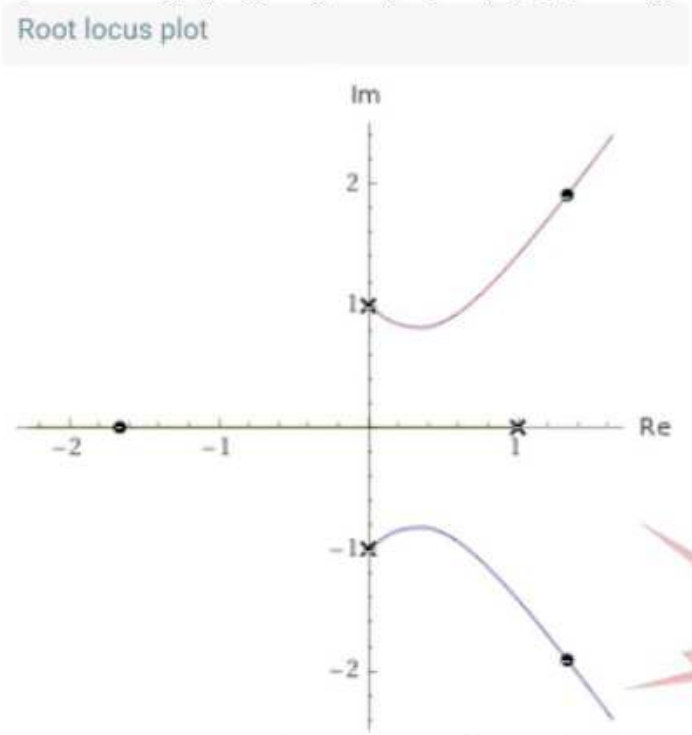
It is an unstable system

Number of right-hand side poles = 3

Polynomial coefficients roots:

$$s_1 = 1.0000 + j0.0000$$
$$s_2 = -0.0000 + j1.0000$$
$$s_3 = -0.0000 - j1.0000$$

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل



262. The characteristic equation of a closed loop system is $s^3 + K_1 s^2 + s + K_2 = 0$, this system is:

a. Critically stable
b. Unstable
c. Stable
d. Conditionally stable

Correct answer: Conditionally stable

Very Important Notes (VINs):

1. A necessary (but not sufficient) condition for stability is that all the coefficients of the polynomial characteristic equation be positive.
2. A system is stable if and only if all the elements of the first column of the Routh array are positive.

See Appendix II

s^3:	1.0000	1.0000
s^2:	K_1	K_2
s^1:	$\frac{K_1 - K_2}{K_1}$	0
s^0:	K_2	0

It is a conditionally stable system if $K_2 > 0$ and $K_1 - K_2 > 0$

$$K_1 - K_2 > 0$$

$$K_1 > K_2$$

So, the values of K for which the system is stable is:

$$K_1 > K_2 > 0$$

263. The characteristic equation of a closed loop system is $s^3 + K_1 s^2 - K_2 s + 1 = 0$, this system is:

a. Conditionally stable
b. Unstable

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

c. Stable
d. Critically stable

Correct answer: Conditionally stable

Very Important Notes (VINs):

3. A necessary (but not sufficient) condition for stability is that all the coefficients of the polynomial characteristic equation be positive.
4. A system is stable if and only if all the elements of the first column of the Routh array are positive.

See Appendix II

s^3:	1.0000	$-K_2$
s^2:	K_1	1.0000
s^1:	$\frac{-K_1K_2-1}{K_1}$	0
s^0:	1.0000	0

It is a conditionally stable system if $K_1 > 0$ and $-K_1K_2 - 1 > 0$

$$-1 > K_1K_2$$

So, the values of K for which the system is stable is:

$$K_1 > 0$$

$$K_2 > -\frac{1}{K_1}$$

264. The characteristic equation of a closed loop system is $2s^3 + Ks^2 + 4s + 2 = 0$, this system is:

a. Unstable
b. Critically stable
c. Conditionally stable
d. Stable

Correct answer: Conditionally stable

Very Important Notes (VINs):

5. A necessary (but not sufficient) condition for stability is that all the coefficients of the polynomial characteristic equation be positive.
6. A system is stable if and only if all the elements of the first column of the Routh array are positive.

See Appendix II

s^3:	2.0000	4.0000
s^2:	K	2.0000
s^1:	$\frac{4(K-1)}{K}$	0
s^0:	2.0000	0

It is a conditionally stable system if $K > 0$ and $K - 1 > 0$

$$K - 1 > 0$$

$$K > 1$$

So, the values of K for which the system is stable is:

$$K > 1$$

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

265. The characteristic equation of a closed loop system is $3s^3 - s^2 + 6s + K = 0$, this system is:

a. Critically stable
b. Stable
c. Unstable
d. Conditionally stable

Correct answer: Unstable

Very Important Notes (VINs):

7. A necessary (but not sufficient) condition for stability is that all the coefficients of the polynomial characteristic equation be positive.
8. A system is stable if and only if all the elements of the first column of the Routh array are positive.

See Appendix II

s^3:	3.0000	6.0000
s^2:	-1.0000	K
s^1:	6+3K	0
s^0:	K	0

It is an unstable system because the coefficient of S^2 term is negative

266. The characteristic equation of a closed loop system is $4s^3 + (K-2)s^2 + 3s + K_1 = 0$, this system is stable when:

a. $K_1 > 0, K > \frac{4}{3}K_1 + 2$
b. $K_1 > \frac{4}{3}K_1 + 2, K > 0$
c. $K_1 > 0, K > \frac{1}{3}K_1 - 2$
d. None of the above

Correct answer: $K_1 > 0, K > \frac{4}{3}K_1 + 2$

Very Important Notes (VINs):

1. A necessary (but not sufficient) condition for stability is that all the coefficients of the polynomial characteristic equation be positive.
2. A system is stable if and only if all the elements of the first column of the Routh array are positive.

See Appendix II

s^3:	4	3
s^2:	$K-2$	K_1
s^1:	$\frac{3K-6-4K_1}{K-2}$	0
s^0:	K_1	0

$$K_1 > 0$$
$$\frac{3K-6-4K_1}{K-2} > 0$$
$$K > \frac{4}{3}K_1 + 2$$

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

267. Consider the loop transfer function. $\frac{K(s+6)(s-2)}{[(s+3)(s+5)]}$, In the root locus diagram centroid will be located at t:

a. 4
b. There is no centroid
c. -4
d. None of the above

Correct answer: There is no centroid

$$\alpha = \frac{\sum Real\ part\ of\ finite\ open\ loop\ poles - \sum Real\ part\ of\ finite\ open\ loop\ zeros}{P - Z}$$

$$P = Z$$

Then,

$$P - Z = 0$$

So, there is no centroid

268. Consider the loop transfer function. $\frac{K(s+6)(s-1)}{[s^3(s+3)(s+5)(s-3)]}$, In the root locus diagram centroid will be located at t:

a. There is no centroid
b. -1
c. 1
d. None of the above

Correct answer:

$$\alpha = \frac{\sum Real\ part\ of\ finite\ open\ loop\ poles - \sum Real\ part\ of\ finite\ open\ loop\ zeros}{P - Z}$$

$$\alpha = \frac{(0+0+0-3-5+3)-(-6+1)}{6-2} = 0$$

269. Consider the loop transfer function. $\frac{K(s+6)(s-1)}{[s^3(s+3)(s+5)]}$, In the root locus diagram centroid will be located at t:

a. -1
b. There is no centroid
c. 1
d. None of the above

Correct answer:

$$\alpha = \frac{\sum Real\ part\ of\ finite\ open\ loop\ poles - \sum Real\ part\ of\ finite\ open\ loop\ zeros}{P - Z}$$

$$\alpha = \frac{(0+0+0-3-5)-(-6+1)}{5-2} = -1$$

270. Consider the loop transfer function. $\frac{K(s+6)(s+3)}{[(s+5)]}$, In the root locus diagram centroid will be located at t:

a. There is no centroid
b. -4
c. 4

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

d. None of the above

Correct answer:

$$\alpha = \frac{\sum Real\ part\ of\ finite\ open\ loop\ poles - \sum Real\ part\ of\ finite\ open\ loop\ zeros}{P - Z}$$

$$\alpha = \frac{(-5) - (-6 - 3)}{1 - 2} = -4$$

271. Consider the loop transfer function. $\frac{K(s-4)(s+7)}{[(s+3)(s-5)(s+1)]}$, In the root locus diagram centroid will be located at t:

a. -6
b. 4
c. -4
d. None of the above

Correct answer:

$$\alpha = \frac{\sum Real\ part\ of\ finite\ open\ loop\ poles - \sum Real\ part\ of\ finite\ open\ loop\ zeros}{P - Z}$$

$$\alpha = \frac{(-3 + 5 - 1) - (4 - 7)}{3 - 2} = 4$$

272. All the root locus branches start at finite open loop poles and end at finite open loop zeros if:

a. P≠Z
b. P>Z
c. P<Z
d. P=Z

Correct answer: P=Z

273. Z number of root locus branches start at finite open loop poles and end at finite open loop zeros and P−Z number of root locus branches start at finite open loop poles and end at infinite open loop zeros if:

a. P<Z
b. P=Z
c. P≠Z
d. P>Z

Correct answer: P>Z

274. P number of root locus branches start at finite open loop poles and end at finite open loop zeros and Z−P number of root locus branches start at infinite open loop poles and end at finite open loop zeros if:

a. P<Z
b. P≠Z
c. P=Z
d. P>Z

Correct answer: P<Z

275. Some of the root locus branches approach infinity when P=Z:

a. False
b. True

Correct answer: False

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

Some of the root locus branches approach infinity when P≠Z

276. Roots on the imaginary axis makes the system:

a. Linear
b. Marginally stable
c. Stable
d. Unstable

Correct answer: Marginally stable

Roots on the imaginary axis makes the system marginally stable.

277. Stability of a system implies that:

a. Small changes in the system parameters does not result in large change in system output
b. Small changes in the initial conditions does not result in large change in system output
c. Small changes in the system input does not result in large change in system output
d. All the above mentioned

Correct answer: All the above mentioned

Stability of the system implies that small changes in the system input, initial conditions, and system parameters does not result in large change in system output.

278. For non-linear systems stability cannot be determined due to:

a. Possible existence of multiple equilibrium states
b. Output may be bounded for the particular bounded input but may not be bounded for the bounded inputs
c. No correspondence between bounded input and bounded output stability and asymptotic stability
d. All the mentioned

Correct answer: All of the mentioned

For non-linear systems stability cannot be determined as asymptotic stability and BIBO stability concepts cannot be applied, existence of multiple states and unbounded output for many bounded inputs.

279. Roots with higher multiplicity on the imaginary axis makes the system:

a. Unstable
b. Linear
c. Stable
d. Absolutely stable

Correct answer: Unstable

Repetitive roots on the imaginary axis makes the system unstable.

280. If the roots of the transfer function of the system have negative real parts then the response is _____:

a. Unstable
b. Stable
c. Bounded
d. Marginally stable

Correct answer: Bounded

If the roots of the have negative real parts then the response is bounded and eventually decreases to zero.

281. If root of the characteristic equation has positive real part the system is:

a. Unstable
b. Stable
c. Linear

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

d. Marginally stable

Correct answer: Unstable

The impulse response of the system is infinite when the roots of the characteristic equation has positive real part.

282. A linear system can be classified as:

a. Conditionally stable
b. Unstable
c. Absolutely stable
d. All the mentioned

Correct answer: All the mentioned

A system can be stable, unstable and conditionally stable also.

283. The roots of the transfer function do not have any effect on the stability of the system.

a. True
b. False

Correct answer: False

The roots of transfer function also determine the stability of system as they may be real, complex and may have multiplicity of various order.

Spark POLYTECHNIC

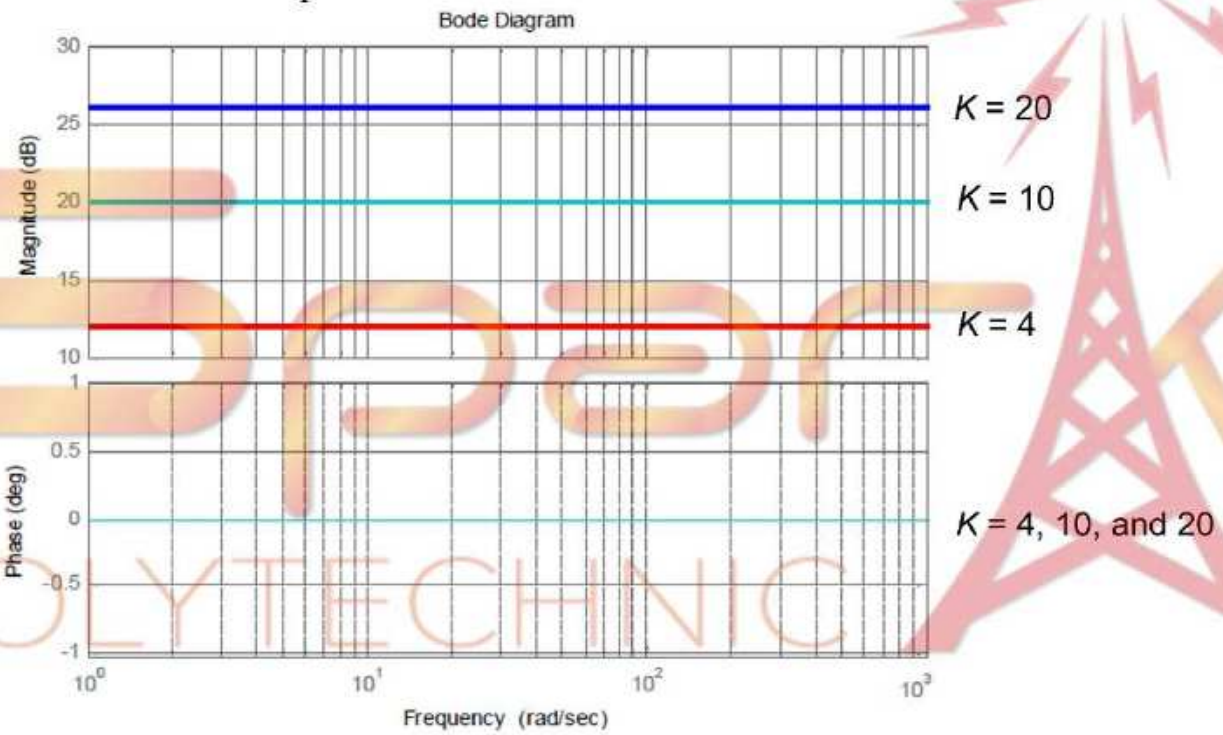
ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

Drawing Bode Plot and Nyquist Plot for a Given Transfer Function

284. The slope of logarithm of magnitude vs. frequency of a constant gain (factor/ elements) (K) is:

a. -20dB/ dec at all frequencies
b. Zero dB at all frequencies
c. +20dB/dec at all frequencies
d. None of the above

Correct answer: Zero dB at all frequencies



285. If a zero is located at origin, how does it get represented on the magnitude plot?

a. $40\ \log(w)\ dB$
b. $10\ \log(w)\ dB$
c. $60\ \log(w)\ dB$
d. $20\ \log(w)\ dB$

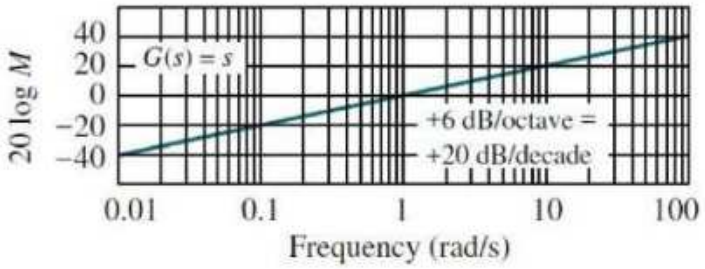
Correct answer: $20\ \log(w)\ dB$

بسواعدنا نبنيها

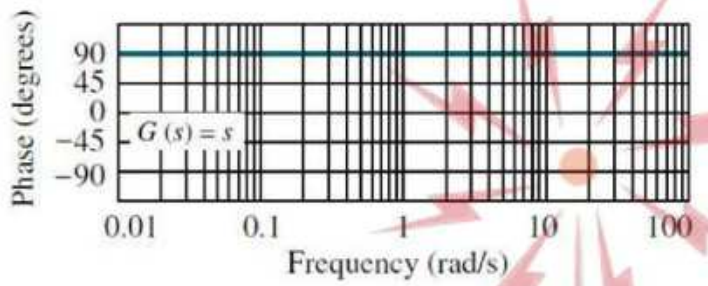
ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

Since it's a zero, then this means that it's in the numerator because zeros are the roots of the numerator, and as it is located at the origin, then $s = 0$, in other words, it is s in the numerator of the transfer function, which is equivalent to the differentiator (derivative factor/ $j\omega$ factor).

The Bode magnitude plot is a straight line with a +20 dB/decade slope passing through 0 dB at ω = 1.



The Bode phase plot is equal to a constant +90°.



286. If two zeros are located at origin, how do they get represented on the magnitude plot?

a. $20\ \log(w)\ dB$
b. $10\ \log(w)\ dB$
c. $60\ \log(w)\ dB$
d. $40\ \log(w)\ dB$

Correct answer: $40\ \log(w)\ dB$

Since they're zeros, then this means that they're in the numerator because zeros are the roots of the numerator, and as they're located at the origin, then $s = 0$, in other words, they're s in the numerator of the transfer function for each pole, which is equivalent to differentiator (derivative factor/ $j\omega$ factor) for each pole.

287. If three zeros are located at origin, how do they get represented on the magnitude plot?

a. $60\ \log(w)\ dB$
b. $40\ \log(w)\ dB$
c. $10\ \log(w)\ dB$
d. $20\ \log(w)\ dB$

Correct answer: $60\ \log(w)\ dB$

Since they're zeros, then this means that they're in the numerator because zeros are the roots of the numerator, and as they're located at the origin, then $s = 0$, in other words, they're s in the numerator of the transfer function for each pole, which is equivalent to differentiator (derivative factor/ $j\omega$ factor) for each pole.

288. If P zeros are located at origin, how do they get represented on the magnitude plot?

a. $20 * P\ \log(w)\ dB$

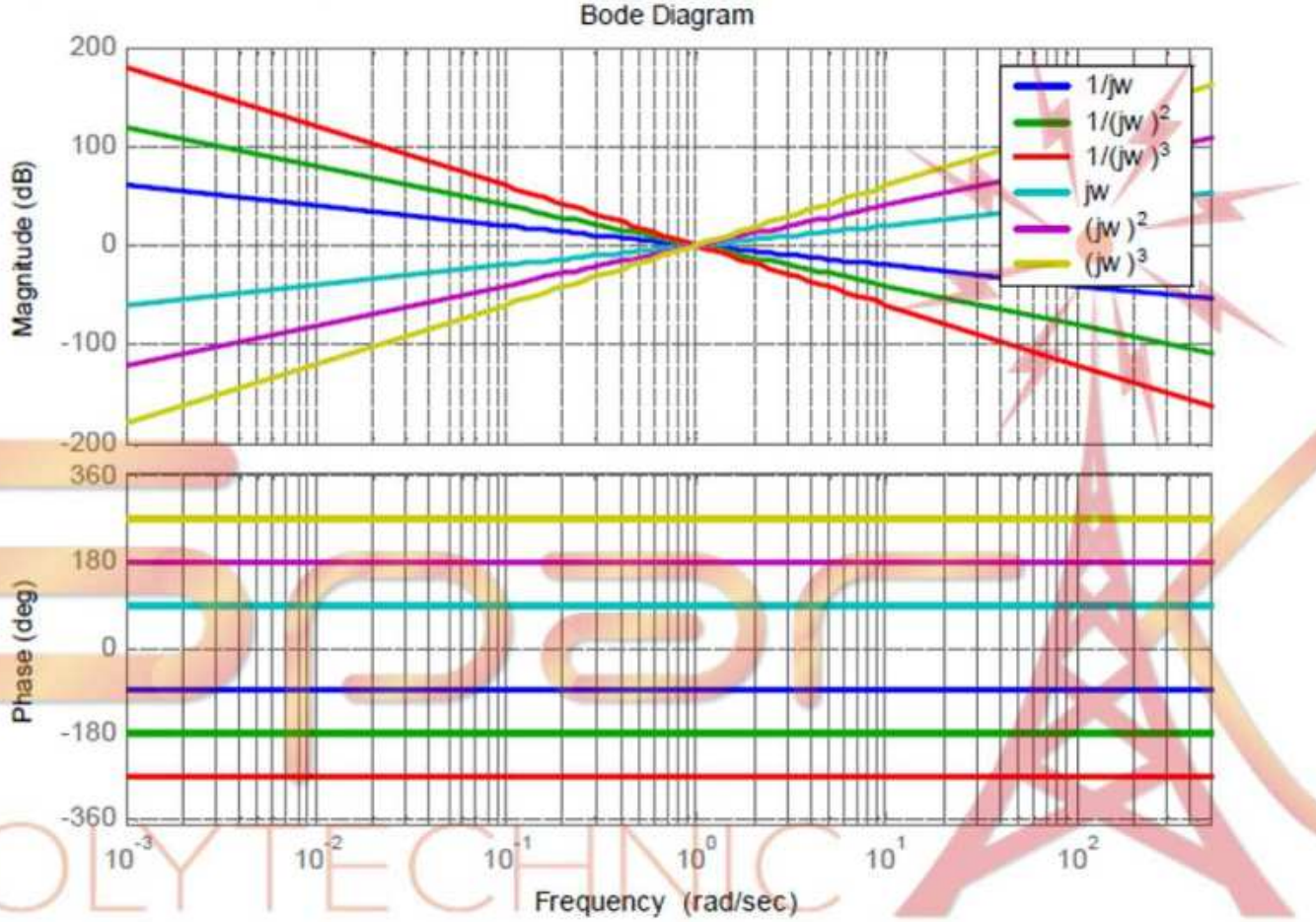
بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

b. $60 * P\ \log(w)\ dB$
c. $10 * P\ \log(w)\ dB$
d. $40 * P\ \log(w)\ dB$

Correct answer: $20 * P\ \log(w)\ dB$

Since they're zeros, then this means that they're in the numerator because zeros are the roots of the numerator, and as they're located at the origin, then $s = 0$, in other words, they're s in the numerator of the transfer function for each pole, which is equivalent to differentiator (derivative factor/ $j\omega$ factor) for each pole.



289. If two poles are located at origin, how do they get represented on the magnitude plot?

a. $-60\ \log(w)\ dB$
b. $-40\ \log(w)\ dB$
c. $-10\ \log(w)\ dB$
d. $-20\ \log(w)\ dB$

Correct answer: $-40\ \log(w)\ dB$

Since they're poles, then this means that they're in the denominator because poles are the roots of the denominator, and as they're located at the origin, then $s = 0$, in other words, they're s in the denominator of the transfer function for each pole, which is equivalent to the integrator (integral factor/ $(j\omega)^{-1}$ factor) for each pole.

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

290. If three poles are located at origin, how do they get represented on the magnitude plot?

a. $-60\ \log(w)\ dB$
b. $-20\ \log(w)\ dB$
c. $-40\ \log(w)\ dB$
d. $-10\ \log(w)\ dB$

Correct answer: $-60\ \log(w)\ dB$

Since they're poles, then this means that they're in the denominator because poles are the roots of the denominator, and as they're located at the origin, then $s = 0$, in other words, they're s in the denominator of the transfer function for each pole, which is equivalent to the integrator (integral factor/ $(j\omega)^{-1}$ factor) for each pole.

291. If P poles are located at origin, how do they get represented on the magnitude plot?

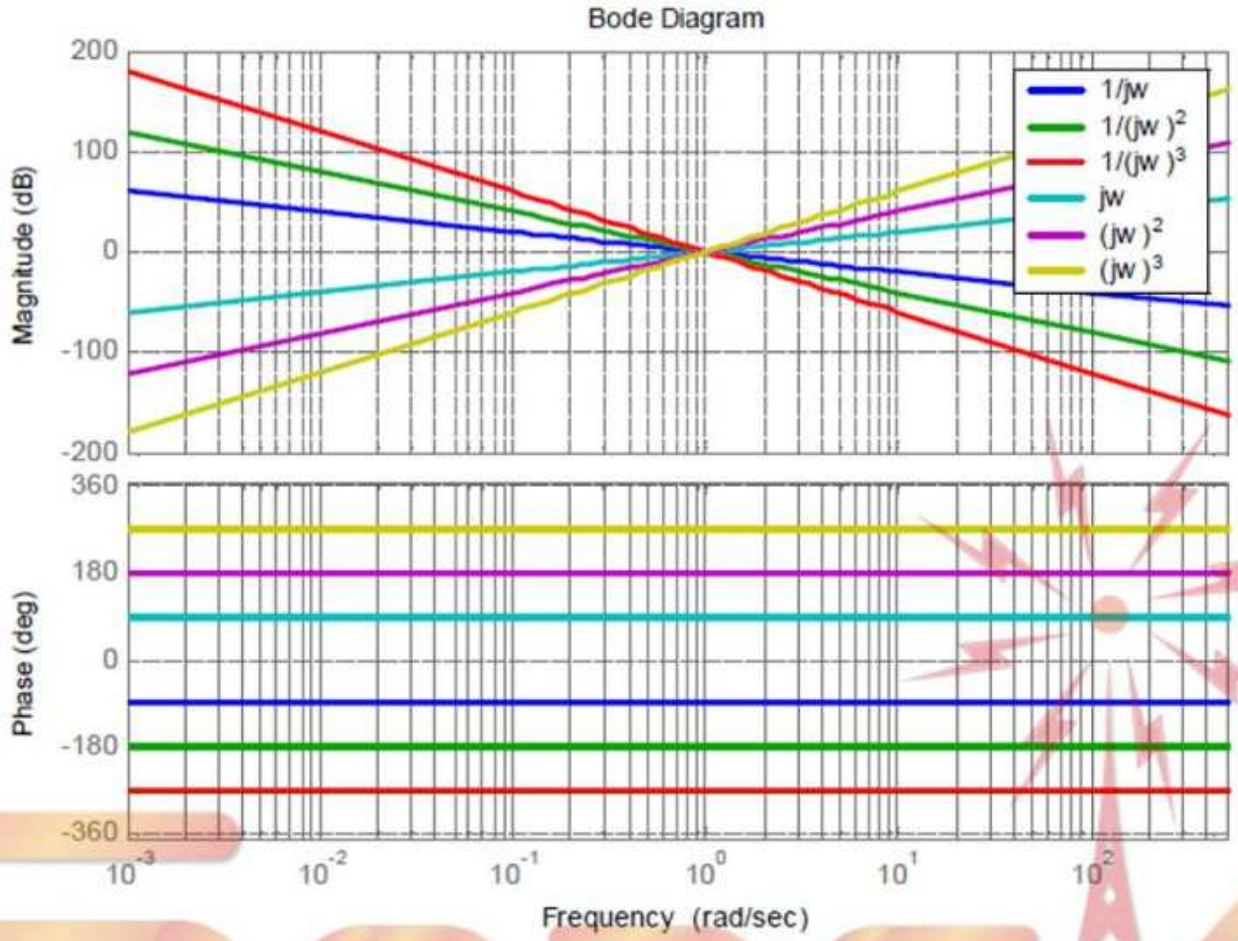
a. $-60 * P\ \log(w)\ dB$
b. $-10\ * P \log(w)\ dB$
c. $-40 * P\ \log(w)\ dB$
d. $-20 * P\ \log(w)\ dB$

Correct answer: $-20 * P\ \log(w)\ dB$

Since they're poles, then this means that they're in the denominator because poles are the roots of the denominator, and as they're located at the origin, then $s = 0$, in other words, they're s in the denominator of the transfer function for each pole, which is equivalent to the integrator (integral factor/ $(j\omega)^{-1}$ factor) for each pole.

spark POLYTECHNIC

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل



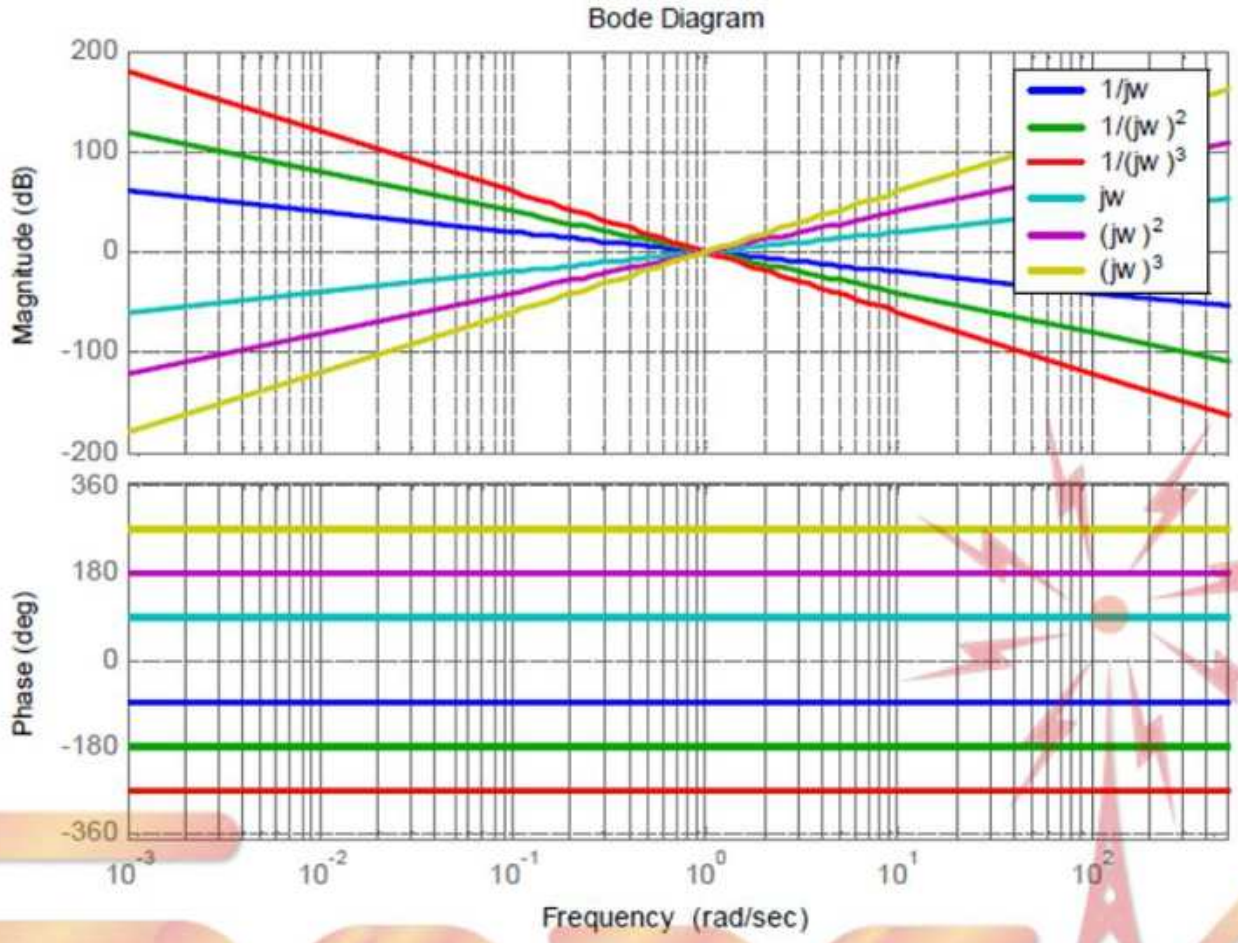
292. The slope of logarithm of magnitude vs. frequency of a $(j\omega)^{\pm P}$ (factor/ elements) is:
Or
The slope of an approximate logarithm of magnitude vs. frequency plot for a $(j\omega)^{\pm P}$ (factor/ elements) is:

a. Zero dB/dec at very low frequencies and $\mp$20dB*P/ dec at very high frequencies
b. ±20dB*P/dec at all frequencies
c. Zero dB at all frequencies
d. $\mp$20dB*P/ dec at all frequencies
e. Zero dB at very low frequencies and $\mp$20dB*P at very high frequencies
f. Zero dB at very low frequencies and ±20dB*P at very high frequencies
g. Zero dB/dec at very low frequencies and ±20dB*P/ dec at very high frequencies

Correct answer: ±20dB*P/dec at all frequencies

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

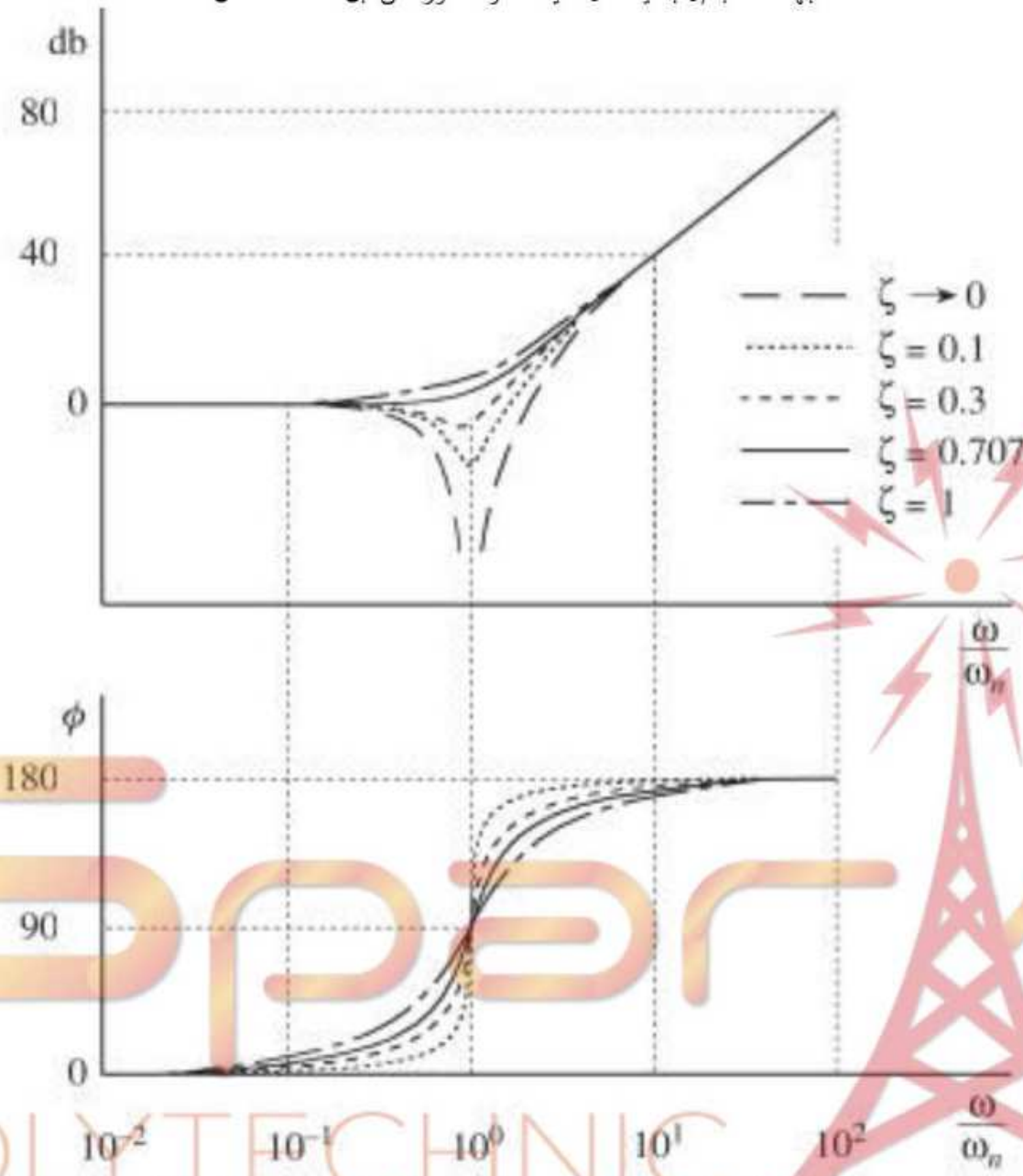


293. The slope of logarithm of magnitude vs. frequency of lead second order (factor/ elements) is:
Or
The slope of an approximate logarithm of magnitude vs. frequency plot for lead second order (factor/ elements) is:

a. Zero dB at all frequencies
b. Zero dB at very low frequencies and +40 dB at very high frequencies
c. -40dB/ dec at all frequencies
d. +40dB/dec at all frequencies
e. Zero dB at very low frequencies and -40 dB at very high frequencies
f. Zero dB/dec at very low frequencies and -40dB/ dec at very high frequencies
g. Zero dB/dec at very low frequencies and +40dB/ dec at very high frequencies

Correct answer: Zero dB/dec at very low frequencies and +40dB/ dec at very high frequencies

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل



294. The phase shift of a constant gain (factor/ elements) (K) when the frequency is changed from zero up to infinity is:

Or

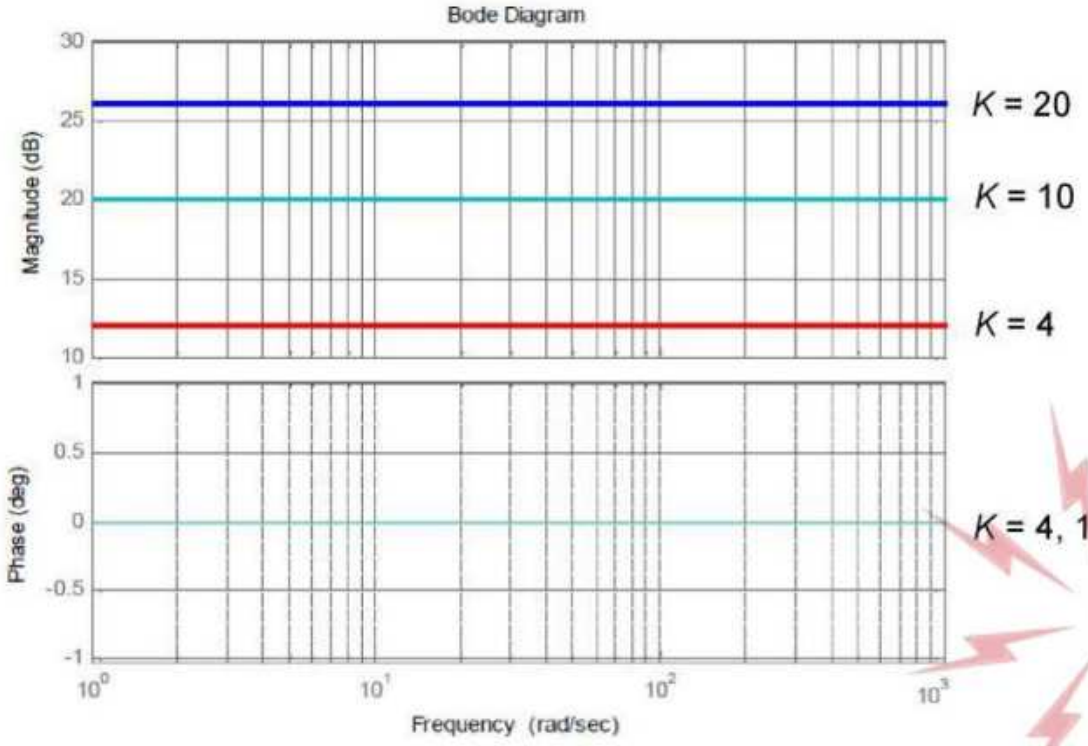
In frequency response, the phase angle of a constant gain (factor/ elements) (K) is:

a. Changed from zero degrees up to -90 degrees
b. Zero degree
c. +90 degree
d. Changed from zero degrees up to +90 degrees
e. -90 degrees
f. +180 degree
g. -180 degrees

Correct answer: Zero degree

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل



295. The phase shift of a $(j\omega)^{\pm P}$ (factor/ elements) when the frequency is changed from zero up to infinity is:
Or
In frequency response, the phase angle of $(j\omega)^{\pm P}$ (factor/ elements) is:

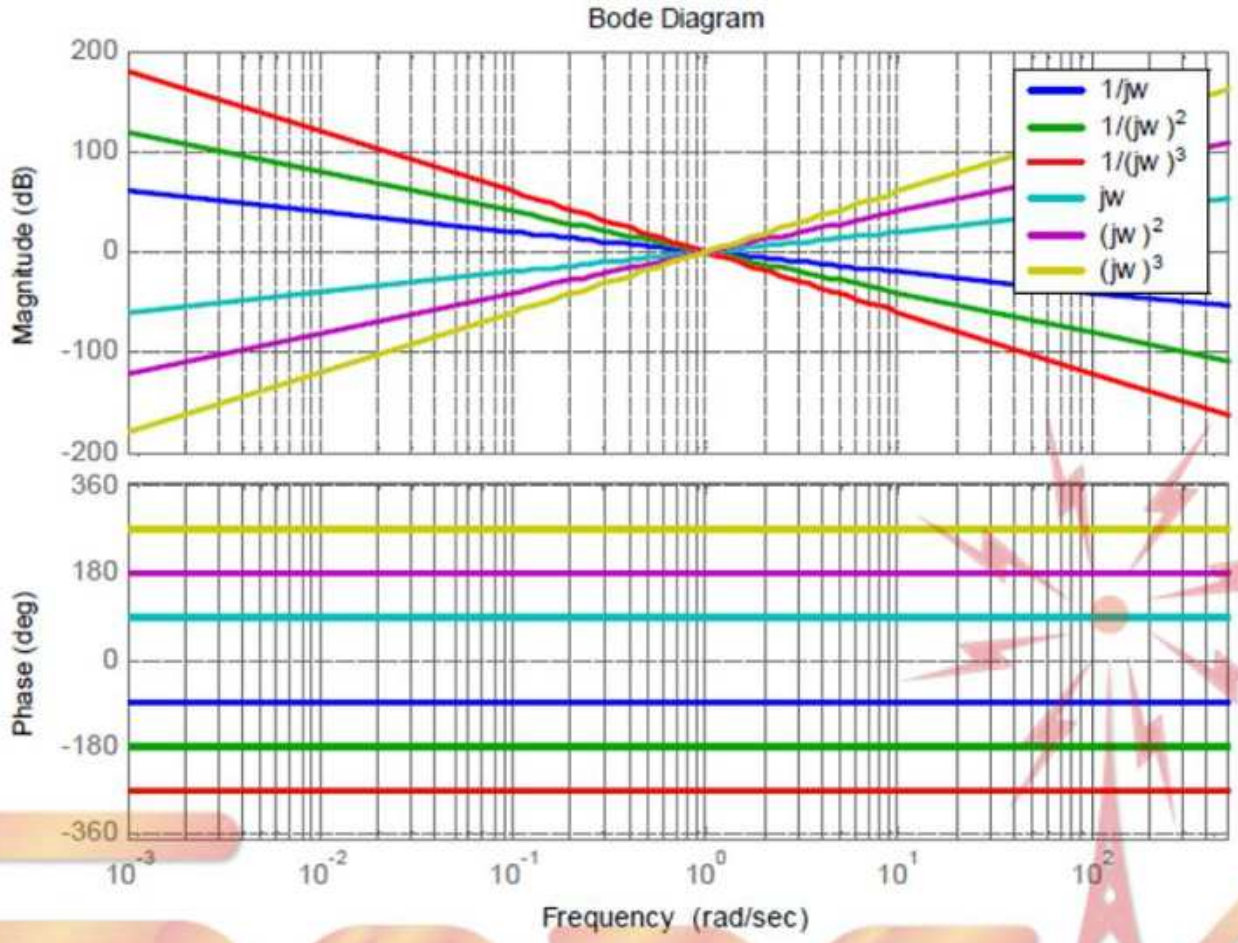
a. Changed from zero degrees up to ±180*P degrees
b. Zero degree
c. ±90*P degrees
d. Changed from zero degrees up to ±90*P degrees
e. +90*P degree

Correct answer: ±90*P degrees

POLYTECHNIC

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل



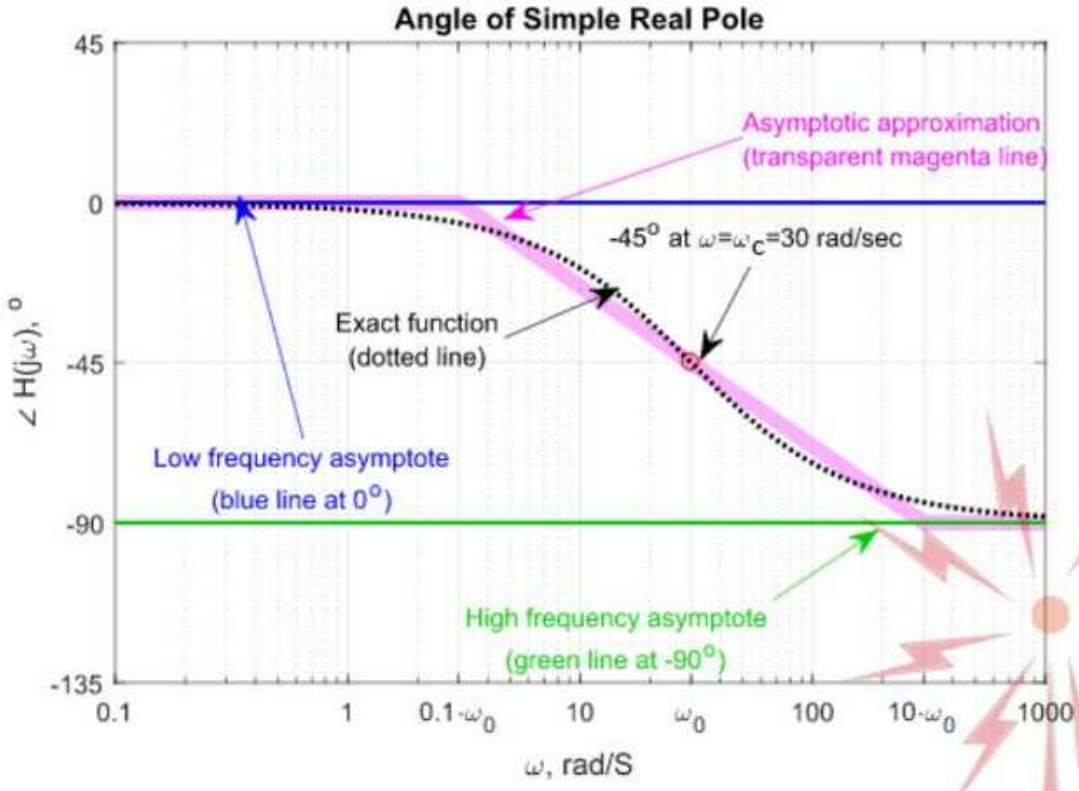
296. The phase shift of a first order lag (factor/ elements) when the frequency is changed from zero up to infinity is:
Or
The phase shift of transfer function $G(s)H(s) = Ts + 1$ when the frequency is changed from zero up to infinity is:
Or
In frequency response, the phase angle of a first order lag (factor/ elements) when the frequency is changed from zero up to infinity is:
Or
In frequency response, the phase angle of transfer function $G(s)H(s) = Ts + 1$ when the frequency is changed from zero up to infinity is:

a. Changed from zero degrees up to +90 degrees
b. Changed from zero degrees up to -180 degrees
c. Changed from zero degrees up to -90 degrees
d. Changed from zero degrees up to +180 degrees

Correct answer: Changed from zero degrees up to -90 degrees

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل



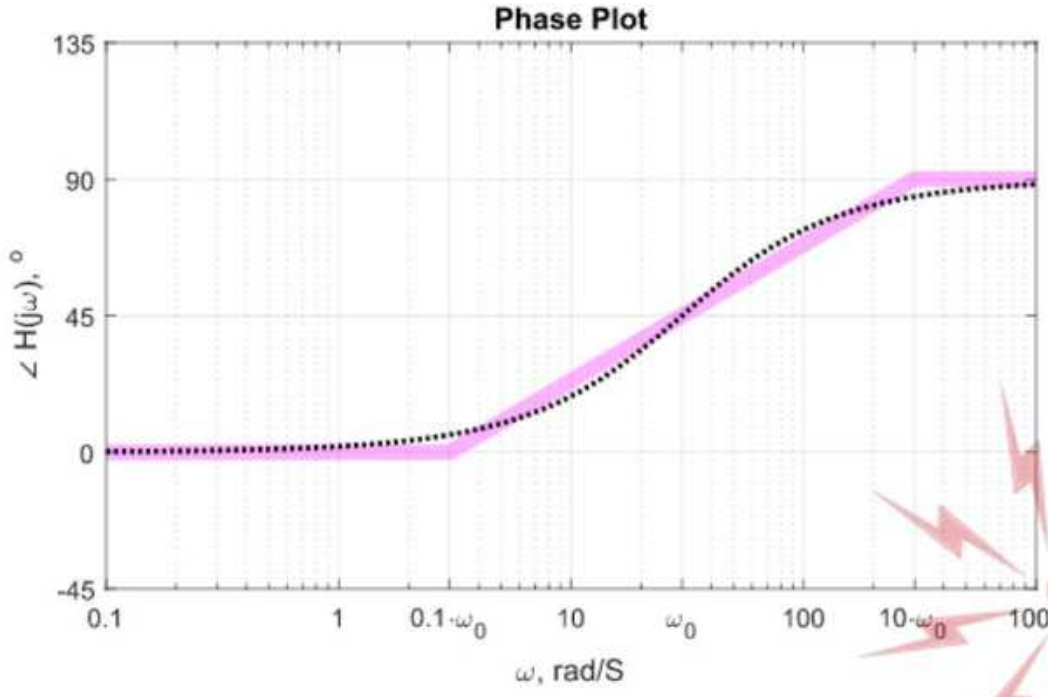
297. The phase shift of a first order lead (factor/ elements) when the frequency is changed from zero up to infinity is:
Or
The phase shift of transfer function $G(s)H(s) = \frac{1}{(Ts+1)}$ when the frequency is changed from zero up to infinity is:
Or
In frequency response, the phase angle of a first order lead (factor/ elements) when the frequency is changed from zero up to infinity is:
Or
In frequency response, the phase angle of transfer function $G(s)H(s) = \frac{1}{(Ts+1)}$ when the frequency is changed from zero up to infinity is:

a. Changed from zero degrees up to -180 degrees
b. Changed from zero degrees up to +90 degrees
c. Changed from zero degrees up to +180 degrees
d. Changed from zero degrees up to -90 degrees

Correct answer: Changed from zero degrees up to +90 degrees

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل



298. The phase shift of a second order lead (factor/ elements) when the frequency is changed from zero up to infinity is:
Or
In frequency response, the phase angle of a second order lead (factor/ elements) when the frequency is changed from zero up to infinity is:

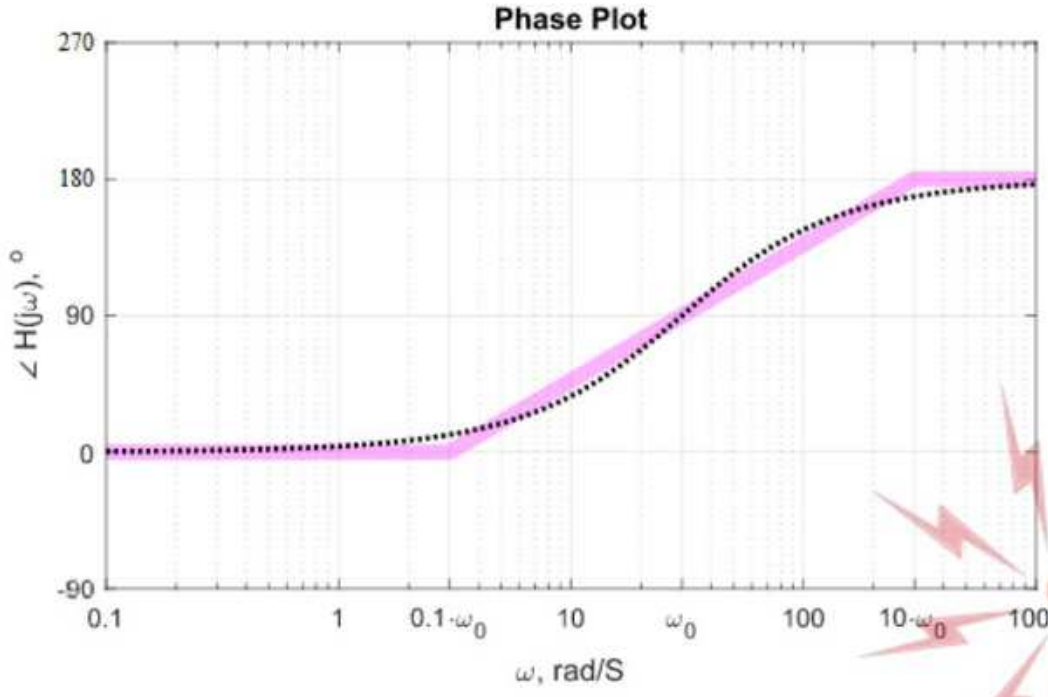
a. Changed from zero degrees up to -90 degrees
b. Changed from zero degrees up to -180 degrees
c. Changed from zero degrees up to +90 degrees
d. Changed from zero degrees up to +180 degrees

Correct answer: Changed from zero degrees up to +180 degrees

POLYTECHNIC

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل



299. Consider the system represented by the equation given below. The phase shift when the frequency is changed from zero up to infinity is:

$$G(s)H(s) = \frac{250}{s^5\,(s-1)(s\,+\,10)(s\,+\,3)}$$

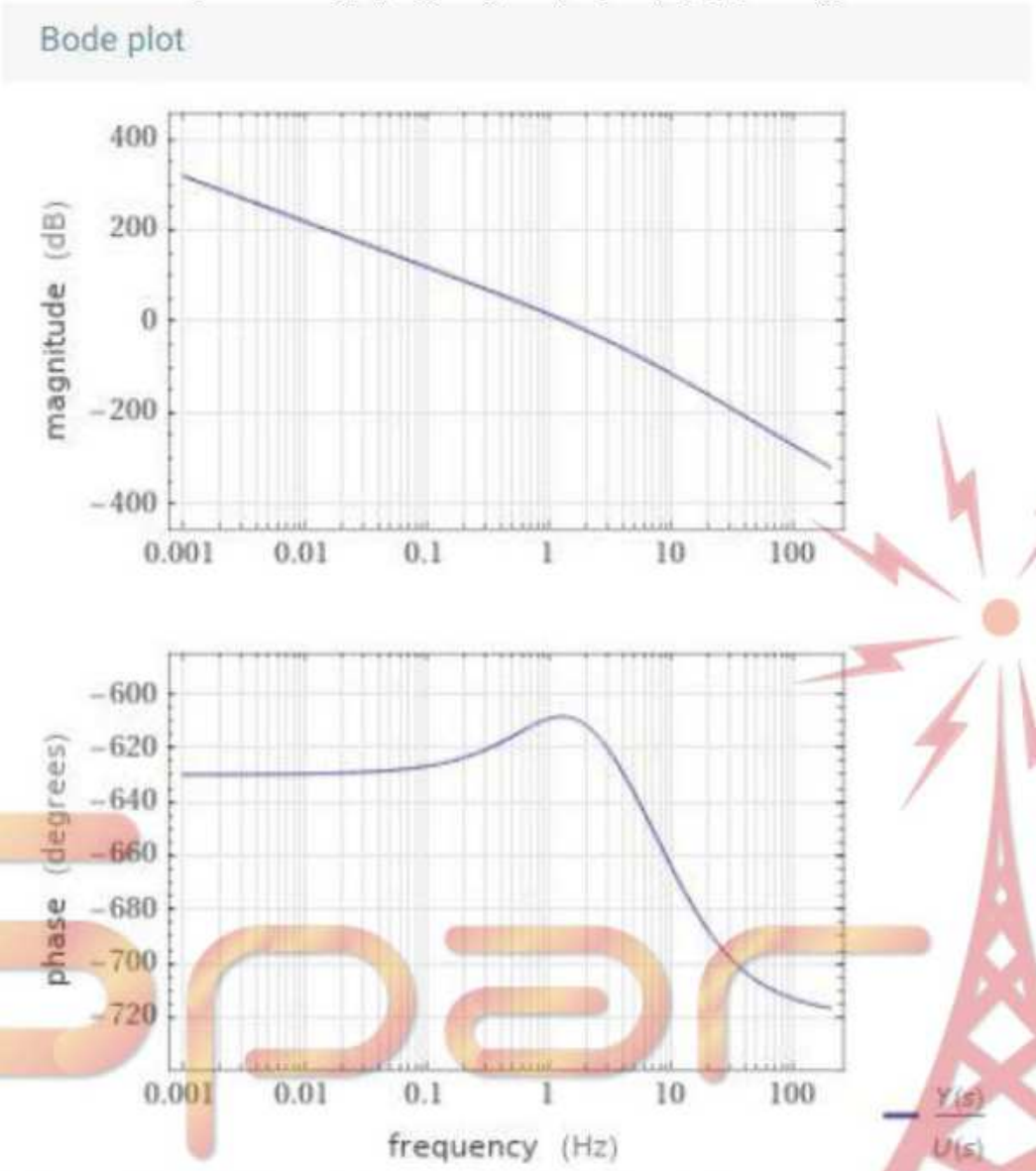
a. Changed from -630 degrees up to -720 degrees
b. Changed from zero degrees up to -720 degrees
c. Changed from -450 degrees up to -720 degrees
d. None of the above

Correct answer: Changed from -630 degrees up to -720 degrees

POLYTECHNIC

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل



300. Consider the system represented by the equation given below. The phase shift when the frequency is changed from zero up to infinity is:

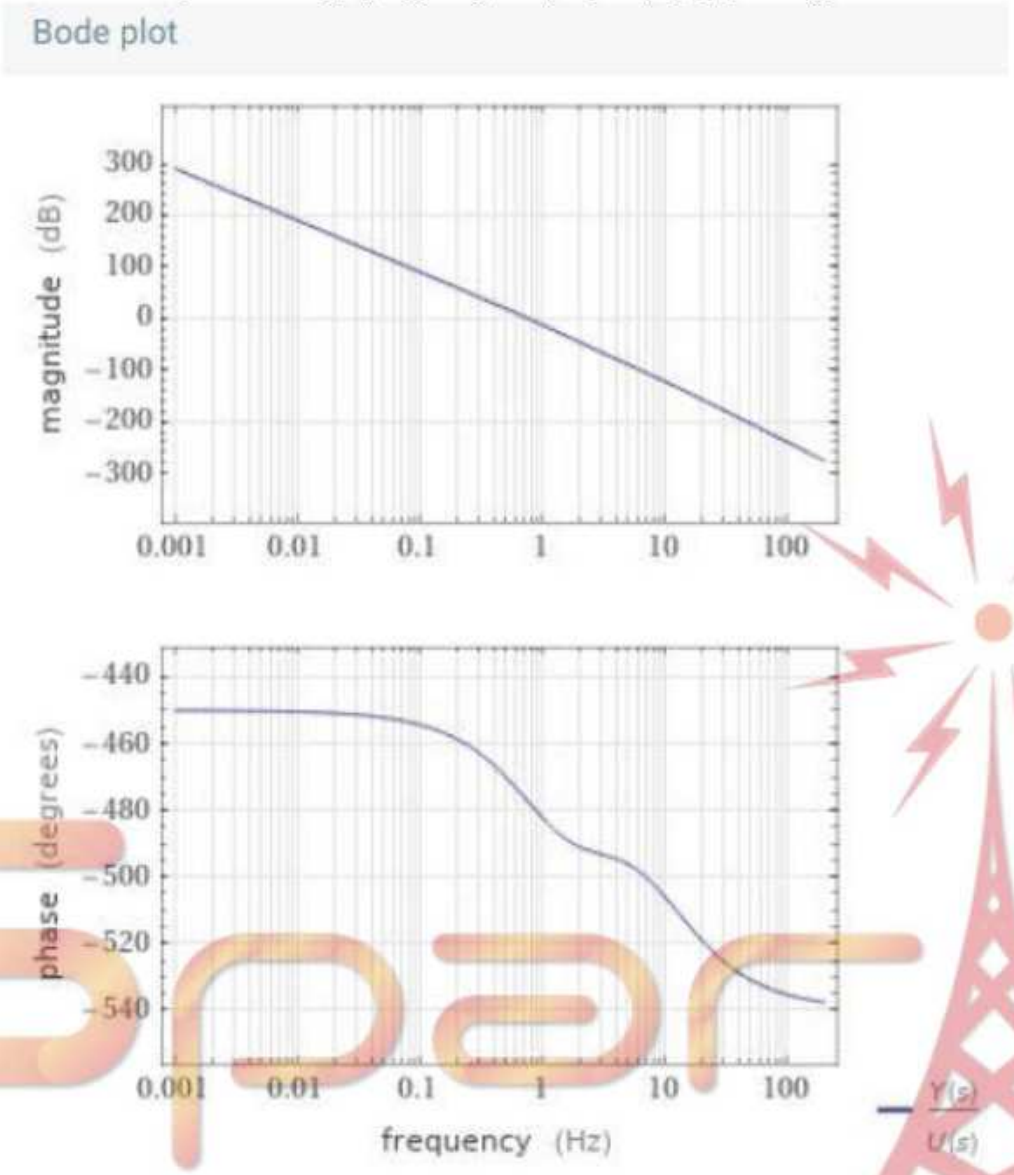
$$G(s)H(s) = \frac{(s + 3)}{s^5\,(s + 1)(s + 10)}$$

a. Changed from -450 degrees up to -630 degrees
b. Changed from zero degrees up to -630 degrees
c. Changed from -450 degrees up to -540 degrees
d. None of the above

Correct answer: Changed from -450 degrees up to -540 degrees

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل



301. The corner frequency of lead first order (factor/ elements) is:

a. 0
b. $\frac{1}{T}$
c. inf
d. T

Correct answer: $\frac{1}{T}$

302. What is the frequency at which the phase angle of the open-loop transfer function equals -180°?

a. Resonant frequency
b. Phase crossover frequency
c. Cut-off frequency
d. Gain crossover frequency

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

Correct answer: Phase crossover frequency

303. The frequency at which the phase of the system acquires ____ is known as 'Phase crossover frequency'.

a. $-90°$
b. $90°$
c. $-180°$
d. $180°$

Correct answer: $-180°$

304. What is the reciprocal of the magnitude of $G(jw)$ at the phase cross over frequency?

a. Phase Crossover Frequency
b. Gain margin
c. Phase margin
d. Gain Crossover Frequency

Correct answer: Gain margin

305. What is that amount of additional phase lag at the gain crossover frequency required to bring the system to the verge of instability.

a. Phase margin
b. Phase Crossover Frequency
c. Gain margin
d. Gain Crossover Frequency

Correct answer: Phase margin

Spark POLYTECHNIC

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

Applying Root-Locus Technique for Control System Analysis

306. For root locus which of the following are the end points:

a. Open loop poles
b. Closed loop zeros
c. Closed loop poles
d. Open loop zeros

Correct answer: Open loop zeros

307. What is the number of the root locus segments which do not terminate on zeroes?

a. The number of poles
b. The sum of the number of poles and the number of the zeroes
c. The number of zeroes
d. The difference between the number of poles and zeroes

Correct answer: The difference between the number of poles and zeroes

Spark POLYTECHNIC

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

Appendix I

Key Concept: Analogous Quantities

Electrical Quantity	Mechanical Analog I (Force-Current)	Mechanical Analog II (Force-Voltage)
Voltage, e	Velocity, v	Force, f
Applied voltage, e(t)		Applied force, $f(t)$
Current, i	Force, f	Velocity, v
Mesh current, $i(t)$		Velocity, $v(t)$
Resistance, R	Lubricity, 1/B (Inverse friction)	Friction, B
Resistor, f_v $Ohms$		Viscous damper, f_v
Capacitance, C	Mass, M	Compliance, 1/K (Inverse spring constant)
Inductance, L	Compliance, 1/K (Inverse spring constant)	Mass, M
Transformer, N1:N2	Lever, L1:L2	Lever, L1:L2

Appendix II

Routh-Hurwitz stability criterion Matlab code (the code has been taken from mathworks website)
Code link: https://www.mathworks.com/matlabcentral/fileexchange/17483-routh-hurwitz-stability-criterion
The code finds:

1. **Routh-Hurwitz Table**
2. **If the system is stable or not**
3. **Number of right-hand side poles**
4. **Polynomial coefficients roots**

SPARK POLYTECHNIC

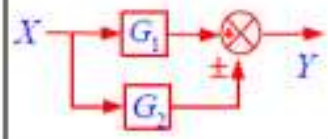
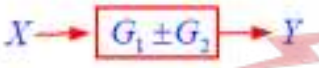
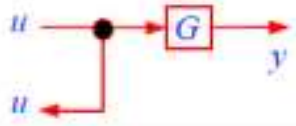
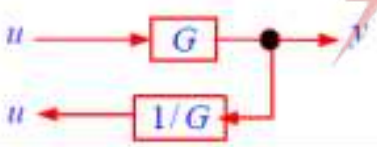
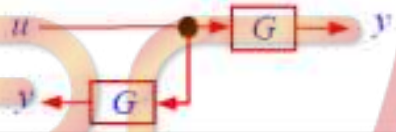
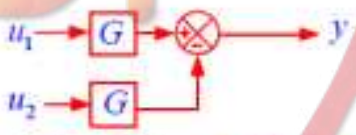
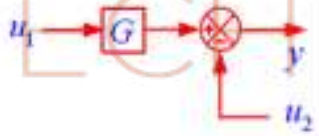
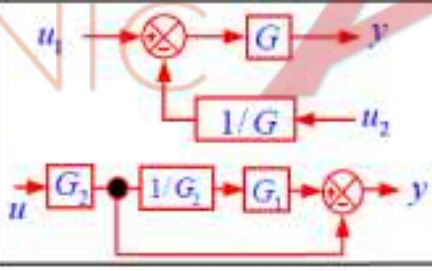
بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

Appendix III

For more examples about Block Diagram Reduction & Routh's Stability Criterion:
https://uotechnology.edu.iq/dep-MechanicsandEquipment/Lectures%20and%20Syllabus/Lectures/Same/Fourth%20Grade/Control%20and%20Instrumentation.pdf.pdf

Basic rules with block diagram transformation:

	Manipulation	Original Block Diagram	Equivalent Block Diagram	Equation
1	Combining Blocks in Cascade			$Y=(G_1G_2)X$
2	Combining Blocks in Parallel; or Eliminating a Forward Loop			$Y=(G_1\pm G_2)X$
3	Moving a pickoff point behind a block			$y = Gu$ $u = \frac{1}{G}y$
4	Moving a pickoff point ahead of a block			$y = Gu$
5	Moving a summing point behind a block			$e_2 = G(u_1 - u_2)$
6	Moving a summing point ahead of a block			$y = Gu_1 - u_2$ $y = (G_1 - G_2)u$

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

References

[1] Control Theory Slides for Prof. Ibrahim Khalaf Al-Abbas
[2] Modern Control Engineering Katsuhiko Ogata 5th ed
[3] Control Systems Engineering Norman S. Nise 6th ed
[4] Matlab Program
[5] Mathworks Website:
https://www.mathworks.com/
[6] https://www.wolframalpha.com/
[7] https://electronicscoach.com/block-diagram-reduction-rules.html
[8] https://www.careerride.com/mcq-tag-wise.aspx?Key=Signal%20Flow%20Graphs&Id=21
[9] https://instrumentationtools.com/root-locus-control-systems/
[10] https://www.sanfoundry.com/control-systems-questions-answers-root-locus-concepts-2/
[11] https://www.careerride.com/mcq-tag-wise.aspx?Key=Bode%20Plots&Id=21
[12] https://uotechnology.edu.iq/dep-MechanicsandEquipment/Lectures%20and%20Syllabus/Lectures/Same/Fourth%20Grade/Control%20and%20Instrumentation.pdf.pdf
[13] https://www.mathworks.com/matlabcentral/fileexchange/17483-routh-hurwitz-stability-criterion
[14] https://pt.slideshare.net/sagarkuntmal/block-diagram-examples
[15] https://web.engr.oregonstate.edu/~webbky/MAE4421_files/Section%202%20Block%20Diagrams%20&%20Signal%20Flow%20Graphs.pdf
[16] http://www.cs.mun.ca/~av/courses/7825-s16/manual_uploads/time2_quad.pdf
[17] https://deltamotion.com/support/webhelp/rmctools/Controller_Features/Control_Modes/Advanced_Control/Cascaded_Loops.htm
[18] https://www.sanfoundry.com/control-systems-questions-answers-signal-flow-graphs/
[19] https://slideplayer.com/slide/16305450/
[20] https://electronicscoach.com/stability-of-control-system.html
[21] https://www.mechanicaltutorial.com/control-system-multiple-choice-questions-and-answers-02-10

SPARK POLYTECHNIC

بسواعدنا نبنيها